

Optimizing Incomplete, Large-Scale and Sparse Multi-Graph Matching in Bioimaging

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Abstract—Multi-graph matching is a fundamental problem in computer vision. Our work is motivated by a challenging application in bioimaging, where *dozens* or even *hundreds* of 3D microscopy images of worms must be brought into correspondence. Existing datasets do not cover this large-scale regime, and virtually all existing methods are inapplicable because they assume a *complete* or *dense* problem setting. To support further research, our first contribution is a new large-scale dataset based on problem instances from bioimaging. Our second contribution is a comprehensive analysis of the two main multi-graph matching paradigms: *direct* and *permutation synchronization*-based formulations. We argue, in part by proof, that practical large-scale methods must explicitly address problem *sparsity* and *incompleteness*. Since standard permutation synchronization approaches fail in this setting, we further introduce a *sparse* permutation synchronization paradigm. Our final contribution is GREEDA, a general method for sparse and incomplete problems that can be instantiated across cost orders and paradigms. While our paper focuses on objective functions up to quadratic order, GREEDA is inherently generalizable to arbitrary orders. On larger, sparse instances, GREEDA outperforms competing methods in both objective value and runtime. For example, for moderately-sized problems based on 30 worm images GREEDA produces a high-quality solution within 2 minutes, whereas competitors require at least half an hour and yield far worse results. On smaller dense problems, GREEDA remains on par with leading methods while being an order of magnitude faster.

Index Terms—multi-graph matching, assignment problem, combinatorial optimization

I. INTRODUCTION

ESTABLISHING correspondences across multiple finite sets is a fundamental combinatorial problem important for, e.g., 3D model retrieval [1], shape matching [2, 3], statistical shape modeling [4, 5], federated learning [6], and genomic data analysis [7]. Typically, each *object*, such as an image or shape, is represented by a set of *keypoints*, which stand for semantically meaningful parts of an object. The task is to bring these keypoints into correspondence, or *match* them. This matching must satisfy several conditions, see Fig. 1:

- *Uniqueness*. Each keypoint of a given object can be matched to *at most* one keypoint of any other object. If “at most” is substituted by “exactly”, one speaks of a *complete* and otherwise of an *incomplete* (or *partial*) matching. Due to occlusions or noise during the keypoint extraction process, the incomplete setting is prevalent in practice.

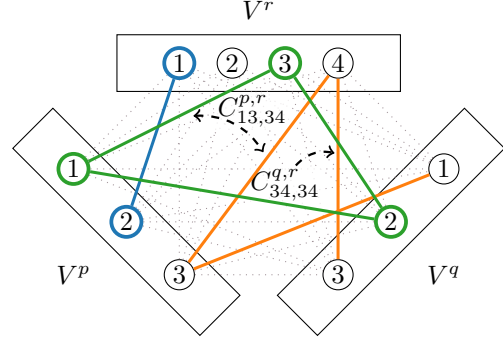


Fig. 1. **Multi-graph matching and cycle consistency.** Shown are three objects (solid rectangles) V^p, V^q, V^r , whose keypoints (circles) must be matched (edges). By construction, the setting is incomplete since V^r has 4 keypoints, whereas V^p, V^q only 3. The blue and green matchings are cycle-consistent and form cliques in the shown 3-partite graph. The orange matching is not cycle-consistent and as such does not form a clique. Examples of linear $C_{34,34}^{q,r}$ and quadratic $C_{13,34}^{p,r}$ costs (dashed arrows) and their associated matchings are also shown.

- *Cycle consistency*. If keypoint 1 in object V^p is matched to keypoint 2 in object V^q and keypoint 3 in object V^r , then keypoint 2 must be matched to keypoint 3. Similar transitivity conditions must hold for all matching cycles across arbitrary object subsets.
- *Costs*. Matchings must be *minimal* w.r.t. the given costs, quantifying the similarity between all keypoints. Costs for d objects decompose into a sum of costs for each of the $d(d-1)/2$ object pairs. The latter, in turn, are sums of *linear*, keypoint-to-keypoint costs and *quadratic*, keypoint-pair-to-keypoint-pair costs. Quadratic costs allow the modeling of mutual geometric relations between keypoints, considerably improving matching accuracy [8].

The problem combining these conditions is known as *multi-graph matching (MGM)*, and it is NP-hard [9, 10] in general. Note, the term *graph* in multi-graph matching originally stems from interpreting objects as graphs and deriving quadratic costs from differences of respective edge attributes. We call an MGM problem *sparse* if most keypoint-to-keypoint matchings are *forbidden* (otherwise *allowed*). An MGM problem where all matchings are allowed is called *dense*.

Fig. 2 illustrates the primary application of our work, where 100 *C. elegans* worms have to be matched, and the keypoints are candidates for nuclei. This is a large-scale problem with about 45M possible keypoint correspondences, a factor of 90 more than previously considered.

Given a matching, biologist can extract various statis-

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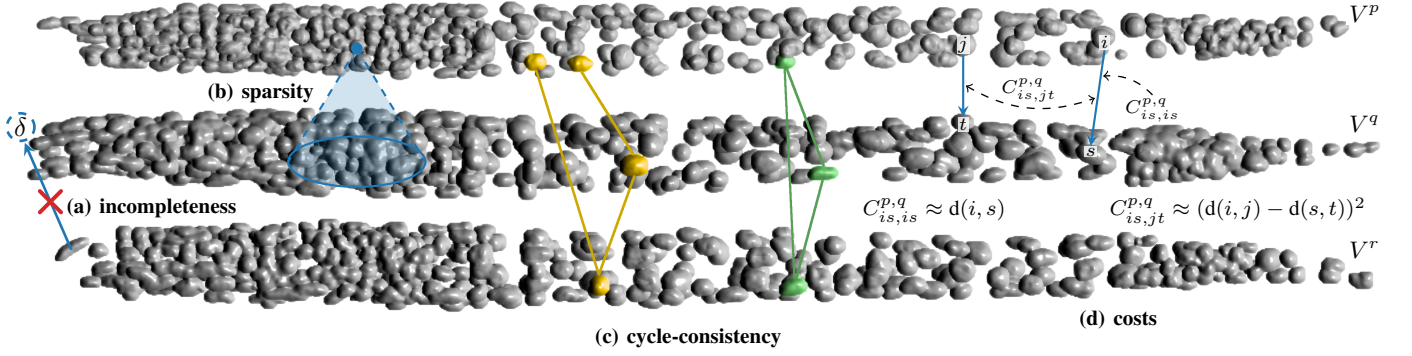


Fig. 2. **Three exemplary worms** from the considered datasets, shown from the side. Depicted are 3D renderings of the corresponding nuclei segmentation masks. Since *C. elegans* has a fixed number of cells in a given developmental stage (558 in the L1 larval stage [11]), there exists a 1-to-1 mapping between the cells of any two worms. However, due to occlusions and noise, the number of segmented nuclei varies across masks (top to bottom: 554, 532, 539), making correct correspondences challenging even for humans. **(a)** Because the number of segmented cells varies across worms, the formulation must be *incomplete*, *i.e.*, not every cell is required to be matched. This can be handled either *explicitly*, by adding a dummy node δ for each cell, or *implicitly*, by the underlying matching algorithm. We argue in Sec. VI that the former is prohibitively expensive in practice. **(b)** Due to the large number of cells, only *sparse* formulations allow for tractable optimization, see Sec. II. We restrict each cell to a local set of candidate matches, which both sparsifies the problem and prevents implausible long-range, *e.g.*, head-to-tail, assignments. In practice, only about 3% of all possible keypoint-to-keypoint matchings are allowed, hence, have finite costs. **(c)** Any valid solution must be *cycle-consistent*. The (Green) matching is cycle-consistent. The (Orange) one is cycle-inconsistent. **(d)** Costs for the corresponding multi-graph matching problem are computed from the segmentation masks. After centering each worm in a common coordinate system, linear terms $C_{is,ts}^{p,q}$ are defined from the distance $d(i, s)$ between the coordinates of cells i and s , favoring matchings at similar absolute positions. Additional linear terms are derived from cell shape similarity (not shown). Quadratic terms $C_{is,jt}^{p,q}$ are defined from the difference between the distances $d(i, j)$ and $d(s, t)$ between cell pairs and favor matchings that preserve relative geometry. That is, when matching i to s and j to t , the distance $d(i, j)$ should equal the distance $d(s, t)$. For details on the cost definitions and pre-processing steps such as straightening, see [12].

tics [13] about the worm. The three exemplary worms shown in Fig. 2 have different numbers of keypoints, and only a small fraction of keypoint-to-keypoint matchings are allowed. Hence, by construction, we are given an incomplete MGM problem with a sparse cost structure.

Despite finding a good matching, certainly runtime also plays a key role. Firstly, a fast method may scale better for problems with a larger number of objects. Secondly, better applied results can be obtained when the costs are learned, *i.e.*, the matchings are forced to fit ground truth in supervised (*e.g.*, [14]) or certain structural criteria in unsupervised cases (*e.g.*, [15, 16]). This means that the MGM task is inside a learning-loop and hence must be solved in reasonable runtime. Thirdly, MGM has so far been used rarely in practical applications, apart from bioimaging. We conjecture that the reason is a lack of good and fast MGM solvers.

Existing methods mainly follow two paradigms: direct and permutation synchronization-based formulations. Direct formulations address the quadratic MGM problem itself, whereas synchronization approaches first solve all pairwise GM problems independently and then find a cycle-consistent approximation by changing a minimal number of matchings. In the following, we refer to these as *synchronization* methods and omit the word *permutation*.

Our contributions are as follows:

- We analyze direct and synchronization-based multi-graph matching w.r.t. *sparsity* and *incompleteness*. In particular, we adapt the synchronization paradigm to *sparse* problems and prove that methods designed for the *complete* multi-graph matching problem are not practically applicable in *incomplete* settings due to prohibitive runtime

scaling.

- We present a new algorithm, *GREEDA*, for efficiently solving the incomplete multi-graph matching problem under sparse costs. It handles linear, quadratic, and, theoretically, higher-order objective functions, and can be instantiated as both a direct and a synchronization method through formulation-specific subroutines. Across both paradigms, it considerably and consistently outperforms the state-of-the-art on large-scale datasets in both runtime and objective value. On small-scale data, it remains competitive with or superior to leading methods in terms of objective value while being an order of magnitude faster.
- We construct and publish a new large-scale dataset based on existing problem instances [17]. Unlike previous datasets featuring at most 10 worms, our dataset contains up to 100 worms, contains 11 problem instances and is available at <https://doi.org/10.11588/DATA/RTINUU>.

This work builds upon the previous conference paper [18]. The central addition is the extension of our theoretical and experimental analysis to the synchronization paradigm, establishing *GREEDA* as a general-purpose method for multi-graph matching across problem sizes, sparsity regimes, and both direct and synchronization formulations. Additionally, we improve our swap local search, the main bottleneck of the earlier version, additionally evaluate on 11 new large-scale instances, and publish the related dataset. Furthermore, we now provide our algorithm as an easy-to-use Python package to facilitate the adoption of our method. Our code is available at <https://pypi.org/project/pylibmgm/>.

II. REAL-WORLD APPLICABILITY OF MGM METHODS

Large-scale MGMs in bioimaging applications, as in Fig. 2, have two decisive properties hardly covered by existing MGM algorithms: *Incompleteness* and *sparsity*. Whereas incomplete MGMs naturally arise due to noise in images and pre-processing steps like segmentation, sparsity allows to limit the size of the problem and encode important spatial relationships. Limiting the size is important as the number of quadratic costs grows as $O(n^4 d^2)$, where n is the number of keypoints and d the number of objects. Hence, dense MGMs with $n > 100$ become practically infeasible. Additionally, *e.g.*, forbidding the matching of nuclei in the head of a worm to those in its tail is an important natural prior (see Fig. 2).

Complete vs. incomplete MGM. Many approaches [10, 19–22] only treat complete MGM. Despite popular claims to the contrary, there is no straightforward and efficient way to apply them to the incomplete case. While a polynomial transformation from incomplete to complete graph matching exists [8], its often-mentioned generalization to MGM, *e.g.*, [20–22], is prohibitively expensive. Transforming an *incomplete* MGM problem with d objects, each with n keypoints, into a *complete* one results in nd keypoints in *each* of the d objects, see Sec. VI. This makes any complete MGM algorithm impractical. The following theorem proven in Sec. VI essentially states that there exists *no significantly better* transformation.

Theorem 1. *Let B be a complete MGM problem instance that is a clique-wise transformation of an incomplete MGM problem instance A . Let A have $|V|$ vertices in total. Then each object in B has at least $|V|$ vertices.*

In turn, our approach applies to incomplete MGMs *directly*.

Sparse problems. As defined above, an MGM problem is sparse if most keypoint-to-keypoint matchings are forbidden. In practice, forbidden matchings can be modeled by *infinite* costs that are only implicitly present in the problem description [8, 23, 24], rendering the latter *sparse*. However, many methods, especially the solution construction and synchronization ones, discussed in the following paragraph, do not apply to infinite costs. In particular, synchronization methods often fail to reconcile infinite costs with their assumptions, *e.g.*, by heavily relying on spectral methods [25–27]. In Sec. V, we discuss this incompatibility with infinite costs in more detail and propose the *sparse synchronization* task as a remedy. Overall, our method guarantees to return only *allowed*, *finite-cost* matchings for *sparse* problems, across direct and synchronization formulations.

III. RELATED WORK

Graph matching (GM) refers to the well-studied [8] special case of matching *two* objects. Similar to MGM, one distinguishes *complete* and *incomplete* GM. Complete GM is also known as the NP-hard [28] *quadratic assignment problem (QAP)* [29] or, if quadratic costs are zero, the polynomially solvable [30] *linear assignment problem (LAP)* [31]. MGM can be viewed as $d(d-1)/2$ GM problems between all object pairs, coupled via cycle consistency constraints. If we

wish to distinguish between cost orders similar to GM, we refer to (complete or incomplete) MGM as Multi-QAP and, respectively, as Multi-LAP if quadratic costs are zero. Unlike the LAP, the Multi-LAP is NP-hard [9, 10]. As a result of this coupled structure, GM solvers are an essential but interchangeable component of many MGM methods, including those considered in this work. We utilize the GM solver [32] as a GM subroutine since it shows the best results in the recent GM benchmark [8].

Multi-graph matching. Although MGM has not yet received the same attention as GM, several types of solvers have been proposed in recent years. In the following we review their ability to deal with incomplete and sparse MGMs. The works most closely related to our algorithm are additionally addressed in Sec. IV.

GM-solver-based primal heuristics is probably the largest class of MGM algorithms including ours. These methods usually consist of *construction* and *local search* subroutines that employ a GM solver as their decisive component. Construction methods are usually based on the *composition principle*: They assign a matching between two objects by composition via a third one. This third object is either iteratively chosen w.r.t. a metric combining costs and a (pairwise) consistency measure [33, 34] or is a node in a *spanning tree* [10, 20, 22] of the complete graph connecting all objects as its nodes. Whereas the first subclass does not even guarantee cycle consistency of the result, the second cannot address sparse problems as forbidden matchings are often selected through composition. Moreover, existing methods of this type consider only complete MGMs.

In contrast, our approach extends a feasible solution consisting of $k \leq d$ objects by one object in each iteration until the solution includes all objects, *i.e.*, $k = d$. In combination with the incomplete GM solver [32], it guarantees a cycle-consistent and allowed matching.

Local search subroutines are usually based on the observation that cycle-consistent matchings can be improved by re-matching one object to the remaining, already matched $(d-1)$ objects, see Fig. 4 in Sec. IV. It turns out that this re-matching constitutes a GM problem. Hence, one can iteratively re-match different objects utilizing a GM solver. We refer to this algorithm as *GM local search* (GM-LS), see Sec. IV-C for details. This idea was initially proposed by [35] for the closely related *multi-dimensional assignment problem (MDAP)* and then independently for different MGM variants [10, 20–22]. Similarly to the construction subroutine, existing works explicitly consider local search only for the complete problem.

In this work we *parallelize* GM local search and make it applicable to incomplete sparse MGMs. We also propose another local search method, referred to as (*clique*) *swap local search*, and so far unknown in the computer vision literature. The related work from the operations research is discussed in Sec. IV-D.

Synchronization approaches [36–38] first independently solve all pairwise GM problems, which induces a *cycle-inconsistent* matching. Afterward, they try to find a *cycle-*

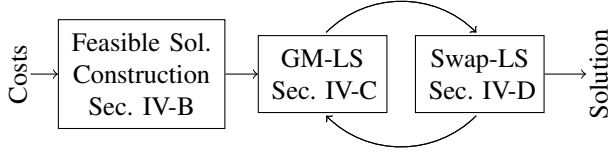


Fig. 3. **Conceptual diagram of our MGM method GREEDA.** *GREEDA* is composed of three parts – a construction heuristic (Sec. IV-B) and two local search heuristics *GM-LS* (Sec. IV-C), *SWAP-LS* (Sec. IV-D).

consistent approximation by changing a *minimal* number of matchings [25, 26, 39]. Such synchronization methods often serve as a subroutine for convex-relaxation-based MGM solvers, e.g., [19, 40]. However, they are *expensive* and prone to *suboptimal* decisions as they require the solution of $d(d-1)/2$ GM problems and ignore costs during the approximation. Due to the latter, existing synchronization methods also fail to deal with sparse problems because they often introduce “*blunders*” in choosing forbidden matchings. In contrast, our algorithm considers the problem costs in any of its stages and thus avoids blunders. At the same time, if required, it can also be used as a sparsity-aware synchronization method, see Sec. V.

Convex relaxation-based methods is probably the smallest subclass among existing MGM techniques. The work [19] considers lifting-free quadratic relaxation, but addresses complete problems only. The work [41] proposes a powerful semi-definite relaxation for incomplete MGMs, but its scalability is strongly limited due to variable lifting, as mentioned by the authors. The closest competitor for our method is the Lagrange relaxation-based method [40], able to deal with incomplete sparse problems. In Sec. VII we demonstrate, however, that our algorithm significantly outperforms [40] in terms of runtime and objective value.

IV. OUR METHOD: GREEDA

In Sec. IV-A, we formalize the MGM problem. In Sections IV-B to IV-D, we describe the individual building blocks of our method, summarized in Fig. 3, together with the corresponding technical contributions. We term our method *GREEDA* alluding to the greedy nature of our construction algorithm, see Sec. IV-B.

A. Formal problem definition

Graph matching concerns itself with matching two finite keypoint sets V^1 and V^2 , further referred to as *vertex sets* or *objects*. It considers the undirected complete bipartite graph $G = (V := V^1 \cup V^2, \bar{E} := V^1 \times V^2)$ with objects V^1 and V^2 as independent sets, where an edge $ij := (i, j) \in \bar{E}$ corresponds to matching vertex $i \in V^1$ to vertex $j \in V^2$. Although discussing undirected graphs, we write the edge set $\bar{E}$ as a Cartesian product $V^1 \times V^2$ to emphasize the two independent sets V^1 and V^2 . Due to their independence, we can always identify directed with undirected edges. An *incomplete* matching between object V^1 and V^2 is defined as subset of

edges $E \subset \bar{E}$ containing *at most one* incident edge for each vertex. *Complete* matchings, conversely, contain *exactly one* incident edge for each vertex, which demands equal cardinalities of both objects $|V^1| = |V^2|$. The goal of GM is to find minimal matchings w.r.t. given costs $C : (V^1 \times V^2)^2 \mapsto \mathbb{R}$. In the matrix identification $C(i, s, j, t) = C_{is,jt}$, diagonal entries $C_{is,is}$ describe *linear* costs, and off-diagonal entries *quadratic* costs, see Fig. 1. Linear costs $C_{is,is}$ penalize *vertex-to-vertex* correspondences, i.e., matching vertex $i \in V^1$ to vertex $s \in V^2$, whereby infinite costs $C_{is,is} = \infty$ *forbid* such a matching. Quadratic costs $C_{is,jt}$, in turn, penalize *vertex-pair-to-vertex-pair* correspondences, i.e., matching the pair $(i, j) \in V^1 \times V^1$ to the pair $(s, t) \in V^2 \times V^2$.

Multi-graph matching generalizes GM by matching multiple, $d \in \mathbb{N}_{\geq 3}$ objects V^p , $p \in [d] := [1, d] \cap \mathbb{N}$, w.r.t. costs between all pairs $C^{p,q} : (V^p \times V^q)^2 \mapsto \mathbb{R}$, $p \neq q$, $p, q \in [d]$. Instead of a bipartite graph, it considers the undirected complete d -partite graph $\bar{G} = (V := \cup_{p \in [d]} V^p, \bar{E})$ with objects V^p as independent sets. *Incomplete* multi-matchings are subsets of edges $E \subset \bar{E}$ s.t. any vertex is incident to *at most one* edge connecting the same objects. Similar to GM, *complete* multi-matchings require *exactly one* such edge and equal cardinalities of objects.

Cycle consistency. A multi-matching $E \subset \bar{E}$ is *cycle-consistent* if each *path* in E can be extended within E to a *cycle* with at most one vertex per object. As shown in [40], enforcing this for all 3-cycles is sufficient. That is, a multi-matching $E \subset \bar{E}$ is cycle-consistent iff $ij \in E$ and $jk \in E$ imply $ik \in E$ for all $i, j, k \in V$, see Fig. 1.

Clique representation. In [42], it was shown that a multi-matching $E \subset \bar{E}$ is cycle-consistent iff the corresponding subgraph $G = (V, E)$ is a *union of cliques*, i.e., there exist partitions $\{Q_l\}_{l \in L}$ of vertices V and $\{E_l\}_{l \in L}$ of edges E indexed via the same finite set L , s.t. for each index $l \in L$ the subgraph $G|_{Q_l}$ restricted to part Q_l is a *clique* with edges E_l . Therefore, cycle-consistent multi-matchings are induced by vertex partitions $\mathcal{Q} \subset 2^V$ where any part $Q \in \mathcal{Q}$ contains at most one element per object V^p , i.e., $|Q \cap V^p| \leq 1$, see Fig. 1. They translate to multi-matchings by considering elements of the same part Q as matched to each other. We call such vertex partitions *feasible* and denote the set of feasible vertex partitions or *solutions* as $\mathcal{Q}$. The set of feasible partitions over the vertices $V^D := \cup_{p \in D} V^p$ of an object *subset* $D \subseteq [d]$ is denoted by $\mathcal{Q}^D$. For simplicity, partitions permit the empty set. Abusing terminology, we refer to parts $Q \in \mathcal{Q}$ of a solution as *cliques*. The object subset actually covered by such a clique $Q \in \mathcal{Q}$ is denoted as $D(Q) \subseteq [d]$, and a clique’s vertex belonging to object $p \in D(Q)$ as Q^p , i.e., $\{Q^p\} := Q \cap V^p$, see Fig. 4.

The MGM objective is to find cycle-consistent multi-matchings minimizing the sum of all costs, i.e.,

$$\min_{\mathcal{Q} \in \mathcal{Q}} C(\mathcal{Q}) := \sum_{Q, R \in \mathcal{Q}} \sum_{p, q \in D(Q) \cap D(R)} C_{Q^p Q^q, R^p R^q}^{p, q}, \quad (1)$$

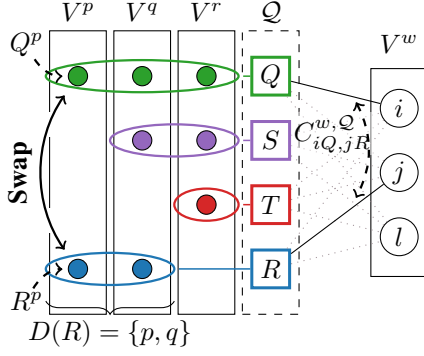


Fig. 4. **Feasible solution construction and swap local search.** Depicted are three cycle-consistently matched objects V^p, V^q, V^r . Matched vertices are of the same color, horizontally aligned, and decomposed into cliques Q, S, T, R (ellipses), yielding a partial solution Q (dashed rectangle). As example, the clique R spans the objects $D(R) = \{p, q\}$, where its vertex R^p belongs to object V^p . The costs $C_{iQ,jR}^{w,Q}$ for matching vertex i and j of object V^w to cliques Q and R of the partial solution Q are also shown. We consider two types of swaps (solid arrow), see Sec. IV-D: The *vertex swap* fixes the object p and interchanges the vertices Q^p and R^p . The *2-clique swap* is more global and jointly optimizes over all possible vertex swaps between cliques Q and R .

where we assume $C^{p,q} = C^{q,p}$ and count this cost for each pair of objects $p, q \in [d]$, only once by requiring $p < q$. Note that the formulation in Eq. (1) implicitly assumes zero costs for *not* matching a vertex.

Alternative formulations. The clique formulation from Eq. (1) is non-standard. Many formulations [38, 40, 43] represent (multi-)matchings $E \subset \bar{E}$ by *partial permutation matrices* X , where $X_{is} = 1$ iff $is \in E$. Others [27, 44, 45] view MGM as a *labeling problem*, where each vertex of the same object must be assigned a different label – commonly called *universe point*. Vertices with the same label are matched to each other, *i.e.*, comprise a clique. We use the clique formulation because it allows the most natural description of our algorithm.

B. Feasible solution construction

Basic algorithm. The basic construction Alg. 1 obtains feasible solutions by solving a *chain* of GM problems. Given a *random ordering* of objects $[d]$, it iteratively extends a *partial solution*, which, in the k -th iteration, matches the objects $V^1, \dots, V^k$. The matching E is the solution of the GM problem with costs $C_{iS,jT}^{k+1,Q}$, $i, j \in V^{k+1}$, $S, T \in Q$, stemming from the *summation* over a clique’s individual elements

$$C_{iS,jT}^{k+1,Q} := \sum_{q \in D(S) \cap D(T)} C_{iS^q,jT^q}^{k+1,q}. \quad (2)$$

The partial solution is extended by adding matched vertices $i \in V^{k+1}, iS \in E$ to their assigned cliques $\{i\} \cup S \in Q'$. If a vertex $i \in V^{k+1}$ is unmatched, *i.e.*, $\forall jS \in E : j \neq i$, it is added as a singleton $\{i\} \in Q'$. If a clique $S \in Q$ is unmatched, *i.e.*, $\forall iT \in E : S \neq T$, it remains unchanged $S \in Q'$. We denote the (partial) solution resulting from this *merge* as $Q' = \text{merge}(V^{k+1}, Q; E)$.

Alg. 1 has six notable properties: (1) It guarantees cycle consistency. (2) It can use virtually any GM solver as a black-box. (3) It is independent of the cost order and thus applicable to higher-order problems as long as the underlying

Algorithm 1: Basic solution construction.

```

 $Q \leftarrow \bigcup_{i \in V^1} \{i\}$ 
for  $k \in \{1, \dots, d-1\}$  do
    // compute matching  $E$  between object  $(k+1)$ 
    // and the already matched  $[k]$ 
     $E \leftarrow (\text{Approx.}) \text{ solve GM with costs } C^{k+1,Q} \text{ (Eq. (2))}$ 
     $Q \leftarrow \text{merge}(V^{k+1}, Q; E)$ 

```

GM solver is applicable. (4) It is parameter-free w.r.t. the number of cliques. (5) It scales linearly in the number of objects $d \in \mathbb{N}_{\geq 3}$. (6) It is randomized, hence the best solution from its multiple parallel runs can be kept.

Extensions and parallelization. We propose two extensions of Alg. 1: *Incremental* and *parallel* construction.

Incremental construction addresses the problem of *error propagation*. Errors during early matchings “propagate” along the chain of pairwise problems, sway later matchings, and worsen the final solution. Because MGM problems’ restrictions to object subsets are again MGM problems, we propose to weaken such propagations by using a *better* (but potentially more expensive) *multi-matching solver* for the first $s \in [d]$ objects $[s]$.

Parallel construction improves the linear scaling w.r.t. the number of objects. It generalizes the chain of pairwise problems to *binary trees*, specifically *leaf-labeled, ordered, binary trees* with *leaf label set* $[d]$. Leaf labels associate leaves with objects. Each tree vertex corresponds to a (partial) solution $Q \in \mathbb{Q}^D$ matching its descendant leaves $D \subseteq [d]$. In analogy to the basic variant, a tree vertex’s (partial) solution is obtained by first solving a GM problem that matches the cliques of its children $\mathcal{A} \in \mathbb{Q}^{D_A}$, $\mathcal{B} \in \mathbb{Q}^{D_B}$ and then merging matched cliques. Pairwise costs $C_{IS,JT}^{A,B}$, $I, J \in \mathcal{A}, S, T \in \mathcal{B}$, are again obtained by summing costs of a clique’s individual elements,

$$C_{IS,JT}^{A,B} := \sum_{p \in D(I) \cap D(J)} \sum_{q \in D(S) \cap D(T)} C_{IP^pS^q,J^pT^q}^{p,q}. \quad (3)$$

Ultimately, the root corresponds to a final solution. GM problems of tree vertices at the same level can be solved in *parallel*. Therefore, *balanced* trees yield the best acceleration – in sequence only $O(\log(d))$ instead of $O(d)$ problems need solving. Additionally, most properties of the basic construction transfer to the parallel version: It is *randomized* w.r.t. the leaf order, allows *plug-&-play* of pairwise solvers, guarantees *cycle consistency*, and is *parameter-free* w.r.t. the number of cliques. Pseudocode and further details are provided in Suppl. A.

Our novelty and related work. The basic solution construction Alg. 1 was first mentioned in [46] and later used in [35] for complete Multi-LAPs seen as a special case of the MDAP. Experiments in [35] also demonstrate its superiority over star-shaped spanning tree approaches. Its application to *Multi-QAPs* and its *parallelization* using binary trees are new. Incremental construction conceptually resembles an idea proposed in [10], where, as part of a spanning tree heuristic, a

Algorithm 2: Basic GM local search.

Input: Solution $\mathcal{Q} \in \mathbb{Q}$, Object Sequence $(p_k)_{k \in \mathbb{N}}$
 $k \leftarrow 1$
while stopping criterion not met **do**
 // split object k from the matching $\mathcal{Q}$
 $\mathcal{Q}' \leftarrow \{Q \setminus V^{p_k} | Q \in \mathcal{Q}\}$
 // compute matching E between object k and the rest
 $E \leftarrow (\text{Approx.}) \text{ solve GM with costs } C^{p_k, \mathcal{Q}'} \text{ (Eq. (2))}$
 $\mathcal{Q}' \leftarrow \text{merge}(V^{p_k}, \mathcal{Q}'; E)$
 if $C(\mathcal{Q}') < C(\mathcal{Q})$ **then**
 $\mathcal{Q} \leftarrow \mathcal{Q}'$ // accept if profitable
 $k \leftarrow k + 1$

local search is run on the hitherto constructed partial solution. To summarize, our contribution is to introduce the prior art from operations research [35, 46], applying it to *Multi-QAPs* instead of Multi-LAPs, and proposing new extension and parallelization schemes.

C. GM local search

Basic algorithm. The basic GM local search step comprises *splitting* and *merging* an object V^p from and to a given solution $\mathcal{Q} \in \mathbb{Q}$ as defined by Alg. 2. An object V^p is split by *subtracting* $Q \setminus V^p$ its vertices from all cliques $Q \in \mathcal{Q}$. Merging it back to the split solution works just as in Sec. IV-B by solving a GM problem with costs from Eq. (2) whose solution dictates the merge. While only accepting improvements, the basic GM local search Alg. 2 repeats this step along an *object sequence* $(p_k)_{k \in \mathbb{N}}, p_k \in [d]$, until a suitable stopping criterion is met. In practice, we run Alg. 2 cyclically along the random object ordering $[d]$ used in Alg. 1 as long as the solution improves. Like Alg. 1, the pairwise solver is *blackboxed* and solutions are always *cycle-consistent*.

Extensions and parallelization. A known extension [35] of this local search is to split a solution $\mathcal{Q} \in \mathbb{Q}$ along a subset of objects $D \subset [d]$. As a result, two partial solutions $\{Q \cap V^D | Q \in \mathcal{Q}\} \in \mathbb{Q}^D$, $\{Q \cap V^{[d] \setminus D} | Q \in \mathcal{Q}\} \in \mathbb{Q}^{[d] \setminus D}$ are merged using Eq. (3). Since these extensions performed worse than the base in preliminary experiments, we did not investigate them further.

Our proposed *parallelized* search step consists of two passes. First, it runs the basic search step for *all* objects $p \in [d]$ in *parallel*, splitting d objects $\mathcal{Q}^p := \{Q \setminus V^p | Q \in \mathcal{Q}\}$, generating d matchings $E^p \subset V^p \times \mathcal{Q}^p$, yielding d proposed solutions, but ultimately leaving the input solution $\mathcal{Q} \in \mathbb{Q}$ untouched. Second, it splits and merges objects according to the collected matchings $E^p, p \in [d]$, in *ascending* order of their proposed solutions' objective value – only accepting profits. The more matchings yield profit despite being “outdated” (because prior matchings changed the solution), the more *acceleration* is generated over the basic search step. Both the basic and parallel search step *sequentially* solve only *one* GM problem. Additionally, the parallelized step always accepts the *most profitable* proposal though this is only a *secondary* source

Algorithm 3: Swap local search.

Input: Solution $\mathcal{Q} \in \mathbb{Q}$
while stopping criterion not met **do**
 for $Q, R \in \mathcal{Q}$ **do**
 $x^* \leftarrow \text{Approximately solve Eq. (8) for cliques } Q, R$
 // swap the vertices according to x^*
 $Q', R' \leftarrow \text{swap}(x^*, Q, R)$
 // update the clique representation
 $\mathcal{Q}' \leftarrow (\mathcal{Q} \setminus \{Q, R\}) \cup \{Q', R'\}$
 if $C(\mathcal{Q}') < C(\mathcal{Q})$ **then**
 $\mathcal{Q} \leftarrow \mathcal{Q}'$ // accept if profitable

of acceleration, as demonstrated in Sec. VII-A. Further details can be found in Suppl. B.

Our novelty and related work. Again, the basic Alg. 2 was first mentioned in [35] targeting complete Multi-LAPs as special case of the MDAP. In [35], its extension to object subsets is investigated and our preliminary findings are confirmed for the Multi-LAP. In computer vision, it was independently proposed in [10, 20] for Multi-LAPs and in [21, 22] for Multi-QAPs, both only complete. Our parallelized extension is novel.

D. Swap local search

Vertex swap. Given a solution $\mathcal{Q} \in \mathbb{Q}$, *swapping* vertices of the same object between two cliques $Q, R \in \mathcal{Q}$ maintains feasibility. While fixing one object $p \in [d]$, a *vertex swap* subtracts a vertex covered by either clique from said clique and adds it to the other, yielding two new cliques, Q' and R' , see Fig. 4. For example, if both cliques Q and R cover the fixed object $p \in [d]$ and the vertices Q^p and R^p are swapped, it results in $Q' = (Q \setminus \{Q^p\}) \cup \{R^p\}$ and $R' = (R \setminus \{R^p\}) \cup \{Q^p\}$. If only one clique covers the fixed object p , only one vertex changes its clique, *i.e.* $Q' = (Q \setminus \{Q^p\})$ and $R' = (R \cup \{Q^p\})$. Finally, both cliques remain unchanged if none covers the object. After a swap, cliques still cover at most one vertex per object. The new solution $\mathcal{Q}' = (\mathcal{Q} \setminus \{Q, R\}) \cup \{Q', R'\}$ is, thus, also feasible. Furthermore, the objective value change induced by a vertex swap decomposes into a sum of objective value changes $\delta C_{\text{swap}}^{p,q}(Q, R)$ between the fixed object $p \in [d]$ and all other objects $q \in [d] \setminus \{p\}$, *i.e.*,

$$C(\mathcal{Q}') - C(\mathcal{Q}) = \sum_{q \in [d] \setminus \{p\}} \delta C_{\text{swap}}^{p,q}(Q, R). \quad (4)$$

For each other object $q \in [d] \setminus \{p\}$, the corresponding objective value change is the difference between the cost terms involving the cliques Q', R' w.r.t. the solution $\mathcal{Q}'$ and the cost terms involving the cliques Q, R w.r.t. the solution $\mathcal{Q}$, *i.e.*,

$$\delta C_{\text{swap}}^{p,q}(Q, R) := C_{\{Q', R'\}}^{p,q}(\mathcal{Q}') - C_{\{Q, R\}}^{p,q}(\mathcal{Q}), \quad (5)$$

where the sum of the respective cost terms $C_{\{Q,R\}}^{p,q}(\mathcal{Q})$ for the solution $\mathcal{Q}$ is given by

$$C_{\{Q,R\}}^{p,q}(\mathcal{Q}) := \sum_{S \in \{Q,R\}} \left(\sum_{T \in \{Q,R\}} C^{p,q}(S,T) + \sum_{T \in \mathcal{Q} \setminus \{Q,R\}} (C^{p,q}(S,T) + C^{p,q}(T,S)) \right). \quad (6)$$

Here, $C^{p,q}(S,T)$ denotes the costs of matching the objects p, q through the cliques $S, T \in \mathcal{Q}$, given by

$$C^{p,q}(S,T) = \begin{cases} C_{S^p S^q, T^p T^q}^{p,q} & ; p < q \wedge p, q \in D(S,T) \\ C_{S^q S^p, T^q T^p}^{q,p} & ; q < p \wedge p, q \in D(S,T) \\ 0 & ; \text{else} \end{cases}, \quad (7)$$

where $D(S,T) := D(S) \cap D(T)$ are all objects covered by the cliques $S, T \in \mathcal{Q}$. A more detailed derivation is given in Suppl. C.

2-clique swap. We leverage the decomposition in Eq. (4) to jointly consider *all possible* swaps between two cliques $Q, R \in \mathcal{Q}$. A 2-clique swap is a set of binary decision variables $x = \{x^p\}_{p \in [d]} \in \{0, 1\}^d$, deciding for every object $p \in [d]$, whether the vertex swap fixing it should be performed ($x^p = 1$), or not ($x^p = 0$). Performing all vertex swaps specified by a 2-clique swap $x \in \{0, 1\}^d$ yields, again, two new cliques Q' and R' , which we denote by $Q', R' = \text{swap}(x; Q, R)$. The optimal 2-clique swap x^* is a solution of the following quadratic pseudo-boolean optimization (QPBO) problem [47]:

$$x^* \in \arg \min_{x \in \{0,1\}^d} \sum_{p \in [d]} x^p \left(\sum_{q \in [d]} \bar{x}^q \cdot \delta C_{\text{swap}}^{p,q}(Q, R) \right), \quad (8)$$

where $\bar{x} = (1 - x)$ denotes negation. The implicit product $x^p \bar{x}^q$ allows to count only those $\delta C_{\text{swap}}^{p,q}(Q, R)$ where exactly one vertex pair, either Q^p, R^p or Q^q, R^q was swapped between them, but not both, as the latter does not change the costs. This problem is NP-hard [48], which is why we address it using a state-of-the-art approximative solver QPBO-I [48]. Our swap local search Alg. 3 iterates over all clique pairs, solving the problem from Eq. (8) with QPBO-I and performing the obtained 2-clique swap if profitable.

Sparsity constraints can couple decision variables in Eq. (8), since forbidden matchings *force* vertices of different graphs to be swapped *jointly*, see Fig. 5. If a vertex swap between two cliques $Q, R \in \mathcal{Q}$ along an object $p \in D(Q)$ results in a forbidden matching, because there exists an object $q \in D(R)$ s.t. either the matching of Q^p to R^q or of R^p to Q^q is forbidden, then the vertex swaps along both objects p, q must be performed jointly. We also say the objects p, q are coupled, i.e., $x^p = x^q$. This relationship is *transitive*. If the objects p, q and the objects q, r are coupled, then the objects p, r are so as well, i.e., the vertex swaps along all three objects p, q, r must be performed jointly. Hence, the set of covered graphs $D(Q) \cup D(R) \subset [d]$ admits a partition $\mathcal{D} \subset 2^{[d]}$ into sets of objects $D \in \mathcal{D}$ whose vertices must be swapped jointly. Instead of

Algorithm 4: Swap partition construction.

Input: Cliques $Q, R \in \mathcal{Q}$
 // initialize as singletons
 $\mathcal{D} \leftarrow \{\{p\} : p \in D(Q) \cap D(R)\}$
for $p \in D(Q) \cap D(R)$ **do**
 $A \leftarrow A \in \mathcal{D}$ s.t. $p \in A$
 for $B \in \mathcal{D}$ **do**
 if $\exists q \in B : C_{Q^p R^q, Q^p R^q}^{p,q} = \infty$ or $C_{R^p Q^q, R^p Q^q}^{p,q} = \infty$
 then
 // merge object sets to be coupled
 $\mathcal{D} \leftarrow \mathcal{D} \setminus \{A, B\} \cup \{A \cup B\}$

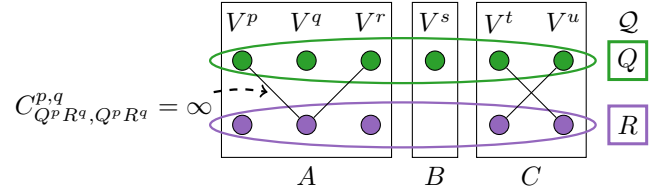


Fig. 5. **Swap partition.** Shown are two cliques $Q, R \in \mathcal{Q}$ spanning six objects V^p to V^u . Solid lines depict sparsity constraints, i.e., infinite linear costs for matching two vertices, e.g., Q^p and R^q . These constraints induce the *swap partition* with object subsets $A = \{p, q, r\}$, $B = \{s\}$, and $C = \{t, u\}$ (solid rectangles). Swaps along objects within each subset must be performed *jointly* to avoid *forbidden* matchings, reducing the problem size from six to three decision variables.

Eq. (8), one can, therefore, solve the *smaller* quadratic pseudo-boolean optimization problem:

$$x^* \in \arg \min_{x \in \{0,1\}^{\mathcal{D}}} \sum_{A \in \mathcal{D}} x^A \left(\sum_{B \in \mathcal{D}} \bar{x}^B \cdot \delta C_{\text{swap}}^{A,B}(Q, R) \right) \quad (9)$$

with costs

$$\delta C_{\text{swap}}^{A,B}(Q, R) = \sum_{p \in A} \sum_{q \in B} \delta C_{\text{swap}}^{p,q}(Q, R). \quad (10)$$

In practice, we build the *swap partition* $\mathcal{D} \subset 2^{[d]}$ from the *bottom up* via Alg. 4. The partition is initialized as singletons and then iteratively updated by merging two parts if each contain objects to be coupled. If *all* covered objects are coupled, problem (9) contains a single decision variable, hence, is *trivially solvable*. In application, this is, nonetheless, a common case, e.g., when one clique exclusively matches vertices in the head and the other in the tail of the worm, see Sec. II. For this reason alone, solving problem (9) instead of (8) *significantly accelerates* our previously proposed swap local search [18].

Our novelty and related work. The vertex swap was originally introduced for the MDAP with three objects [49] and later extended [50–52] to consider multiple swaps at the same time. However, these propositions come without a method to solve the 2-clique swap problem efficiently. Our contributions are: Formulating and solving the 2-clique swap as a quadratic pseudo-boolean optimization problem, and efficiently adapting it to sparse problems.

V. SPARSE SYNCHRONIZATION

GREEDA, as a MGM meta-heuristic, applies to not just quadratic but arbitrary, *e.g.*, linear, costs due to the interchangeability of the GM solver. It is therefore *also* a synchronization method. Synchronization can be seen a constructive heuristic, that reduces MGM from arbitrary cost orders to a special case of the Multi-LAP. It has two stages. First, all $d(d-1)/2$ GM subproblems between all object pairs $p, q \in [d]$ (with costs $C^{p,q}$) are *independently* solved, potentially in parallel, yielding matchings $E^{p,q} \subset V^p \times V^q$. These matchings induce an, a priori, *cycle-inconsistent* multi-matching $E := \bigcup_{p,q \in [d]} E^{p,q}$. Second, a cycle-consistent multi-matching E^* sharing the largest possible number of matchings with E is (approximately) recovered [25, 26, 39], *i.e.*,

$$E^* \in \arg \max_F |E \cap F| \quad \text{s.t. } F \subset \bar{E} \text{ cycle-consistent} . \quad (11)$$

The recovering problem (11) can be formulated as a *synchronization Multi-LAP* between all objects V^p with costs

$$M_{is, is}^{p,q} = \begin{cases} -1, & \text{if } is \in E, \\ 0, & \text{else} . \end{cases} \quad (12)$$

An alternative, which agrees with Eq. (11) for complete problems, is to recover the projection w.r.t. to the *Hamming distance*, *i.e.*, minimizing the symmetric difference $|E \Delta E^*|$ [27, 53]. For details and the corresponding Multi-LAP, see Suppl. E.

Synchronization's Achilles' heel is its *ignorance of MGM costs* during recovery, *i.e.* the second stage (11). Especially for *sparse* models with a majority of *forbidden* matchings of *infinite* costs $C_{is, is}^{p,q} = \infty$, including just one in the final solution E^* renders it forbidden. Instead, we propose to take *forbidden matchings* into account by solving the above mentioned synchronization Multi-LAP with costs

$$N_{is, is}^{p,q} = \begin{cases} \infty & , \text{ if } C_{is, is}^{p,q} = \infty , \\ M_{is, is}^{p,q} & , \text{ else } , \end{cases} \quad (13)$$

instead of (12). When used within our algorithms, the sparsity-aware costs (13) always lead to *feasible* solutions, containing only *allowed* matchings. We refer to this sparse synchronization variant of our algorithm as *GREEDA (Sync)*. In *GREEDA (Sync)*, we use SciPy's [54] modified Jonker-Volgenant algorithm as described in [55] for the LAPs.

Our novelty and related work. Usually, synchronization methods neither discuss sparsity nor apply to infinite costs. The only sparse synchronization method we could find, and this is only through communication with the authors, is a greedy subroutine of [40] not described in their paper. In contrast, our method, applied to the synchronization Multi-LAP, efficiently and accurately handles the costs from Eq. (13) and, therefore, respects forbidden matchings.

VI. INCOMPLETE TO COMPLETE TRANSFORMATION

Here, we formally state and analyze the *polynomial-time transformation* from incomplete to complete MGM, often mentioned without references or further elaboration in related works, *e.g.*, [20–22]. To transform an incomplete problem

with objects V^p , $p \in [d]$, and costs $C^{p,q}$, $p, q \in [d]$, into a complete one with objects of equal cardinalities, we introduce *dummy* vertices, similar to the respective GM transformation in [8]. More precisely, given that $|V|$ stands for the *total* number of vertices in the original incomplete problem, we add $|V| - |V^p| =: |\Delta^p|$ dummy vertices Δ^p to each object V^p , $p \in [d]$. As a result, we construct a complete MGM problem with objects

$$\tilde{V}^p = V^p \cup \Delta^p, \quad p \in [d], \quad (14)$$

and costs

$$\tilde{C}_{is, jt}^{p,q} = \begin{cases} C_{is, jt}^{p,q} & , \text{ if } i, s, j, t \in V \\ 0 & , \text{ else} \end{cases} . \quad (15)$$

In this way, each vertex $i \in V^p$, $p \in [d]$, of the incomplete problem receives a dummy vertex $\delta \in \Delta^q$ in each object $q \in [d] \setminus \{p\}$ of the complete problem. These dummy vertices are used to turn each clique of the incomplete problem maximal *s.t.* they span all objects of the (complete) problem. In turn, omitting matchings to dummy vertices defines the inverse transformation, which leads us to the following

Theorem 2. *Incomplete MGM is polynomial-time transformable to complete MGM.*

The proof is based on the transformation (14)-(15) and given in Suppl. D. While such a transformation could, in principle, be used to apply solvers for complete MGM to incomplete problems, doing so without further reasoning might be overly simplistic. The transformation (14)-(15) adds $\sum_{p \in [d]} (|V| - |V^p|) = (d-1)|V|$ dummy vertices, *i.e.*, $d-1$ times the size $|V|$ of the incomplete problem, which renders most existing algorithms impractical by considerably slowing them down.

Yet, the described transformation is *minimal* within the class of *clique-wise* transformations, that is, the size of respective complete problems cannot be decreased in general:

Definition 1. *A problem class P^A is transformable in polynomial time to the problem class P^B , if (i) there exists a mapping $T_{inst}: P^A \rightarrow P^B$ and (ii) for any $A \in P^A$ there exists a mapping T_{sol} from the optimal solutions of $T_{inst}(A)$ to the optimal solutions of A , where all mappings are computable in polynomial time.*

Henceforth, problem always means *optimization* problem. Being very general, Definition 1 does not consider several practical issues: First, an optimal solution is rarely known a priori. Therefore, the mapping T_{sol} is usually defined for *all* solutions that can potentially be optimal. For MGM problems, however, *any* solution can potentially be optimal, as there always exist costs that make it optimal. Second, if approximate algorithms are used to solve the reduced problem, then the optimum-to-optimum mapping, as required in Definition 1, is insufficient. Instead, the mapping T_{sol} must be *monotone* w.r.t. the objective values of the mapped feasible solutions: Feasible solutions with lower objective values in $T_{inst}(A)$ correspond to feasible solutions with lower objective values in A . These properties are summarized in the following

Definition 2. *We say that there is a robust polynomial time transformation of a problem class P^A to the problem class*

P^B , if (i) there exists a mapping $T_{inst}: P^A \rightarrow P^B$ and (ii) for any $A \in P^A$ there exists a (w.r.t. the objective) monotone mapping T_{sol} from the feasible solutions of $T_{inst}(A)$ to the feasible solutions of A , where all mappings are computable in polynomial time.

Let now P^A and P^B correspond respectively to the problem classes of incomplete and complete MGM. Any feasible solution of an arbitrary MGM problem A' is given by its vertex partition $Q^{A'} \in \mathcal{Q}^{A'}$. In the following, we define a subset of robust transformations, where each clique in the partition Q^A , $A \in P^A$, has its counterpart in the *respective* partition Q^B , $B = T_{inst}(A)$:

Definition 3. Assume $B = T_{inst}(A)$ and $Q^A := T_{sol}(Q^B)$. We call the robust transformation $P^A \rightarrow P^B$ clique-wise, if the mapping T_{sol} is induced by a clique-wise mapping $T_{clique}: Q^B \rightarrow Q^A$ s.t. $T_{clique}|_{Q^B \setminus T_{clique}^{-1}(\emptyset)}$ is injective, which directly implies $|Q^A| \leq |Q^B|$.

As it follows from the proof of Theorem 2, the transformation (14)-(15) is clique-wise. We are not aware of any non-clique-wise transformations, although cannot exclude their existence. Definition 3 leads to the following

Proof of Theorem 1. Consider the feasible multi-matching in A which leaves all vertices unmatched. This has $|Q^A| = |V|$ cliques representing individual vertices. On the other side, the number of (maximal) cliques in any (complete) MGM solution in B is equal to the number of vertices in each of its objects. Hence, the latter cannot be less than $|V|$ due to Definition 3. $\square$

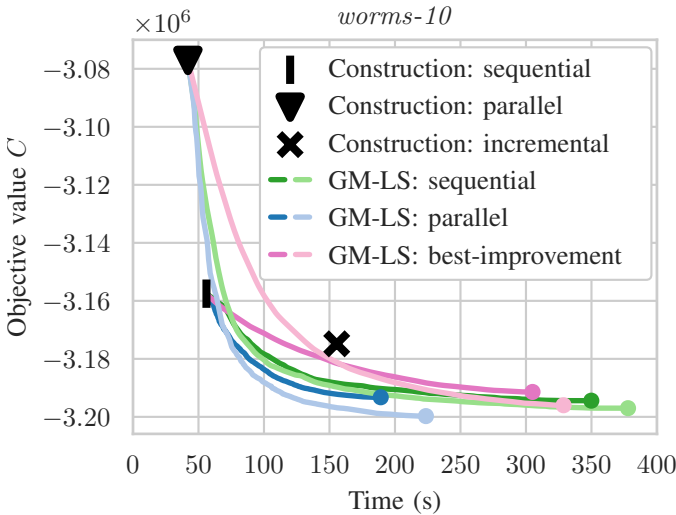


Fig. 6. **Convergence** of sequential and parallel *GM local search* variants after sequential and parallel construction, respectively. For the incremental construction only the objective value is given to keep readability of the plot. Though taking notably more time, this method does not result in a better solution. The result is a single run on the worms-10 dataset and averaged over the first 10 problem instances.

VII. EXPERIMENTAL VALIDATION

Our experiments comprise an *ablation study* (Sec. VII-A) and a *comparison to other methods*, for *direct MGM* (Sec. VII-B) and *sparse synchronization* (Sec. VII-C). Whereas the ablation study compares different variants of *GREEDA*'s *constructive heuristic* and *GM-LS*, see Fig. 3, in the comparison to others, we only utilize *sequential* variants of all algorithms to ensure a fair runtime comparison.

Contrary to most existing works, we only compare *objective values* instead of *ground-truth-based accuracies* since none of the considered algorithms explicitly optimize the latter or has access to the ground truth. We leave accuracy comparisons to the works addressing modeling and learning questions.

Datasets. We evaluate methods on the established datasets: *synthetic*, *CMU hotel* and *house*, and *worms*. All of them are widely used in MGM literature and accessible through the archive accompanying [23].

Additionally, to showcase our approach, we create the *worms-large* dataset [17]. While the original *worms* dataset considers only 3 to 10 worm objects per matching problem, *worms-large* features problems with 20 to 100 objects¹:

- worms-20
 - worms-30
 - worms-50
 - worms-100
- } ~510-570 vertices each
} 558 vertices each

Solving problems of this scale became necessary as part of a related research line [15] focused on the unsupervised training of matching cost hyperparameters via Bayesian optimization. This prompted us to create and publish this dataset to facilitate benchmark comparisons. Compared to the *worms* dataset, the *worms-large* instances are significantly more sparse: on average, each vertex is restricted to 14 potential matches instead of 23. This degree of sparsity was chosen to substantially reduce problem size and runtime without compromising the solution quality of downstream matching tasks. Furthermore, the distinct hyperparameter configurations used for these instances result in different cost values compared to the original *worms* dataset.

With approximately 45M keypoint correspondences (compared to 3M in the original *worms-29* instance from [18]), the *worms-100* instances represent the largest MGM problems considered in computer vision to date. To promote further research in this regime, we publish the *worms-large* dataset by adding it to the benchmark collection of [23]. It is also directly accessible under <https://doi.org/10.11588/DATA/RTINUU>.

A. Ablation study

Since the ablation study focuses on the underlying meta-heuristic rather than a particular instantiation, we restrict our analysis to *GREEDA* and extend conclusions to *GREEDA (Sync)*. We compare all combinations of the *sequential*, *parallel*, and *incremental* solution construction methods from Sec. IV-B with the *sequential* and *parallel GM local search* methods from Sec. IV-C. For lack of a better solver, *incremental* construction uses *GREEDA* for the first 5 objects.

¹We also substituted the worms-29 dataset from [18] with worms-30.

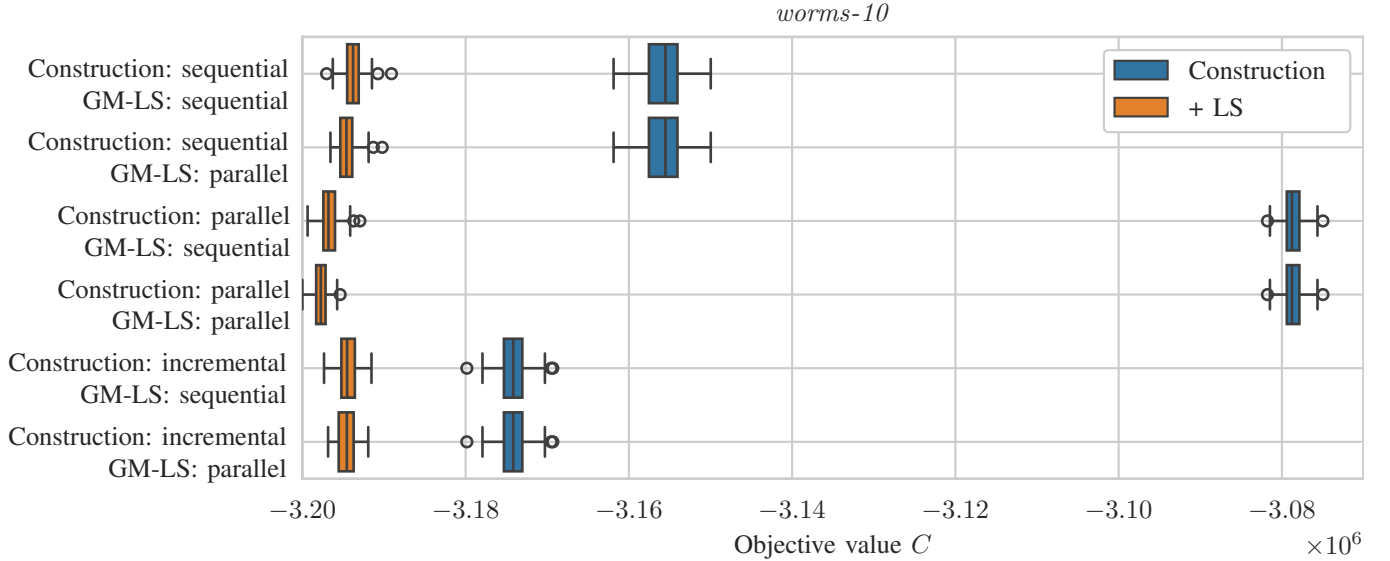


Fig. 7. **Boxplots of cost distributions** from 100 runs averaged over 10 *worms-10* instances after our *sequential*, *parallel*, or *incremental* construction (see Sec. IV-B), and after additionally applying our *sequential* or *parallel* GM local search (+ LS) to each (see Sec. IV-C).

To showcase our parallelization, we additionally consider a straightforward “*best-improvement*” parallelization of the GM local search: Although it performs all GM-LS steps in parallel, it accepts only the most profitable one. In this respect it differs from the parallel local search as described in Sec. IV-C, which aims to accept *all possible* profitable changes. The experiments were run on *worms-10*, containing the largest MGM instances addressed in computer vision [40] so far, using 10 processing cores in the parallel setting. Further details about the experimental setup and solver settings are provided in Suppl. F.

Convergence. Fig. 6 depicts costs with respect to runtime. While all methods converge to virtually the same value, *parallel* GM-LS converges fastest. It also notably outperforms the “*best-improvement*” parallelization.

Cost Distributions. To assess the influence of randomization, we run each algorithm 100 times using different object orderings. Fig. 7 shows box plots of the resulting cost distributions. Differences across construction variants vanish after local search. Hence, the extra runtime of *incremental* construction (see Fig. 6) is hard to justify. In contrast, *parallel* construction’s improved scaling can be fully utilized because the local search compensates the cost difference to the *sequential* construction.

B. Comparison to other methods: Direct MGM

Algorithms. We compare our method *GREEDA* to the state-of-the-art MGM methods *MP-T* [40] and *DS** [19]. The latter only applies to complete and dense MGM problems, while the former also applies to incomplete and sparse MGM problems. We also include the synchronization methods *PPI*, *Stiefel*, and *Spectral*, see Sec. VII-C.

We additionally evaluated the *Cao* [34] and *Floyd* [33] algorithms from the *pygmtools* package [56]. Like *DS**, they are restricted to complete, dense MGM problems. However, their

current implementation does not enforce cycle-consistency and their solutions were, in our experiments, in general cycle-inconsistent. We therefore exclude them from the main comparison and report their results separately in Suppl. H. These results show that their solutions are still inferior to *GREEDA* in terms of objective value despite being cycle-inconsistent.

Implementation Details. We use the original author implementations for both *DS** (Matlab) and *MP-T* (C++). *GREEDA* is also implemented in C++. Details regarding the synchronization methods can be found in Sec. VII-C.

Results. For the *worms-50* and *worms-100* datasets, the only true competitor, *MP-T*, runs out of memory on a machine with 236GB of RAM. By contrast, *GREEDA* finds a feasible solution after 8 minutes on *worms-50* and 30 minutes on *worms-100*, and converges after about 2 hours using 25GB of RAM and after about 16 hours using 100GB of RAM, respectively.

Otherwise, for brevity, we only provide the comparison for three representative datasets, and refer to Suppl. H for a full list of results with all algorithms and datasets (in total 331 problem instances).

Fig. 8 shows the costs-runtime plots for different methods. Its counterpart in the preliminary version [18] mistakenly reports results for *GREEDA* without SWAP-LS. Here, we show results for the full method, including the improved SWAP-LS. For synchronization methods, we consider only the final objective value, as the need to solve $d(d-1)/2$ pairwise problems during pre-processing makes runtime comparisons with *direct* MGM methods difficult. Again, *GREEDA* consistently outperforms or is on par with competitors in terms of final objective value, and is an order of magnitude faster in terms of runtime. In particular, for our new, large-scale *worms-30* dataset, see Fig. 8a, *GREEDA* achieves a better solution much faster than the direct competitor *MP-T*, improving construction time from half an hour to just two

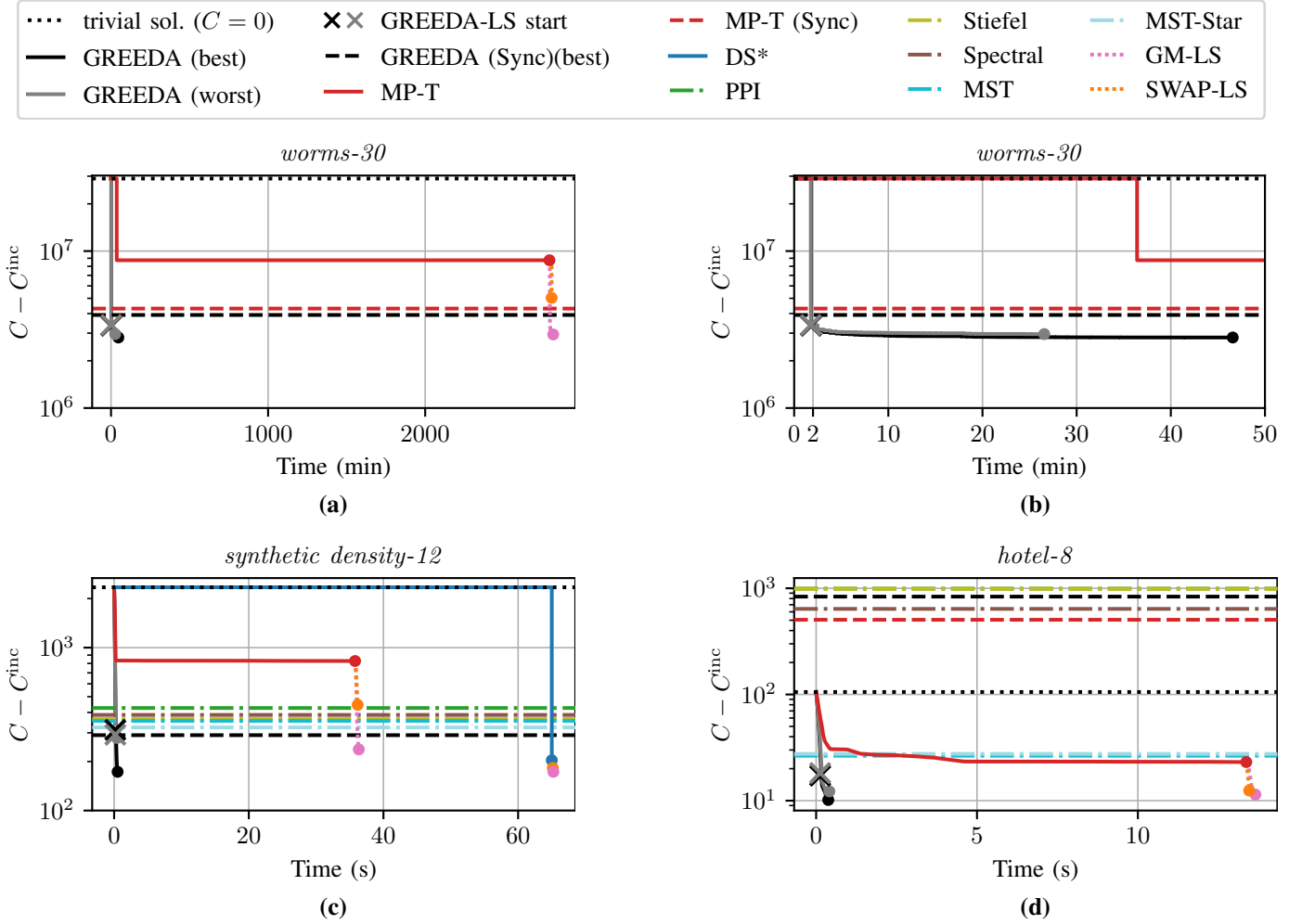


Fig. 8. **Method Comparison.** Comparison of objective value w.r.t. runtime. The objective is plotted on a log scale and offset by the cycle-inconsistent solution’s objective C^{inc} , obtained by independently solving the $d(d-1)/2$ pairwise GM problems. This value approximates, since [32] is itself approximative, a lower bound to Eq. (1) given by the MGM relaxation that ignores cycle consistency. For the sequential variant of *GREEDA*, we report the best and worst result over 10 runs. Fig. 8b shows the first minutes of Fig. 8a, illustrating the considerable difference in construction time between *GREEDA* and *MP-T* on large problems. In both, we only compare methods that can find allowed solutions for sparse problems. Fig. 8c and Fig. 8d report results averaged over all instances of the 12-object *synthetic density* and the 8-object *hotel* datasets, respectively. Since the preprocessing stage of synchronization methods makes runtime comparisons with direct MGM methods difficult, we report only their objective values. *DS** results are shown only in Fig. 8c, since the other datasets contain incomplete problems. Note that even if *GREEDA* is run 10 times sequentially to obtain the result of *GREEDA (best)*, it remains the fastest algorithm.

minutes (see Fig. 8b). While *MP-T* fails to improve upon its initial solution significantly, our local search subroutines *GM-LS* and *SWAP-LS* quickly and significantly improve its result. Fig. 8c includes a comparison to *DS**, which yields high-quality results but is substantially slower than the other methods, partially due to its Matlab implementation. Since it is a solver for complete problems, it cannot be applied to the *hotel*, *house*, or *worms* datasets, as the incomplete to complete problem transformation would increase the runtime even further, see Sec. VI. All plots in Fig. 8 additionally show the application of *GM-LS* and *SWAP-LS* to the results of the other methods. Although the resulting objective values are at times comparable to *GREEDA*, optimization takes significantly longer. As mentioned in Sec. III, synchronization methods may introduce blunders that substantially impact the resulting cost. This can be seen in Fig. 8d, where all synchronization methods return a solution that is worse than the trivial one, that would

leave all vertices unmatched. Note, that the corresponding problem instance is *dense*. For *sparse* synchronization, see Sec. VII-C.

C. Comparison to other methods: Synchronization

Here, we compare algorithms w.r.t. the synchronization objective Eq. (12).

Algorithms. We compare our sparse synchronization algorithm *GREEDA (Sync)* to the following synchronization algorithms: The projected power iteration (*PPI*) [36], *Stiefel* Synchronization [25], and the *Spectral* approach [27] extended by the Successive Block Rotation Algorithm [53] to yield incomplete matchings. We also compare to the conceptually similar constructive heuristics based on arbitrary (*MST*) [10] and star-shaped (*MST-Star*) [22] spanning trees. Additionally, we adapt *MP-T* [40] as a synchronization method *MP-T (Sync)*

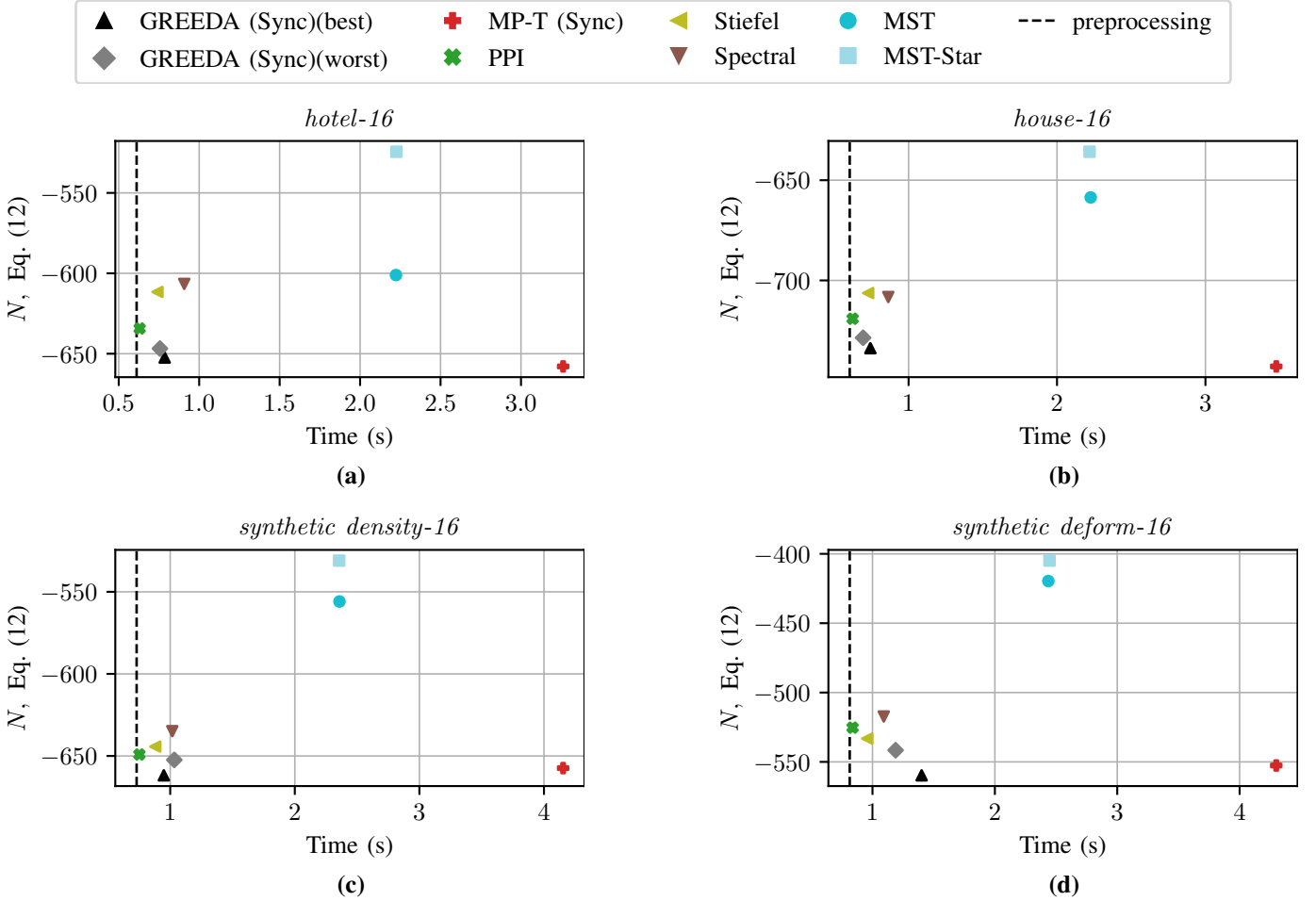


Fig. 9. **Synchronisation Comparison on Smaller, Dense Problems.** Comparison of the *sparse* (13), or equivalently, since all instances are dense, the *standard* (12) synchronization MLAP objective w.r.t. total runtime, averaged over all instances of the respective datasets. All methods except *MST* and *MST-Star* directly optimize the MLAP objective (12). The dashed line marks the end of the preprocessing stage, in which all GM subproblems are solved independently.

by applying it to the Multi-LAP (11) with sparsity-aware costs (13).

Except for *MP-T (Sync)*, no competitor guarantees to return only allowed matchings for sparse problems.

For all synchronization methods, we solve the (potentially sparse) GM subproblems arising in the preprocessing stage using the state-of-the-art GM solver [32, 57], which we turn from a randomized algorithm into a deterministic one by fixing its randomization seed. Further details about the experimental setup and solver settings are given in Suppl. F.

Implementation Details. While *GREEDA (Sync)* and *MP-T (Sync)* are available in C++, *Stiefel*, and *Spectral* are Matlab implementations. We use the original software from the authors for all but *PPI*, *MST*, and *MST-Star*, which we re-implemented ourselves in Python.

Results. Quantitative comparisons on the large-scale, sparse *worms-10* instances are summarized in Tab. I, see Suppl. G for detailed results. *GREEDA (Sync)* achieves the best performance across the reported metrics. The synchronization methods *PPI*, *Spectral*, and *Stiefel* consistently produce forbidden matchings in all evaluated experiments, even when employing soft constraints that penalize infinite costs with a positive

constant $\alpha > 0$. Similarly, the spanning-tree based methods *MST* and *MST-Star* fail to avoid forbidden matchings even for $\alpha = \infty$. Moreover, the objective value attained in this setting is inferior to those of *GREEDA (Sync)* and *MP-T (Sync)*, even with respect to the dense synchronization objective (12).

Additionally, we observed that using synchronization solutions, including those containing forbidden matchings, as starting points for the *GREEDA* local search not only yields *allowed* multi-matchings, but also converges to highly consistent MGM objective values (1), regardless of the initialization. We attribute this robustness to the ability of the local search to efficiently explore the exponentially large neighborhood of a current solution. This effectiveness, in turn, stems from the efficiency of the underlying graph matching solver [32] utilized in our method.

Fig. 9 shows the synchronization Multi-LAP objective (12) against runtime for all methods on the small-scale, dense datasets *house-16*, *hotel-16*, *synthetic deform-16*, and *synthetic density-16*, averaged over all 10 instances of each dataset. Since all problems are dense, the *sparse* synchronization Multi-LAP objective (13) coincides with the *standard* syn-

TABLE I

RESULTS FOR THE SPARSE SYNCHRONIZATION TASK (13) on *worms-10*, averaged over 10 instances, see Sec. V. We report the Multi-LAP (M-LAP), eq. (12), and MGM, eq. (1) objectives, as well as the number of forbidden matchings (#forb.) in the final solution. Since neither *PPI*, *Spectral*, nor *Stiefel* are able to deal with the sparse synchronization problem (13) directly, we resort to *soft constraints* by replacing infinite $N_{is, is}^{p, q}$ values in (13) with a finite value $\alpha > 0$, for $\alpha \in \{1, 10, 50\}$. Among the three, only *PPI* is able to reduce the number of forbidden matchings #forb. significantly. Nevertheless, only *MP-T(Sync)* and *GREEDA(Sync)* are able to obey sparsity constraints, which renders the MGM objective infinite for all other methods. As baselines, we also list results of direct optimization algorithms *MP-T* and *GREEDA*, the latter over 10 runs to account for randomization.

Inf.Value	Method	M-LAP↓	# forb./MGM↓
$\alpha = 0$ (Eq. (12))	PPI [36]	-18428	791 / ∞
	Spectral [27, 53]	-18521	713 / ∞
	Stiefel [25]	-18461	580 / ∞
	MP-T (Sync) [40]	-18497	599 / ∞
	GREEDA (Sync)(worst)	-18622	755 / ∞
	GREEDA (Sync)(best)	-18669	692 / ∞
$\alpha = 1$	PPI [36]	-18390	275 / ∞
	Spectral [27, 53]	-3913	15426/ ∞
	Stiefel [25]	-14213	2493 / ∞
$\alpha = 10$	PPI [36]	-17576	303 / ∞
	Spectral [27, 53]	-1118	17745/ ∞
	Stiefel [25]	-10531	3663 / ∞
$\alpha = 50$	PPI [36]	-16586	389 / ∞
	Spectral [27, 53]	-1061	17731/ ∞
	Stiefel [25]	-10700	3643 / ∞
$\alpha = \infty$ (Eq. (13))	MST [10]	-14740	360 / ∞
	MST-Star [22]	-16349	213 / ∞
	MP-T (Sync)	-18333	0/-3108350
	GREEDA (Sync)(worst)	-18451	0/-3132603
	GREEDA (Sync)(best)	-18510	0/-3136748
baseline	MP-T [40]	-	0/-2393022
	GREEDA (worst)	-	0/-3201256
	GREEDA (best)	-	0/-3208824

chronization objective (12), and all methods² except *MST* and *MST-Star* optimize this objective explicitly. *GREEDA (Sync)* consistently outperforms the dedicated synchronization methods *PPI*, *Stiefel*, and *Spectral* in terms of objective value. The spanning-tree based heuristics *MST* and *MST-Star* underperform with respect to the Multi-LAP objective (12), though they achieve better values for the MGM objective (1), for which they partially optimize, see Suppl. G for details. The only method that provides competitive results w.r.t. the Multi-LAP objective value (12) is the synchronization adaptation of *MP-T*, namely *MP-T (Sync)*. In terms of objective value, *GREEDA (Sync)* outperforms *MP-T (Sync)* on all datasets except *hotel* and *house*, where it is consistently second best and trails *MP-T (Sync)* only by a small margin, while being at least half an order of magnitude faster. For example, on *hotel-16*, the average percentage difference relative to *MP-T* is 0.84% for *GREEDA (Sync)(best)*, 1.69% for *GREEDA (Sync)(worst)*, and 3.59% for the closest remaining competitor, *PPI*. In terms of runtime, *GREEDA (Sync)* outperforms *MP-T (Sync)* by an order of magnitude on most datasets, with exceptions limited to small or simple cases (*hotel-4*, *house-4*, *synthetic outlier-4* & *-8*, and *synthetic complete*). We provide results for all datasets in Suppl. G.

²*Spectral* optimizes the Hamming distance; for dense problems, this is equivalent to (12).

Overall, *GREEDA (Sync)* consistently offers the best trade-off between Multi-LAP objective value (13) and runtime across the evaluated datasets. In particular, on the large *worms* instances, it consistently outperforms its only competitor in terms of objective value while being an order of magnitude faster.

VIII. CONCLUSIONS AND FUTURE WORK

In this paper, we addressed the challenging problem of large-scale multi-graph matching under sparsity and incompleteness constraints, a setting motivated by real-world bioimaging applications. We introduced *GREEDA*, an efficient solver that operates natively on sparse and incomplete problem formulations, as direct or synchronization method. By decomposing the heavy multi-graph assignment task into localized graph matching subroutines, *GREEDA* successfully scales to hundreds of objects while maintaining cycle consistency and yielding state-of-the-art accuracy at an execution speed that is orders of magnitude faster than existing competitors.

A promising direction remains for future exploration. As discussed in Sec. I, a modern approach to MGM includes learning the matching costs directly from data using neural networks. Integrating *GREEDA*'s exceptionally fast solving routine directly within such an end-to-end learning loop represents an exciting path forward. This could unlock more accurate and unsupervised correspondences in biological volumetric scans while operating inside the tight time constraints typically required by training cycles.

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REFERENCES

- [1] W. Nie, A. Liu, Y. Hao, and Y. Su, "View-based 3D model retrieval via multi-graph matching," *Neural Processing Letters*, vol. 48, pp. 1395–1404, 2018.
- [2] Y. Sahillioglu, "Recent advances in shape correspondence," *The Visual Computer*, vol. 36, no. 8, pp. 1705–1721, 2020.
- [3] M. Gao, Z. Lahner, J. Thunberg, D. Cremers, and F. Bernard, "Isometric multi-shape matching," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 14 183–14 193.
- [4] R. Yadav, F.-X. Dupé, S. Takerkart, and G. Auzias, "Population-wise labeling of sulcal graphs using multi-graph matching," *Plos one*, vol. 18, no. 11, p. e0293886, 2023.
- [5] T. Heimann and H.-P. Meinzer, "Statistical shape models for 3D medical image segmentation: A review," *Medical Image Analysis*, 2009.
- [6] C. Liu, C. Lou, R. Wang, A. Y. Xi, L. Shen, and J. Yan, "Deep neural network fusion via graph matching with applications to model ensemble and federated learning," in *International Conference on Machine Learning*. PMLR, 2022, pp. 13 857–13 869.
- [7] J. Chen, G. Han, A. Xu, and H. Cai, "Identification of multidimensional regulatory modules through multi-graph matching with network constraints," *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 4, pp. 987–998, 2019.
- [8] S. Haller, L. Feineis, L. Hutschenreiter, F. Bernard, C. Rother, D. Kainmüller, P. Swoboda, and B. Savchynskyy, "A comparative study of graph matching algorithms in computer vision," in *European Conference on Computer Vision*. Springer, 2022, pp. 636–653.

- [9] Y. Crama and F. C. Spieksma, "Approximation algorithms for three-dimensional assignment problems with triangle inequalities," *European Journal of Operational Research*, vol. 60, no. 3, pp. 273–279, 1992.
- [10] D. Tang and T. Jebara, "Initialization and coordinate optimization for multi-way matching," in *Artificial Intelligence and Statistics*. PMLR, 2017, pp. 1385–1393.
- [11] J. L. Richards, A. L. Zacharias, T. Walton, J. T. Burdick, and J. I. Murray, "A quantitative model of normal *Caenorhabditis elegans* embryogenesis and its disruption after stress," *Developmental biology*, vol. 374, no. 1, pp. 12–23, 2013.
- [12] D. Kainmueller, F. Jug, C. Rother, and G. Myers, "Active graph matching for automatic joint segmentation and annotation of *C. elegans*," in *Proceedings of the International Conference on Medical Image Computing and Computer Assisted Intervention*, 2014.
- [13] F. Long, H. Peng, X. Liu, S. K. Kim, and E. Myers, "A 3D digital atlas of *C. elegans* and its application to single-cell analyses," *Nature methods*, vol. 6, no. 9, pp. 667–672, 2009.
- [14] M. Rolínek, P. Swoboda, D. Zietlow, A. Paulus, V. Musil, and G. Martius, "Deep graph matching via blackbox differentiation of combinatorial solvers," in *European Conference on Computer Vision*. Springer, 2020, pp. 407–424.
- [15] S. Stricker, C. Karg, L. Hutschenreiter, B. Savchynskyy, and D. Kainmueller, "Fully unsupervised annotation of *C. elegans*," in *The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.
- [16] S. Tourani, M. H. Khan, C. Rother, and B. Savchynskyy, "Discrete cycle-consistency based unsupervised deep graph matching," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 6, 2024, pp. 5252–5260.
- [17] S. Stricker, M. Kahl, L. Hutschenreiter, and B. Savchynskyy, "Worms-Large: Large-Scale Multi-Graph Matching Instances for *C. elegans* Cell Matching," 2026. [Online]. Available: <https://doi.org/10.11588/DATA/RTINUU>
- [18] M. Kahl, S. Stricker, L. Hutschenreiter, F. Bernard, C. Rother, and B. Savchynskyy, "Towards optimizing large-scale multi-graph matching in bioimaging," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 11 569–11 578.
- [19] F. Bernard, C. Theobalt, and M. Moeller, "DS*: Tighter lifting-free convex relaxations for quadratic matching problems," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018.
- [20] J. Yan, Z. Ren, H. Zha, and S. Chu, "A constrained clustering based approach for matching a collection of feature sets," in *2016 23rd International Conference on Pattern Recognition (ICPR)*. IEEE, 2016, pp. 3832–3837.
- [21] J. Yan, Y. Tian, H. Zha, X. Yang, Y. Zhang, and S. M. Chu, "Joint optimization for consistent multiple graph matching," in *Proceedings of the IEEE international conference on computer vision*, 2013, pp. 1649–1656.
- [22] J. Yan, J. Wang, H. Zha, X. Yang, and S. Chu, "Consistency-driven alternating optimization for multigraph matching: A unified approach," *IEEE Transactions on Image Processing*, vol. 24, no. 3, pp. 994–1009, 2015.
- [23] P. Swoboda, B. Andres, A. Hornakova, F. Bernard, J. Irmay, P. Roetzer, B. Savchynskyy, D. Stein, and A. Abbas, "Structured prediction problem archive," *arXiv preprint arXiv:2202.03574*, 2022.
- [24] L. Torresani, V. Kolmogorov, and C. Rother, "A dual decomposition approach to feature correspondence," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2013.
- [25] F. Bernard, D. Cremers, and J. Thunberg, "Sparse quadratic optimisation over the Stiefel manifold with application to permutation synchronisation," *Advances in Neural Information Processing Systems*, vol. 34, pp. 25 256–25 266, 2021.
- [26] F. Arrigoni, E. Maset, and A. Fusiello, "Synchronization in the symmetric inverse semigroup," in *Image Analysis and Processing-ICIAP 2017: 19th International Conference, Catania, Italy, September 11-15, 2017, Proceedings, Part II 19*. Springer, 2017, pp. 70–81.
- [27] D. Pachauri, R. Kondor, and V. Singh, "Solving the multi-way matching problem by permutation synchronization," in *Advances in Neural Information Processing Systems*. Citeseer, 2013, pp. 1860–1868.
- [28] P. M. Pardalos, F. Rendl, and H. Wolkowicz, "The quadratic assignment problem - a survey and recent developments," *Quadratic Assignment and Related Problems*, 1993.
- [29] E. L. Lawler, "The quadratic assignment problem," *Management Science*, 1963.
- [30] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, 1955.
- [31] R. Burkard, M. Dell'Amico, and S. Martello, *Assignment Problems*. SIAM, 2009.
- [32] L. Hutschenreiter, S. Haller, L. Feineis, C. Rother, D. Kainmüller, and B. Savchynskyy, "Fusion moves for graph matching," in *Proceedings of the IEEE International Conference on Computer Vision*, 2021.
- [33] Z. Jiang, T. Wang, and J. Yan, "Unifying offline and online multi-graph matching via finding shortest paths on supergraph," *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 10, pp. 3648–3663, 2020.
- [34] J. Yan, M. Cho, H. Zha, X. Yang, and S. M. Chu, "Multi-graph matching via affinity optimization with graduated consistency regularization," *IEEE transactions on pattern analysis and machine intelligence*, vol. 38, no. 6, pp. 1228–1242, 2015.
- [35] H.-J. Bandelt, A. Maas, and F. C. Spieksma, "Local search heuristics for multi-index assignment problems with decomposable costs," *Journal of the Operational Research Society*, vol. 55, no. 7, pp. 694–704, 2004.
- [36] Y. Chen and E. J. Candès, "The projected power method: An efficient algorithm for joint alignment from pairwise differences," *Communications on Pure and Applied Mathematics*, vol. 71, no. 8, pp. 1648–1714, 2018.
- [37] Y. Shen, Q. Huang, N. Srebro, and S. Sanghavi, "Normalized spectral map synchronization," *Advances in neural information processing systems*, vol. 29, 2016.
- [38] Y. Chen, L. J. Guibas, and Q.-X. Huang, "Near-optimal joint object matching via convex relaxation," *arXiv preprint arXiv:1402.1473*, 2014.
- [39] E. Maset, F. Arrigoni, and A. Fusiello, "Practical and efficient multi-view matching," in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 4568–4576.
- [40] P. Swoboda, D. Kainmüller, A. Mokarian, C. Theobalt, and F. Bernard, "A convex relaxation for multi-graph matching," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 11 156–11 165.
- [41] I. Kezurer, S. Z. Kovalsky, R. Basri, and Y. Lipman, "Tight relaxation of quadratic matching," in *Computer graphics forum*, vol. 34, no. 5. Wiley Online Library, 2015, pp. 115–128.
- [42] R. Tron, X. Zhou, C. Esteves, and K. Daniilidis, "Fast multi-image matching via density-based clustering," in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 4057–4066.
- [43] X. Zhou, M. Zhu, and K. Daniilidis, "Multi-image matching via fast alternating minimization," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 4032–4040.
- [44] Z. Nurlanov, F. R. Schmidt, and F. Bernard, "Universe points representation learning for partial multi-graph matching," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 2, 2023, pp. 1984–1992.
- [45] F. Bernard, J. Thunberg, P. Swoboda, and C. Theobalt, "HiPPI: Higher-order projected power iterations for scalable multi-matching," in *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 10 284–10 293.
- [46] H.-J. Bandelt, Y. Crama, and F. C. Spieksma, "Approximation algorithms for multi-dimensional assignment problems with decomposable costs," *Discrete Applied Mathematics*, vol. 49, no. 1–3, pp. 25–50, 1994.
- [47] E. Boros and P. Hammer, "Pseudo-boolean optimization," *Discrete Applied Mathematics*, 2002.
- [48] C. Rother, V. Kolmogorov, V. S. Lempitsky, and M. Szummer, "Optimizing binary MRFs via extended roof duality," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2007.
- [49] E. Balas and M. J. Saltzman, "An algorithm for the three-index assignment problem," *Operations Research*, vol. 39, no. 1, pp. 150–161, 1991.
- [50] R. A. Murphey, P. M. Pardalos, and L. S. Pitsoulis, "A greedy randomized adaptive search procedure for the multitarget multisensor tracking problem," *Network design: Connectivity and facilities location*, vol. 40, pp. 277–302, 1997.
- [51] A. J. Robertson, "A set of greedy randomized adaptive local search procedure (GRASP) implementations for the multidimensional assignment problem," *Computational Optimization and Applications*, vol. 19, no. 2, pp. 145–164, 2001.
- [52] D. Karapetyan and G. Gutin, "Local search heuristics for the multi-dimensional assignment problem," *Journal of Heuristics*, vol. 17, pp. 201–249, 2011.
- [53] F. Bernard, J. Thunberg, J. Goncalves, and C. Theobalt, "Synchronisation of partial multi-matchings via non-negative factorisations," *Pattern Recognition*, vol. 92, pp. 146–155, 2019.

- [54] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.
- [55] D. F. Crouse, “On implementing 2D rectangular assignment algorithms,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 52, no. 4, pp. 1679–1696, 2016.
- [56] R. Wang, Z. Guo, W. Pan, J. Ma, Y. Zhang, N. Yang, Q. Liu, L. Wei, H. Zhang, C. Liu, Z. Jiang, X. Yang, and J. Yan, “Pygmtools: A python graph matching toolkit,” *Journal of Machine Learning Research*, vol. 25, no. 33, pp. 1–7, 2024. [Online]. Available: <https://jmlr.org/papers/v25/23-0572.html>
- [57] L. Hutschenreiter, S. Haller, L. Feineis, C. Rother, D. Kainmüller, and B. Savchynskyy, “Fusion moves for graph matching website,” 2021, <https://vislearn.github.io/libmpopt/iccv2021/>.

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Supplementary Material

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APPENDIX A

FEASIBLE SOLUTION CONSTRUCTION: DETAILS

Merge operation. The merge operation between two partial solutions $\mathcal{A} \in \mathbb{Q}^{D_A}$ and $\mathcal{B} \in \mathbb{Q}^{D_B}$ of disjoint $D_A \cap D_B = \emptyset$ object subsets $D_A, D_B \subset [d]$ specified via the matching $E \in \mathcal{A} \times \mathcal{B}$ is formally defined as

$$\text{merge}(\mathcal{A}, \mathcal{B}; E) := \{A \cup B \mid AB \in E\} \cup \{A \mid \forall B \in \mathcal{B} : AB \notin E\} \cup \{B \mid \forall A \in \mathcal{A} : AB \notin E\}. \quad (16)$$

The merge between an object V^p and a partial solution $\mathcal{Q} \in \mathbb{Q}^D$, $p \notin D \subset [d]$, specified by the matching $E \subset V^p \times \mathcal{Q}$ is a special case

$$\text{merge}(V^p, \mathcal{Q}; E) := \text{merge}(\mathcal{V}^p, \mathcal{Q}; E'), \quad (17)$$

where $\mathcal{V}^p = \{\{i\} \mid i \in V^p\}$ and $E' = \{\{i\} \cup Q \mid iQ \in E\}$.

Extensions and parallelization. At the heart of both extensions lie *leaf-labeled, ordered trees* with *label set* $[d]$, further referred to as *construction trees*, see Fig. A.1. As described in Sec. IV-B, each vertex of a construction tree can be associated with a partial solution (vertex.solution) for the objects labeled by its descendant leaves. While leaves are always associated with the trivial solution $\mathcal{V}^p = \{\{i\} \mid i \in V^p\}$ of the object they label V^p , $p \in [d]$, solutions are generally the result of (matching and) merging the solutions of their children.

Our *parallel construction* limits itself to *binary* construction trees (see Fig. A.1), which is why it can be implemented with only a GM solver, see Alg. A.1 for pseudocode. The basic construction Alg. 1 can be identified as the special case of binary construction trees with height $d - 1$, see Fig. A.1. It allows for no parallelization as only matchings for solutions of vertices at the same level of a construction tree can be computed in parallel.

Arbitrary construction trees are the idea behind *incremental construction*, where solutions for vertices with more than two children require an *MGM solver*, see Fig. A.1. However, in Sec. IV-B and Sec. VII, we limit ourselves to trees with only one such vertex at level 2, see Fig. A.1. A promising use case is to use a (potentially) better MGM solver for the first few objects to warm-start the construction with a better partial solution.

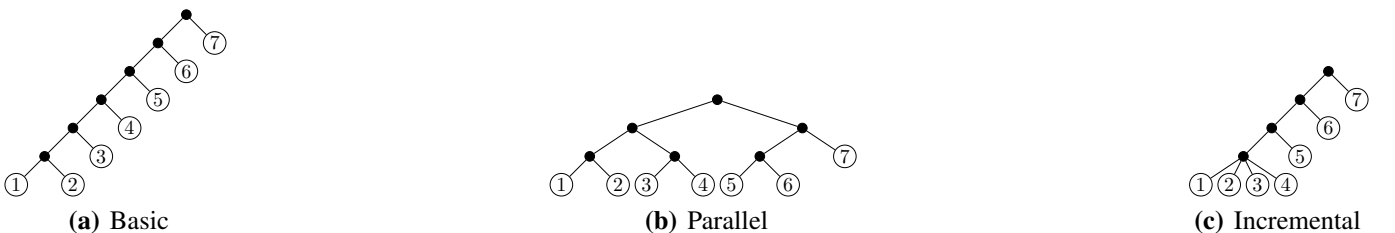


Fig. A.1. **Construction Trees.** All three of our proposed construction variants, *basic* (a), *parallel* (b), and *incremental* (c) construction, can be described via construction trees. Shown are example trees for each variant and $d = 7$ objects.

Algorithm A.1: Parallel solution construction.

Input: Binary Construction Tree T

```

for  $l \in \{T.\text{height}, T.\text{height}-1, \dots, 1\}$  do
  for  $\text{vertex} \in T.\text{level}[l]$  do in parallel
     $\mathcal{A} \leftarrow \text{vertex}.\text{left}.\text{solution}$ 
     $\mathcal{B} \leftarrow \text{vertex}.\text{right}.\text{solution}$ 
     $E \leftarrow \text{Solve GM with costs } C^{\mathcal{A}, \mathcal{B}} \text{ from Eq. (3)}$ 
     $\text{vertex}.\text{solution} \leftarrow \text{merge}(\mathcal{A}, \mathcal{B}; E)$ 

```

APPENDIX B

GM LOCAL SEARCH: DETAILS

Alg. B.1 provides pseudocode for the *parallel* GM local search.

Algorithm B.1: Parallel GM local search.

Input: Solution $\mathcal{Q} \in \mathcal{Q}$

```

while stopping criterion not met do
  for  $p \in [d]$  do in parallel
     $\mathcal{Q}' \leftarrow \{Q \setminus V^p \mid Q \in \mathcal{Q}\}$ 
     $E^p \leftarrow \text{Solve GM with costs } C^{p, \mathcal{Q}'} \text{ from Eq. (2)}$ 
     $c^p \leftarrow C(\text{merge}(V^p, \mathcal{Q}'; E^p))$ 
  for  $p \in [d]$  in ascending order of  $c^p$  do
     $\mathcal{Q}' \leftarrow \{Q \setminus V^p \mid Q \in \mathcal{Q}\}$ 
     $\mathcal{Q}'' \leftarrow \text{merge}(V^p, \mathcal{Q}'; E^p)$ 
    if  $C(\mathcal{Q}'') < C(\mathcal{Q})$  then
       $\mathcal{Q} \leftarrow \mathcal{Q}''$ 

```

APPENDIX C

SWAP LOCAL SEARCH: DETAILS

In the following, we derive the swap energy change formula Eq. (4). The derivation uses a decomposition of a double sum over a subset of the summation domain as specified in Lemma 1.

Lemma 1. *Let S be a finite set and $f : S \times S \rightarrow \mathbb{R}$ a real valued function defined on the cartesian product $S \times S$. For the double sum over S , we have the following decomposition w.r.t. a subset $T \subset S$*

$$\sum_{a,b \in S} f(a,b) = \sum_{a,b \in T} f(a,b) + \sum_{\substack{a \in T \\ b \in S \setminus T}} (f(a,b) + f(b,a)) + \sum_{a,b \in S \setminus T} f(a,b) \quad (18)$$

Proof. Fix $a \in S$, then we have by commutativity and associativity of the sum

$$\sum_{b \in S} f(a,b) = \sum_{b \in T} f(a,b) + \sum_{b \in S \setminus T} f(a,b) . \quad (19)$$

Applying this relation twice to the left hand side of Eq. (18) and using again the commutativity and associativity of the sum after renaming summation indices yields the right hand side. $\square$

Now, consider a single swap (Sec. IV-D) performed over a fixed object $p \in [d]$ and two cliques $Q, R \in \mathcal{Q}$, yielding a new solution $\mathcal{Q}'$. First, we rewrite the MGM objective *s.t.* the cost terms involving the fixed object $p \in [d]$ and cliques $Q, R \in \mathcal{Q}$ can be singled out, see Lemma 2.

Lemma 2. *Let $\mathcal{Q} \in \mathcal{Q}$ be a solution. Fix an object $p \in [d]$ and two cliques $Q, R \in \mathcal{Q}$. We can rewrite the MGM objective (1) as follows*

$$C(\mathcal{Q}) = \sum_{q \in [d] \setminus \{p\}} C_{\{Q,R\}}^{p,q}(\mathcal{Q}) + \mathfrak{C}_{\neg\{Q,R\}}^p(\mathcal{Q}) + \mathfrak{C}^{-p}(\mathcal{Q}) , \quad (20)$$

where the term $\mathfrak{C}_{\neg\{Q,R\}}^p(\mathcal{Q})$ does not depend on the cliques $Q, R \in \mathcal{Q}$, the term $\mathfrak{C}^{-p}(\mathcal{Q})$ does not depend on the object $p \in [d]$, and all terms involving the fixed object or cliques are given by

$$C_{\{Q,R\}}^{p,q}(\mathcal{Q}) = \sum_{S,T \in \{Q,R\}} C^{p,q}(S,T) + \sum_{\substack{S \in \{Q,R\} \\ T \in \mathcal{Q} \setminus \{Q,R\}}} (C^{p,q}(S,T) + C^{p,q}(T,S)) \quad (21)$$

with the symmetrized extended costs

$$C^{p,q}(S, T) := \begin{cases} C_{S^p S^q, T^p T^q}^{p,q} & ; p < q \wedge p, q \in D(S) \cap D(T) \\ C_{S^q S^p, T^q T^p}^{q,p} & ; q < p \wedge p, q \in D(S) \cap D(T) \\ 0 & ; \text{else} \end{cases} . \quad (22)$$

Proof. The MGM objective Eq. (1) is given by

$$C(\mathcal{Q}) := \sum_{S, T \in \mathcal{Q}} \sum_{\substack{u, v \in D(S) \cap D(T) \\ u < v}} C_{S^u S^v, T^u T^v}^{u,v} . \quad (23)$$

We will rewrite it as two double sums over the objects $[d]$ and cliques $\mathcal{Q}$, *s.t.*, we can single out the fixed object $p \in [d]$ and cliques $Q, R \in \mathcal{Q}$ by applying Lemma 1 twice. First, rewrite the MGM objective as two double sums, *i.e.*,

$$C(\mathcal{Q}) = \sum_{S, T \in \mathcal{Q}} \sum_{u, v \in [d]} \tilde{C}^{u,v}(S, T) \quad (24)$$

where $\tilde{C}^{u,v}(S, T)$ are extended costs defined by

$$\tilde{C}^{u,v}(S, T) := \begin{cases} C_{S^u S^v, T^u T^v}^{u,v} & ; u < v \wedge u, v \in D(S) \cap D(T) \\ 0 & ; \text{else} \end{cases} \quad (25)$$

We further define $C^{u,v}(\mathcal{Q}) := \sum_{S, T \in \mathcal{Q}} \tilde{C}^{u,v}(S, T)$ as all costs involving the two objects $u, v \in [d]$. Splitting the double sum over $[d]$ w.r.t. the subset $\{p\} \subset [d]$ according to Lemma 1 yields

$$C(\mathcal{Q}) = \underbrace{C^{p,p}(\mathcal{Q})}_{=0} + \sum_{v \in [d] \setminus \{p\}} (C^{p,v}(\mathcal{Q}) + C^{v,p}(\mathcal{Q})) + \underbrace{\sum_{u, v \in [d] \setminus \{p\}} C^{u,v}(\mathcal{Q})}_{=: \mathfrak{C}^{-p}(\mathcal{Q})} \quad (26)$$

where $\mathfrak{C}^{-p}(\mathcal{Q})$ only sums terms that do not depend on the object $p \in [d]$. Now, the symmetrized extended costs $C^{u,v}(S, T)$ ($= \tilde{C}^{u,v}(S, T) + \tilde{C}^{v,u}(S, T)$) from Eq. (22) allow to write the symmetric summand of Eq. (26) as a double sum over all cliques $\mathcal{Q}$. Again, splitting this double sum w.r.t. the fixed cliques $\{Q, R\}$ yields according to Lemma 1

$$\begin{aligned} C^{p,v}(\mathcal{Q}) + C^{v,p}(\mathcal{Q}) &= \sum_{S, T \in \mathcal{Q}} C^{p,v}(S, T) \\ &= \underbrace{\sum_{S, T \in \{Q, R\}} C^{p,v}(S, T) + \sum_{\substack{S \in \{Q, R\} \\ T \in \mathcal{Q} \setminus \{Q, R\}}} (C^{p,v}(S, T) + C^{p,v}(T, S))}_{=: C_{\{Q, R\}}^{p,v}(\mathcal{Q})} + \underbrace{\sum_{S, T \in \mathcal{Q} \setminus \{Q, R\}} C^{p,v}(S, T)}_{=: \mathfrak{C}_{\neg\{Q, R\}}^{p,v}(\mathcal{Q})} . \end{aligned} \quad (27)$$

Finally, substituting Eq. (27) into Eq. (26) yields

$$C(\mathcal{Q}) = \sum_{v \in [d] \setminus \{p\}} C_{\{Q, R\}}^{p,v}(\mathcal{Q}) + \underbrace{\sum_{v \in [d] \setminus \{p\}} \mathfrak{C}_{\neg\{Q, R\}}^{p,v}(\mathcal{Q})}_{=: \mathfrak{C}_{\neg\{Q, R\}}^p(\mathcal{Q})} + \mathfrak{C}^{-p}(\mathcal{Q}) , \quad (28)$$

where $\mathfrak{C}_{\neg\{Q, R\}}^p(\mathcal{Q})$ only sums terms that do not depend on the cliques $Q, R \in \mathcal{Q}$. $\square$

The decomposition in Lemma 2 allows us to derive the objective change induced by a single swap as described in Sec. IV-D, see Theorem 3.

Theorem 3. Consider a single swap performed over a fixed object $p \in [d]$ and two cliques $Q, R \in \mathcal{Q}$, yielding a new solution $\mathcal{Q}'$ and assume the definitions of Lemma 2. The objective value change induced by this single swap decomposes into a sum of objective value changes between the fixed object $p \in [d]$ and all other objects $q \in [d] \setminus \{p\}$, *i.e.*,

$$C(\mathcal{Q}') - C(\mathcal{Q}) = \sum_{q \in [d] \setminus \{p\}} \delta C_{\text{swap}}^{p,q}(Q, R) \quad \text{with} \quad \delta C_{\text{swap}}^{p,q}(Q, R) = C_{\{Q', R'\}}^{p,q}(\mathcal{Q}') - C_{\{Q, R\}}^{p,q}(\mathcal{Q}) . \quad (29)$$

Proof. According to Lemma 2, we have

$$C(\mathcal{Q}') - C(\mathcal{Q}) = \sum_{q \in [d] \setminus \{p\}} \left(C_{\{Q', R'\}}^{p,q}(\mathcal{Q}') - C_{\{Q, R\}}^{p,q}(\mathcal{Q}) \right) + \underbrace{\mathfrak{C}_{\neg\{Q', R'\}}^p(\mathcal{Q}') - \mathfrak{C}_{\neg\{Q, R\}}^p(\mathcal{Q})}_{=0} + \underbrace{\mathfrak{C}^{-p}(\mathcal{Q}') - \mathfrak{C}^{-p}(\mathcal{Q})}_{=0} , \quad (30)$$

where the last two terms vanish because the swap affects a cost term only if it involves both the fixed object $p \in [d]$ and cliques $Q, R \in \mathcal{Q}$ (or $Q', R' \in \mathcal{Q}'$), which does not hold for $\mathfrak{C}_{\neg\{Q, R\}}^p(\mathcal{Q})$ (or $\mathfrak{C}_{\neg\{Q', R'\}}^p(\mathcal{Q}')$) and $\mathfrak{C}^{-p}(\mathcal{Q})$ (or $\mathfrak{C}^{-p}(\mathcal{Q}')$). $\square$

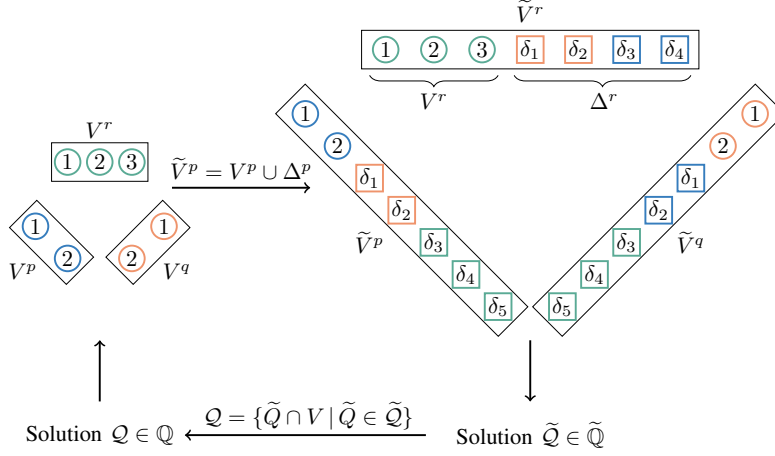


Fig. D.1. **Reducing incomplete to complete MGM.** To reduce an incomplete problem with objects V^p, V^q, V^r to a complete problem, one adds dummy vertices Δ^p to each object V^p . Solutions to the incomplete problem can be recovered from solutions to the complete one by removing dummy vertices from cliques. However, this transformation is, without further reasoning, *impractical* since *each* object $\tilde{V}^p$ of the complete problem is the same size as the *entire* incomplete problem $|V|$.

APPENDIX D

INCOMPLETE TO COMPLETE TRANSFORMATION: DETAILS

A. Proof for polynomial transformation from incomplete to complete MGM

Before we prove the polynomial-time transformation from incomplete to complete MGM, we formalize complete MGM analogously to incomplete MGM in Sec. IV-A, denoting complete pendants to incomplete quantities with a tilde, e.g., V^p and $\tilde{V}^p$.

Instances, hereafter referred to as problems, of incomplete (or complete) problems are fully defined via their objects V^p , $p \in [d]$, (or $\tilde{V}^p$, $p \in [\tilde{d}]$), and costs $C^{p,q}$, $p, q \in [d]$, (or $\tilde{C}^{p,q}$, $p, q \in [\tilde{d}]$), with the difference that objects of the complete instance must have equal cardinalities $\tilde{V}^p = \tilde{V}^q =: \tilde{n} \quad \forall p, q \in [\tilde{d}]$. We denote the set of feasible vertex partitions or solutions to the complete problem with $\tilde{\mathbb{Q}}$. All cliques $\tilde{Q} \in \tilde{\mathbb{Q}}$ of solutions $\tilde{Q} \in \tilde{\mathbb{Q}}$ to the complete problem must contain *exactly one* element per object $p \in [\tilde{d}]$, i.e., $|\tilde{Q} \cap \tilde{V}^p| = 1$, which implies that cliques are *maximal* $D(\tilde{Q}) = [\tilde{d}]$. Furthermore, since all vertices must be matched, solutions $\tilde{Q} \in \tilde{\mathbb{Q}}$ contain exactly $\tilde{n}$ cliques $|\tilde{\mathcal{Q}}| = \tilde{n}$. The objective for complete MGM is defined in analogy to Eq. (1)

$$\min_{\tilde{\mathcal{Q}} \in \tilde{\mathbb{Q}}} \left[\tilde{C}(\tilde{\mathcal{Q}}) := \sum_{\tilde{Q}, \tilde{R} \in \tilde{\mathbb{Q}}} \sum_{p, q \in [\tilde{d}]} \tilde{C}_{\tilde{Q}^p \tilde{Q}^q, \tilde{R}^p \tilde{R}^q}^{p, q} \right]. \quad (31)$$

Proof of Theorem 2. As described in Eq. (14) and (15), given an incomplete problem with objects V^p , $p \in [d]$, and costs $C^{p,q}$, $p, q \in [d]$, we construct a complete problem with objects

$$\tilde{V}^p = V^p \cup \Delta^p, \quad p \in [d], \quad (32)$$

and costs

$$\tilde{C}_{is, jt}^{p, q} = \begin{cases} C_{is, jt}^{p, q} & , \text{ if } i, s, j, t \in V \\ 0 & , \text{ else} \end{cases}, \quad (33)$$

where Δ^p , $p \in [d]$ are *dummy vertices* s.t. $|\Delta^p| = |V| - |V^p|$, see Fig. D.1. This problem is indeed *complete* since $|\tilde{V}^p| = |V^p| + |V| - |V^p| = |V| \quad \forall p \in [d]$. It is also constructed in *polynomial time* since $\sum_{p \in [d]} |V| - |V^p| = (d-1)|V|$ dummy vertices and $\sum_{p \in [d]} \sum_{q \in [d] \setminus \{p\}} (|V|^4 - |V^p|^2 |V^q|^2)$ costs are added to the incomplete problem.

What remains to prove is that the optimal solution of the incomplete problem can be determined in polynomial time by solving the complete one. This follows from the fact that we can *translate* solutions without changing their objective between the incomplete and complete problem by *adding* and *removing* dummy vertices. A solution $\tilde{\mathcal{Q}} \in \tilde{\mathbb{Q}}$ for the complete problem is translated to the solution

$$\mathcal{Q} := \{\tilde{Q} \cap V \mid \tilde{Q} \in \tilde{\mathcal{Q}}\}, \quad (34)$$

for the incomplete problem. We relate the cliques of both solutions via the *well-defined* function $\tau : \tilde{\mathcal{Q}} \rightarrow \mathcal{Q}, \tilde{Q} \mapsto \tilde{Q} \cap V$. The solution to the incomplete problem is actually a solution, i.e., $\mathcal{Q} \in \mathbb{Q}$, because it only contains *non-dummy vertices* by

Algorithm D.1: Incomplete to complete solution translation

Input: Incomplete Solution $\mathcal{Q} \in \mathbb{Q}$
 $\mathcal{Q} \leftarrow \mathcal{Q} \setminus \{\emptyset\}$

// enumerate cliques

 $\{Q_l\}_{l \in [|\mathcal{Q}|]} \leftarrow \mathcal{Q}$
 $\tilde{\mathcal{Q}} \leftarrow \{\tilde{Q}_l\}_{l \in [|\mathcal{Q}|]}$ where $\tilde{Q}_l = Q_l$ if $l \leq |\mathcal{Q}|$ and $\tilde{Q}_l = \emptyset$ else

for $p \in [d]$ **do**

// enumerate dummy vertices

 $\{\delta_l\}_{l \in [|\mathcal{Q}| - |\mathcal{V}^p|]} \leftarrow \Delta^p$

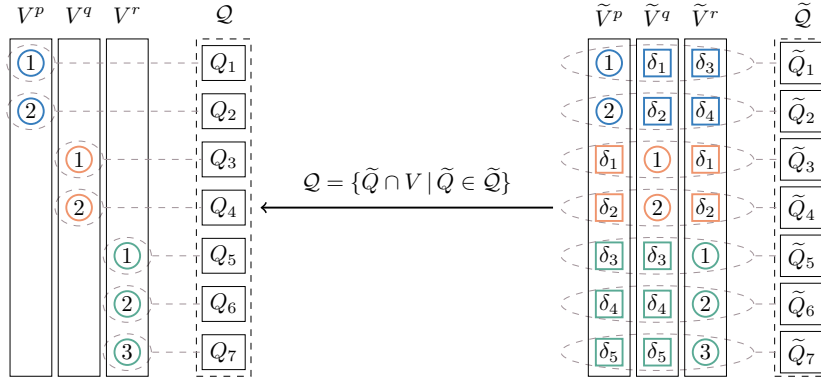
 Reorder $\tilde{\mathcal{Q}}$ s.t. $|\tilde{Q}_l \cap \mathcal{V}^p| = 1$ if $l \leq |\mathcal{V}^p|$
for $l \in [|\mathcal{Q}| - |\mathcal{V}^p|]$ **do**
 $\tilde{Q}_{l+|\mathcal{V}^p|} \leftarrow \tilde{Q}_{l+|\mathcal{V}^p|} \cup \delta_l$


Fig. D.2. **Requiring all dummy vertices.** In the transformation from Theorem 2, to obtain the incomplete solution that does not match any vertex (left), all dummy vertices are required by the complete solution (right). This example uses the setting of Fig. D.1. Solutions are illustrated identically to Fig. 4.

construction and *exhaustiveness*, *pairwise disjointness*, and *feasibility* carry over from the solution to the complete problem. Moreover, both objectives are equal

$$\begin{aligned}
 \tilde{C}(\tilde{\mathcal{Q}}) &= \sum_{\tilde{Q}, \tilde{R} \in \tilde{\mathcal{Q}}} \sum_{p, q \in [d]} \tilde{C}_{\tilde{Q}^p \tilde{Q}^q, \tilde{R}^p \tilde{R}^q}^{p, q} \\
 &\stackrel{\text{Eq. (33)}}{=} \sum_{\tilde{Q}, \tilde{R} \in \tilde{\mathcal{Q}}} \sum_{p, q \in D(\tau(\tilde{Q})) \cap D(\tau(\tilde{R}))} C_{\tau(\tilde{Q})^p \tau(\tilde{Q})^q, \tau(\tilde{R})^p \tau(\tilde{R})^q}^{p, q} \\
 &= \sum_{Q, R \in \mathcal{Q}} \sum_{p, q \in D(Q) \cap D(R)} C_{Q^p Q^q, R^p R^q}^{p, q} \\
 &= C(\mathcal{Q}).
 \end{aligned} \tag{35}$$

For the other direction, Alg. D.1 can be used to translate an incomplete solution $\mathcal{R} \in \mathbb{Q}$ to a complete solution $\tilde{\mathcal{R}} \in \tilde{\mathbb{Q}}$, which is indeed a solution because each dummy vertex is added to *exactly one* (possibly empty) clique out of $|\mathcal{V}|$ in total, which means cliques are *maximal* and *exhaustiveness*, *pairwise disjointness*, and *feasibility* carry over from the solution to the incomplete problem. Note, however, that this solution is generally not unique because the enumeration of dummy vertices in Alg. D.1 is an unspecified degree of freedom. Now, it holds by construction that $\mathcal{R} = \{\tilde{R} \cap \mathcal{V} \mid \tilde{R} \in \tilde{\mathcal{R}}\}$, hence, $C(\mathcal{R}) = \tilde{C}(\tilde{\mathcal{R}})$ by Eq. (35).

Therefore, translating any optimal solution to the complete problem via Eq. (34) yields an optimal solution to the incomplete one, which concludes the proof because this translation can be computed in polynomial time. $\square$

B. Minimality w.r.t. Number of Dummy Vertices

As discussed in Sec. VI, the number of dummy vertices required in Eq. (32) is $d - 1$ times the size of the entire incomplete problem and yet *minimal* because the incomplete solution not matching any vertex can only be obtained through a complete solution matching each non-dummy vertex to $(d - 1)$ dummy vertices, see Fig. D.2 for an example in the setting of Fig. D.1.

APPENDIX E

HAMMING DISTANCE AS SYNCHRONIZATION OBJECTIVE

Let $E \subset \bar{E}$ be a possibly inconsistent multi-matching. Some synchronization works, *e.g.* [27, 53], formulate the projection onto the set of cycle-consistent multi-matchings not by maximizing the overlap, or equivalently the intersection, as in Eq. (11), but instead by minimizing the *Hamming distance*, or equivalently the *symmetric difference*, *i.e.*,

$$E^* \in \arg \min_F |E \Delta F| \quad \text{s.t. } F \subset \bar{E} \text{ cycle-consistent} . \quad (36)$$

As in the overlap formulation from Eq. (12), the Hamming distance formulation from Eq. (36) can be written as a Multi-LAP over all objects $p \in [d]$ with costs

$$\widetilde{M}_{is, is}^{p, q} = \begin{cases} -1, & \text{if } is \in E, \\ +1, & \text{else} . \end{cases} \quad (37)$$

Indeed, for any feasible cycle-consistent multi-matching $F \subset \bar{E}$, the corresponding Multi-LAP cost is given by

$$C_{\text{MLAP}}(F) = \sum_{e \in F} -I(e \in E) + I(e \notin E) = -|F \cap E| + |F \setminus E|, \quad (38)$$

where I is the indicator function. Adding the constant $|E|$, which is independent of F , yields

$$|E| + C_{\text{MLAP}}(F) = |E| - |E \cap F| + |F \setminus E| = |E \setminus F| + |F \setminus E| = |E \Delta F|. \quad (39)$$

Thus, minimizing the Multi-LAP with costs Eq. (37) is equivalent to minimizing the symmetric difference synchronization objective Eq. (36). In the complete case, the overlap (11) and Hamming distance (36) formulations coincide, since on the feasible set their objectives differ only by an additive constant and a negative multiplicative factor, *i.e.*,

$$|E \Delta F| = |E \setminus F| + |F \setminus E| = |E| - |E \cap F| + |F| - |E \cap F| = 2(d^2 n. - |E \cap F|) = \text{const.} - 2|E \cap F|, \quad (40)$$

where $n. = |V^p|$ is the number keypoints in each object $p \in [d]$.

APPENDIX F

EXPERIMENT SETUP: DETAILS

Setup Details. All experiments in Sec. VII-B were run with a single core on an AMD Milan EPYC 7513. Parallel methods in Sec. VII-A were granted 10 processing cores. As mentioned in Sec. VII, we solve GM subproblems of *GREEDA* using the state-of-the-art GM solver [32, 57]. We run it with the following otherwise default parameters: Batch size 10, 10 greedy generations per batch, number of batches 10. When calling the GM solver [32, 57] directly from our *pylibmgm* python package, the stopping criterion k should be set to a high number, *i.e.* $k = 100000$, to yield equivalent results. Additionally, we start with a *fixed* randomization seed 42, which turns the otherwise randomized solver [32, 57] into a deterministic one. This is needed to separate the randomization effects of our construction heuristic (see Sec. IV-B) from those of the underlying GM solver. It follows, that we can attribute any randomization effects in our experiments to *GREEDA* itself.

For all synchronization methods, the $d(d-1)/2$ GM subproblems arising in preprocessing are solved using the same solver [32, 57] with identical settings. To ensure a fair comparison, each problem instance is solved only once, and the resulting solution is used as the common initialization for all algorithms.

APPENDIX G

DETAILED AVERAGED SYNCHRONIZATION RESULTS

We provide here detailed results of the synchronization algorithms considered in Sec. VII-C for every problem instance. We report the Multi-LAP (*M-LAP*), eq. (12), and *MGM*, eq. (1) objectives, as well as the number of forbidden matchings (*#forb.*) in the final solution and the runtime of each algorithm. The *preprocessing time*, the time required to calculate the cycle-inconsistent solution by solving the $d(d-1)/2$ independent GM subproblems, is given in the caption of each table. On the *worms* and *worms-large* datasets, most synchronization algorithms were evaluated only on *worms-10*, as they in general produce infeasible solutions with infinite cost.

TABLE G.1

SYNCHRONIZATION RESULTS FOR SYNTHETIC COMPLETE-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.04 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-60.00	0.00	-552.83	0.007
GREEDA (Sync)(worst)	-60.00	0.00	-552.83	0.007
MP-T (Sync)	-60.00	0.00	-552.83	0.003
MST	-60.00	0.00	-552.83	0.087
MST-Star	-60.00	0.00	-552.83	0.086
PPI	-60.00	0.00	-552.83	0.012
Spectral	-60.00	0.00	-552.83	0.086
Stiefel	-60.00	0.00	-552.83	0.275

TABLE G.2

SYNCHRONIZATION RESULTS FOR SYNTHETIC COMPLETE-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.15 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-280.00	0.00	-2580.57	0.012
GREEDA (Sync)(worst)	-280.00	0.00	-2580.57	0.012
MP-T (Sync)	-280.00	0.00	-2580.57	0.009
MST	-280.00	0.00	-2580.57	0.393
MST-Star	-280.00	0.00	-2580.57	0.391
PPI	-280.00	0.00	-2580.57	0.014
Spectral	-280.00	0.00	-2580.57	0.085
Stiefel	-280.00	0.00	-2580.57	0.256

TABLE G.3

SYNCHRONIZATION RESULTS FOR SYNTHETIC COMPLETE-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.34 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-660.00	0.00	-6076.54	0.019
GREEDA (Sync)(worst)	-660.00	0.00	-6076.54	0.019
MP-T (Sync)	-660.00	0.00	-6076.54	0.018
MST	-660.00	0.00	-6076.54	0.916
MST-Star	-660.00	0.00	-6076.54	0.914
PPI	-660.00	0.00	-6076.54	0.014
Spectral	-660.00	0.00	-6076.54	0.096
Stiefel	-660.00	0.00	-6076.54	0.278

TABLE G.4

SYNCHRONIZATION RESULTS FOR SYNTHETIC COMPLETE-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.61 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-1200.00	0.00	-11048.86	0.029
GREEDA (Sync)(worst)	-1200.00	0.00	-11048.86	0.029
MP-T (Sync)	-1200.00	0.00	-11048.86	0.031
MST	-1200.00	0.00	-11048.86	1.624
MST-Star	-1200.00	0.00	-11048.86	1.634
PPI	-1200.00	0.00	-11048.86	0.015
Spectral	-1200.00	0.00	-11048.86	0.088
Stiefel	-1200.00	0.00	-11048.86	0.269

TABLE G.5

SYNCHRONIZATION RESULTS FOR SYNTHETIC DEFORM-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.05 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-38.50	0.00	-284.85	0.009
GREEDA (Sync)(worst)	-37.10	0.00	-281.61	0.010
MP-T (Sync)	-36.90	0.00	-243.67	0.120
MST	-35.40	0.00	-308.42	0.086
MST-Star	-35.30	0.00	-308.73	0.086
PPI	-32.60	0.00	-255.42	0.013
Spectral	-37.60	0.00	-280.14	0.117
Stiefel	-36.10	0.00	-294.34	0.259

TABLE G.6

SYNCHRONIZATION RESULTS FOR SYNTHETIC DEFORM-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.19 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-144.90	0.00	-1303.02	0.064
GREEDA (Sync)(worst)	-138.00	0.00	-1227.35	0.066
MP-T (Sync)	-139.80	0.00	-1156.88	0.692
MST	-117.70	0.00	-1345.97	0.390
MST-Star	-120.00	0.00	-1374.90	0.393
PPI	-131.40	0.00	-1201.48	0.016
Spectral	-135.20	0.00	-1226.57	0.134
Stiefel	-127.20	0.00	-1190.27	0.262

TABLE G.7

SYNCHRONIZATION RESULTS FOR SYNTHETIC DEFORM-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.45 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-329.50	0.00	-3160.21	0.279
GREEDA (Sync)(worst)	-317.50	0.00	-3044.68	0.129
MP-T (Sync)	-322.30	0.00	-3088.39	1.782
MST	-246.00	0.00	-3072.96	0.912
MST-Star	-253.50	0.00	-3208.23	0.919
PPI	-308.60	0.00	-2998.04	0.021
Spectral	-318.00	0.00	-3136.55	0.150
Stiefel	-304.40	0.00	-3018.41	0.269

TABLE G.8

SYNCHRONIZATION RESULTS FOR SYNTHETIC DEFORM-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.81 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-559.70	0.00	-5794.40	0.588
GREEDA (Sync)(worst)	-541.60	0.00	-5620.65	0.375
MP-T (Sync)	-552.60	0.00	-5713.54	3.485
MST	-419.60	0.00	-5659.78	1.623
MST-Star	-404.90	0.00	-5798.54	1.632
PPI	-525.30	0.00	-5516.54	0.024
Spectral	-533.20	0.00	-5653.30	0.143
Stiefel	-517.30	0.00	-5536.65	0.278

TABLE G.9

SYNCHRONIZATION RESULTS FOR SYNTHETIC DENSITY-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.04 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-42.40	0.00	-189.88	0.009
GREEDA (Sync)(worst)	-41.90	0.00	-184.95	0.008
MP-T (Sync)	-41.70	0.00	-184.50	0.104
MST	-39.20	0.00	-198.15	0.086
MST-Star	-39.20	0.00	-199.97	0.087
PPI	-39.30	0.00	-179.14	0.013
Spectral	-41.60	0.00	-184.99	0.102
Stiefel	-40.90	0.00	-192.39	0.265

TABLE G.10

SYNCHRONIZATION RESULTS FOR SYNTHETIC DENSITY-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.18 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-163.20	0.00	-832.43	0.040
GREEDA (Sync)(worst)	-157.50	0.00	-808.54	0.049
MP-T (Sync)	-158.80	0.00	-780.07	0.647
MST	-144.50	0.00	-870.00	0.388
MST-Star	-142.90	0.00	-867.31	0.390
PPI	-143.40	0.00	-763.99	0.016
Spectral	-157.90	0.00	-808.39	0.123
Stiefel	-155.40	0.00	-810.87	0.274

TABLE G.11

SYNCHRONIZATION RESULTS FOR SYNTHETIC DENSITY-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.40 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-364.40	0.00	-2056.16	0.160
GREEDA (Sync)(worst)	-357.10	0.00	-1975.51	0.134
MP-T (Sync)	-357.70	0.00	-1989.32	1.757
MST	-311.10	0.00	-1991.32	0.913
MST-Star	-301.30	0.00	-2022.73	0.906
PPI	-344.60	0.00	-1920.39	0.019
Spectral	-354.50	0.00	-1981.91	0.133
Stiefel	-343.20	0.00	-1960.10	0.257

TABLE G.12

SYNCHRONIZATION RESULTS FOR SYNTHETIC DENSITY-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.73 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-661.80	0.00	-3837.84	0.218
GREEDA (Sync)(worst)	-652.40	0.00	-3731.19	0.302
MP-T (Sync)	-657.40	0.00	-3767.17	3.423
MST	-555.90	0.00	-3618.30	1.628
MST-Star	-531.00	0.00	-3672.19	1.625
PPI	-649.10	0.00	-3725.49	0.022
Spectral	-644.30	0.00	-3724.86	0.150
Stiefel	-634.90	0.00	-3674.14	0.286

TABLE G.13

SYNCHRONIZATION RESULTS FOR SYNTHETIC OUTLIER-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.04 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-36.40	0.00	109.10	0.007
GREEDA (Sync)(worst)	-36.40	0.00	109.10	0.007
MP-T (Sync)	-36.40	0.00	100.51	0.003
MST	-36.40	0.00	-43.36	0.085
MST-Star	-36.40	0.00	-43.36	0.085
PPI	-34.70	0.00	117.38	0.011
Spectral	-36.40	0.00	115.14	0.086
Stiefel	-36.40	0.00	60.17	0.252

TABLE G.14

SYNCHRONIZATION RESULTS FOR SYNTHETIC OUTLIER-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.16 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-169.40	0.00	530.29	0.013
GREEDA (Sync)(worst)	-169.40	0.00	530.29	0.013
MP-T (Sync)	-169.40	0.00	491.36	0.009
MST	-169.40	0.00	-202.44	0.388
MST-Star	-168.90	0.00	-202.31	0.393
PPI	-167.30	0.00	553.44	0.013
Spectral	-169.40	0.00	516.19	0.089
Stiefel	-169.20	0.00	274.18	0.256

TABLE G.15

SYNCHRONIZATION RESULTS FOR SYNTHETIC OUTLIER-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.35 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-397.70	0.00	1243.78	0.022
GREEDA (Sync)(worst)	-397.70	0.00	1243.78	0.022
MP-T (Sync)	-397.70	0.00	1020.35	0.067
MST	-397.50	0.00	-476.33	0.913
MST-Star	-396.10	0.00	-476.08	0.910
PPI	-393.50	0.00	1292.49	0.014
Spectral	-394.80	0.00	1261.04	0.091
Stiefel	-397.10	0.00	722.74	0.260

TABLE G.16

SYNCHRONIZATION RESULTS FOR SYNTHETIC OUTLIER-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.64 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-721.60	0.00	2278.62	0.035
GREEDA (Sync)(worst)	-721.60	0.00	2278.62	0.035
MP-T (Sync)	-721.60	0.00	1816.98	0.287
MST	-721.10	0.00	-866.38	1.640
MST-Star	-718.50	0.00	-865.78	1.620
PPI	-719.30	0.00	2269.22	0.014
Spectral	-710.80	0.00	2340.42	0.095
Stiefel	-721.00	0.00	1211.50	0.248

TABLE G.17

SYNCHRONIZATION RESULTS FOR HOTEL-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.04 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-32.50	0.00	181.41	0.008
GREEDA (Sync)(worst)	-32.10	0.00	178.84	0.007
MP-T (Sync)	-32.50	0.00	126.65	0.055
MST	-29.60	0.00	-19.05	0.086
MST-Star	-28.90	0.00	-18.36	0.086
PPI	-27.50	0.00	219.93	0.012
Spectral	-31.40	0.00	205.32	0.105
Stiefel	-30.90	0.00	144.78	0.276

TABLE G.18

SYNCHRONIZATION RESULTS FOR HOTEL-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.15 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-150.30	0.00	729.11	0.024
GREEDA (Sync)(worst)	-148.80	0.00	753.05	0.021
MP-T (Sync)	-151.10	0.00	399.68	0.375
MST	-139.70	0.00	-79.01	0.389
MST-Star	-127.20	0.00	-77.93	0.388
PPI	-139.50	0.00	886.68	0.014
Spectral	-140.80	0.00	882.17	0.117
Stiefel	-138.50	0.00	534.18	0.273

TABLE G.19

SYNCHRONIZATION RESULTS FOR HOTEL-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.34 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-358.70	0.00	1718.04	0.080
GREEDA (Sync)(worst)	-355.10	0.00	1734.70	0.053
MP-T (Sync)	-360.70	0.00	652.15	1.191
MST	-334.00	0.00	-171.31	0.913
MST-Star	-297.50	0.00	-177.42	0.904
PPI	-335.90	0.00	1977.17	0.016
Spectral	-329.40	0.00	2087.22	0.141
Stiefel	-327.30	0.00	1363.31	0.292

TABLE G.20

SYNCHRONIZATION RESULTS FOR HOTEL-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.61 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-652.40	0.00	3115.66	0.174
GREEDA (Sync)(worst)	-646.80	0.00	3214.51	0.145
MP-T (Sync)	-657.90	0.00	1037.56	2.651
MST	-601.10	0.00	-301.61	1.612
MST-Star	-524.40	0.00	-315.56	1.614
PPI	-634.30	0.00	3272.17	0.019
Spectral	-611.60	0.00	3588.96	0.129
Stiefel	-606.60	0.00	2214.57	0.296

TABLE G.21

SYNCHRONIZATION RESULTS FOR HOUSE-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.03 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-37.70	0.00	126.99	0.007
GREEDA (Sync)(worst)	-36.90	0.00	137.41	0.007
MP-T (Sync)	-37.70	0.00	89.06	0.044
MST	-35.50	0.00	-24.50	0.087
MST-Star	-35.30	0.00	-23.81	0.085
PPI	-32.80	0.00	181.14	0.012
Spectral	-36.30	0.00	150.10	0.103
Stiefel	-36.20	0.00	101.88	0.280

TABLE G.22

SYNCHRONIZATION RESULTS FOR HOUSE-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.14 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-172.70	0.00	618.13	0.020
GREEDA (Sync)(worst)	-170.80	0.00	620.82	0.018
MP-T (Sync)	-173.70	0.00	274.25	0.292
MST	-159.30	0.00	-103.69	0.395
MST-Star	-157.30	0.00	-101.73	0.396
PPI	-165.90	0.00	702.60	0.014
Spectral	-166.50	0.00	718.55	0.113
Stiefel	-163.80	0.00	344.20	0.281

TABLE G.23

SYNCHRONIZATION RESULTS FOR HOUSE-12. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.33 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-406.40	0.00	1462.16	0.060
GREEDA (Sync)(worst)	-402.80	0.00	1442.94	0.060
MP-T (Sync)	-410.10	0.00	457.94	1.268
MST	-367.60	0.00	-244.94	0.912
MST-Star	-363.40	0.00	-245.03	0.914
PPI	-393.50	0.00	1588.60	0.016
Spectral	-385.10	0.00	1688.97	0.141
Stiefel	-387.80	0.00	843.99	0.269

TABLE G.24

SYNCHRONIZATION RESULTS FOR HOUSE-16. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 0.60 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-733.80	0.00	2588.66	0.139
GREEDA (Sync)(worst)	-728.60	0.00	2653.82	0.090
MP-T (Sync)	-742.90	0.00	634.54	2.873
MST	-658.60	0.00	-440.94	1.623
MST-Star	-635.80	0.00	-429.01	1.616
PPI	-719.10	0.00	2771.18	0.020
Spectral	-706.30	0.00	2895.95	0.124
Stiefel	-708.30	0.00	1681.84	0.260

TABLE G.25

SYNCHRONIZATION RESULTS FOR WORMS-3. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 39.26 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-1379.50	0.00	-209313.79	0.284
GREEDA (Sync)(worst)	-1377.50	0.00	-209081.76	0.282
MP-T (Sync)	-1350.60	0.00	-202457.88	129.673
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.26

SYNCHRONIZATION RESULTS FOR WORMS-4. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 77.57 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-2687.30	0.00	-417463.71	0.879
GREEDA (Sync)(worst)	-2678.20	0.00	-416305.14	0.869
MP-T (Sync)	-2655.30	0.00	-409829.74	269.629
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.27

SYNCHRONIZATION RESULTS FOR WORMS-5. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 131.42 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-4383.60	0.00	-699850.20	1.436
GREEDA (Sync)(worst)	-4366.80	0.00	-699342.74	1.491
MP-T (Sync)	-4334.40	0.00	-688762.44	487.991
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.28

SYNCHRONIZATION RESULTS FOR WORMS-6. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 191.34 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-6446.20	0.00	-1050112.44	2.376
GREEDA (Sync)(worst)	-6421.30	0.00	-1046558.05	2.169
MP-T (Sync)	-6394.00	0.00	-1035444.15	801.740
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.29

SYNCHRONIZATION RESULTS FOR WORMS-7. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 264.08 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-8899.30	0.00	-1465515.57	3.455
GREEDA (Sync)(worst)	-8861.10	0.00	-1462942.47	3.183
MP-T (Sync)	-8816.20	0.00	-1448013.99	1183.435
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.30

SYNCHRONIZATION RESULTS FOR WORMS-8. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 352.87 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-11704.80	0.00	-1951795.10	4.551
GREEDA (Sync)(worst)	-11659.20	0.00	-1947156.36	4.277
MP-T (Sync)	-11606.70	0.00	-1933383.35	1677.381
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.31
SYNCHRONIZATION RESULTS FOR WORMS-9. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 454.05 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-14952.80	0.00	-2512570.77	6.433
GREEDA (Sync)(worst)	-14901.30	0.00	-2510013.86	5.668
MP-T (Sync)	-14832.10	0.00	-2490777.10	2280.563
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.32
SYNCHRONIZATION RESULTS FOR WORMS-10. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 569.02 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-18510.40	0.00	-3136748.13	8.357
GREEDA (Sync)(worst)	-18450.70	0.00	-3132603.41	6.961
MP-T (Sync)	-18332.50	0.00	-3108349.88	3008.551
MST	-14740.20	360.20	inf	113.575
MST-Star	-16349.00	213.30	inf	113.726
PPI	-18427.80	791.40	inf	3.600
Spectral	-18521.00	712.60	inf	25.089
Stiefel	-18461.40	579.50	inf	6.105
MP-T (Sync)($\alpha = 0$)	-18496.60	598.80	inf	30591.339
GREEDA (Sync)(best)($\alpha = 0$)	-18669.20	691.50	inf	189.399
GREEDA (Sync)(worst)($\alpha = 0$)	-18621.70	755.00	inf	202.476

TABLE G.33
SYNCHRONIZATION RESULTS FOR WORMS-20. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 879.33 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-57597.75	0.00	-11136025.34	37.722
GREEDA (Sync)(worst)	-57115.25	0.00	-11090742.15	36.263
MP-T (Sync)	-56568.00	0.00	-10996530.65	18413.977
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.34
SYNCHRONIZATION RESULTS FOR WORMS-30. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 2025.94 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-126659.00	0.00	-25051626.27	95.565
GREEDA (Sync)(worst)	-125874.00	0.00	-25021733.44	121.404
MP-T (Sync)	-123966.00	0.00	-24666921.25	56867.457
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.35
SYNCHRONIZATION RESULTS FOR WORMS-50. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 7667.12 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-544317.50	0.00	-20893113.44	234.614
GREEDA (Sync)(worst)	-543133.00	0.00	-20840082.94	242.706
MP-T (Sync)	-542699.75	0.00	-20897573.95	241885.890
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

TABLE G.36
SYNCHRONIZATION RESULTS FOR WORMS-100. AVERAGED OVER ALL INSTANCES. PREPROCESSING TIME: 30520.96 s.

Method	M-LAP obj. ↓	#Forb.↓	MGM obj. ↓	Runtime Sync ↓
GREEDA (Sync)(best)	-2184187.00	0.00	-84047397.13	1736.458
GREEDA (Sync)(worst)	-2179132.00	0.00	-83951621.19	1798.240
MST	—	—	—	—
MST-Star	—	—	—	—
PPI	—	—	—	—
Spectral	—	—	—	—
Stiefel	—	—	—	—

APPENDIX H

DETAILED OBJECTIVE VALUES PER INSTANCE

In most cases the best and the second best results have been obtained by *GREEDA (best)* and *GREEDA (worst)* computed based on 10 runs of our sequential variant of *GREEDA*. Note that these runs can be executed in parallel in the same time and even their sequential execution often takes less time than a single execution of the competing algorithms. This, in turn, means that achieving even *GREEDA (best)* results is computationally cheaper than execution of the competing algorithms.

In the tables, *Our-C* refers to the intermediate result of *GREEDA*, obtained by our sequential construction heuristic of Sec. IV-B and serves as a point of reference for the influence of the *GM local search* (see Sec. IV-B) and *SWAP local search* (see Sec. IV-D) on the final results of *GREEDA*.

The algorithms *DS** [19] and the *pygmttools* [56] implementations of *Cao* [34] and *Floyd* [33] are designed for complete MGM problems and were therefore evaluated only on the *synthetic complete/density/deform* datasets. On all other datasets, transforming the problem into a complete MGM as described in Sec. VI and Suppl. D makes these algorithms impractical, both in terms of runtime and memory required. Furthermore, *Cao* and *Floyd* do not guarantee cycle-consistent solutions. Results that are cycle-inconsistent are marked with a dagger (†) and excluded from the selection of the best result, which is indicated in bold.

On the *worms* and *worms-large* datasets, most synchronization algorithms were evaluated only on *worms-10*, as they in general produce infeasible solutions with infinite cost.

Lastly, we report out of memory (OOM) if an algorithm exceeds the evaluation machine’s memory limit of 236 GB.

TABLE H.4
SYNTHETIC COMPLETE (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
GREEDA (worst)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
GREEDA (Sync)(best)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
GREEDA (Sync)(worst)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
GREEDA-C (best)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
GREEDA-C (worst)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
MP-T [40]	-11080.96	-11021.58	-10614.98	-10521.96	-11142.22	-10815.80	-10965.60	-10478.03	-11095.05	-10696.66
MP-T (Sync)(best)	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
DS* [19]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Cao (pygmtools) [34, 56]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Floyd (pygmtools) [33, 56]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
MST [10]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
MST-Star [22]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
PPI ($\alpha = -1$) [36]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
PPI ($\alpha = -10$) [36]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
PPI ($\alpha = -50$) [36]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
PPI ($\alpha = 0$) [36]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Stiefel ($\alpha = -1$) [25]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Stiefel ($\alpha = -10$) [25]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Stiefel ($\alpha = -50$) [25]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Stiefel ($\alpha = 0$) [25]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Spectral ($\alpha = -1$) [27, 53]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Spectral ($\alpha = -10$) [27, 53]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Spectral ($\alpha = -50$) [27, 53]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11
Spectral ($\alpha = 0$) [27, 53]	-11080.96	-11021.58	-11059.12	-10953.02	-11142.22	-11079.59	-10965.60	-11096.36	-11095.05	-10995.11

TABLE H.5
SYNTHETIC DEFORM (4 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-342.83	-321.97	-292.49	-337.38	-320.10	-324.16	-328.66	-342.18	-315.66	-333.06
GREEDA (worst)	-335.20	-321.97	-289.34	-325.09	-310.72	-317.34	-320.85	-331.55	-307.92	-309.55
GREEDA (Sync)(best)	-276.88	-312.33	-238.74	-301.58	-237.11	-262.74	-301.77	-292.78	-293.50	-331.06
GREEDA (Sync)(worst)	-277.00	-289.34	-214.11	-301.58	-265.82	-278.22	-294.38	-299.98	-264.62	-331.06
GREEDA-C (best)	-342.83	-321.97	-291.64	-334.25	-313.84	-324.16	-327.82	-336.27	-314.52	-333.06
GREEDA-C (worst)	-323.13	-303.98	-288.49	-321.49	-307.63	-315.68	-308.08	-331.31	-301.24	-309.55
MP-T [40]	-259.75	-221.72	-243.01	-273.21	-244.24	-252.82	-249.54	-261.09	-254.38	-251.08
MP-T (Sync)(best)	-235.19	-257.81	-179.15	-300.89	-245.88	-235.78	-248.76	-189.25	-212.91	-331.06
DS* [19]	-300.55	-258.25	-201.20	-335.48	-229.73	-216.31	-252.86	-272.90	-256.74	-333.06
Cao (pygmtools) [34, 56]	-264.15 [†]	-275.53 [†]	-221.79 [†]	-269.68 [†]	-243.37 [†]	-249.82 [†]	-271.62 [†]	-297.74 [†]	-243.92 [†]	-275.86 [†]
Floyd (pygmtools) [33, 56]	-264.15 [†]	-275.53 [†]	-221.79 [†]	-269.68 [†]	-243.37 [†]	-249.82 [†]	-271.62 [†]	-297.74 [†]	-243.92 [†]	-275.86 [†]
MST [10]	-323.76	-296.22	-286.16	-307.18	-300.12	-311.86	-313.07	-312.51	-302.27	-331.06
MST-Star [22]	-323.76	-302.02	-275.60	-323.77	-307.08	-311.86	-310.72	-317.30	-302.27	-312.95
PPI ($\alpha = -1$) [36]	-246.02	-275.12	-224.91	-271.76	-230.45	-230.30	-253.50	-263.17	-233.62	-325.36
PPI ($\alpha = -10$) [36]	-246.02	-275.12	-224.91	-271.76	-230.45	-230.30	-253.50	-263.17	-233.62	-325.36
PPI ($\alpha = -50$) [36]	-246.02	-275.12	-224.91	-271.76	-230.45	-230.30	-253.50	-263.17	-233.62	-325.36
PPI ($\alpha = 0$) [36]	-246.02	-275.12	-224.91	-271.76	-230.45	-230.30	-253.50	-263.17	-233.62	-325.36
Stiefel ($\alpha = -1$) [25]	-316.22	-308.17	-214.10	-300.89	-243.80	-266.55	-271.89	-273.62	-275.13	-331.06
Stiefel ($\alpha = -10$) [25]	-316.22	-308.17	-214.10	-300.89	-243.80	-266.55	-271.89	-273.62	-275.13	-331.06
Stiefel ($\alpha = -50$) [25]	-316.22	-308.17	-214.10	-300.89	-243.80	-266.55	-271.89	-273.62	-275.13	-331.06
Stiefel ($\alpha = 0$) [25]	-316.22	-308.17	-214.10	-300.89	-243.80	-266.55	-271.89	-273.62	-275.13	-331.06
Spectral ($\alpha = -1$) [27, 53]	-302.62	-305.97	-255.17	-306.34	-262.02	-270.83	-313.60	-323.15	-272.69	-331.06
Spectral ($\alpha = -10$) [27, 53]	-302.62	-305.97	-255.17	-306.34	-262.02	-270.83	-313.60	-323.15	-272.69	-331.06
Spectral ($\alpha = -50$) [27, 53]	-302.62	-305.97	-255.17	-306.34	-262.02	-270.83	-313.60	-323.15	-272.69	-331.06
Spectral ($\alpha = 0$) [27, 53]	-302.62	-305.97	-255.17	-306.34	-262.02	-270.83	-313.60	-323.15	-272.69	-331.06

TABLE H.6
SYNTHETIC DEFORM (8 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-1523.81	-1456.37	-1370.43	-1458.59	-1510.40	-1478.13	-1501.71	-1544.77	-1536.48	-1484.89
GREEDA (worst)	-1387.04	-1456.37	-1329.65	-1405.29	-1378.37	-1426.58	-1501.71	-1459.14	-1406.44	-1442.26
GREEDA (Sync)(best)	-1338.05	-1305.97	-1203.33	-1274.75	-1240.10	-1188.23	-1352.27	-1291.82	-1461.57	-1374.09
GREEDA (Sync)(worst)	-1223.47	-1264.77	-1060.45	-1325.60	-1206.95	-1123.49	-1264.02	-1220.99	-1337.67	-1246.10
GREEDA-C (best)	-1523.81	-1450.39	-1342.89	-1456.61	-1510.40	-1478.13	-1488.93	-1540.45	-1510.82	-1440.19
GREEDA-C (worst)	-1381.87	-1319.96	-1305.38	-1368.23	-1365.10	-1421.98	-1377.28	-1436.76	-1395.48	-1348.48
MP-T [40]	-1077.48	-1073.27	-1038.39	-1091.66	-1056.98	-1094.43	-1126.61	-1124.24	-1143.33	-1065.24
MP-T (Sync)(best)	-1280.34	-1263.18	-1076.12	-1101.21	-1235.66	-1000.81	-1297.39	-1092.95	-1173.80	-1047.33
DS* [19]	-1359.40	-1451.00	-1102.54	-1381.85	-1510.40	-1438.76	-1379.50	-1518.18	-1536.48	-1473.00
Cao (pygmtools) [34, 56]	-1192.05 [†]	-1286.82 [†]	-1119.02 [†]	-1183.62 [†]	-1297.33 [†]	-1197.64 [†]	-1266.73 [†]	-1238.26 [†]	-1313.85 [†]	-1205.06 [†]
Floyd (pygmtools) [33, 56]	-1192.05 [†]	-1286.82 [†]	-1119.02 [†]	-1183.62 [†]	-1297.33 [†]	-1197.64 [†]	-1266.73 [†]	-1238.26 [†]	-1313.85 [†]	-1205.06 [†]
MST [10]	-1428.33	-1380.78	-1199.10	-1308.09	-1197.93	-1326.87	-1414.29	-1359.59	-1466.25	-1378.46
MST-Star [22]	-1428.33	-1368.54	-1244.92	-1340.12	-1343.51	-1371.75	-1427.70	-1402.02	-1437.75	-1384.39
PPI ($\alpha = -1$) [36]	-1141.52	-1320.08	-997.08	-1191.42	-1074.56	-1173.50	-1287.75	-1278.51	-1316.33	-1234.06
PPI ($\alpha = -10$) [36]	-1141.52	-1320.08	-997.08	-1191.42	-1074.56	-1173.50	-1287.75	-1278.51	-1316.33	-1234.06
PPI ($\alpha = -50$) [36]	-1141.52	-1320.08	-997.08	-1191.42	-1074.56	-1173.50	-1287.75	-1278.51	-1316.33	-1234.06
PPI ($\alpha = 0$) [36]	-1141.52	-1320.08	-997.08	-1191.42	-1074.56	-1173.50	-1287.75	-1278.51	-1316.33	-1234.06
Stiefel ($\alpha = -1$) [25]	-1174.68	-1225.39	-1073.42	-1320.62	-1232.88	-1095.85	-1282.15	-1171.64	-1429.36	-1259.67
Stiefel ($\alpha = -10$) [25]	-1174.68	-1225.39	-1073.42	-1320.62	-1232.88	-1095.85	-1282.15	-1171.64	-1429.36	-1259.67
Stiefel ($\alpha = -50$) [25]	-1174.68	-1225.39	-1073.42	-1320.62	-1232.88	-1095.85	-1282.15	-1171.64	-1429.36	-1259.67
Stiefel ($\alpha = 0$) [25]	-1174.68	-1225.39	-1073.42	-1320.62	-1232.88	-1095.85	-1282.15	-1171.64	-1429.36	-1259.67
Spectral ($\alpha = -1$) [27, 53]	-1063.79	-1278.72	-1043.35	-1234.60	-1049.63	-1171.76	-1350.71	-1160.19	-1274.98	-1274.97
Spectral ($\alpha = -10$) [27, 53]	-1063.79	-1278.72	-1043.35	-1234.60	-1049.63	-1171.76	-1350.71	-1160.19	-1274.98	-1274.97
Spectral ($\alpha = -50$) [27, 53]	-1063.79	-1278.72	-1043.35	-1234.60	-1049.63	-1171.76	-1350.71	-1160.19	-1274.98	-1274.97
Spectral ($\alpha = 0$) [27, 53]	-1063.79	-1278.72	-1043.35	-1234.60	-1049.63	-1171.76	-1350.71	-1160.19	-1274.98	-1274.97

TABLE H.7
SYNTHETIC DEFORM (12 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-3642.08	-3477.52	-3336.99	-3467.06	-3498.74	-3478.99	-3480.48	-3650.50	-3522.58	-3528.56
GREEDA (worst)	-3342.53	-3156.30	-3100.01	-3222.46	-3118.22	-3332.47	-3328.68	-3328.68	-3239.89	-3385.65
GREEDA (Sync)(best)	-3151.71	-3216.88	-2607.96	-3013.25	-3355.47	-2905.49	-3360.98	-3390.49	-3178.79	-3421.03
GREEDA (Sync)(worst)	-3283.79	-3177.51	-2609.88	-2795.71	-3349.43	-2615.74	-2993.56	-3253.29	-3042.21	-3325.71
GREEDA-C (best)	-3637.43	-3474.32	-3164.47	-3374.96	-3423.41	-3422.84	-3451.42	-3644.58	-3418.37	-3444.29
GREEDA-C (worst)	-3321.13	-3123.26	-3080.38	-3160.84	-3070.59	-3311.89	-3293.50	-3262.86	-3186.95	-3227.73
MP-T [40]	-2501.12	-2553.97	-2463.41	-2502.23	-2338.36	-2531.84	-2623.14	-2587.31	-2657.60	-2504.11
MP-T (Sync)(best)	-3082.02	-3227.93	-2577.48	-2813.64	-3355.47	-2701.78	-3406.26	-3319.78	-3220.70	-3178.88
DS* [19]	-3472.17	-3390.14	-3313.19	-3467.06	-3498.74	-3423.23	-3412.43	-3546.99	-3519.20	-3489.32
Cao (pygmtools) [34, 56]	-2926.98 [†]	-3283.94 [†]	-2755.15 [†]	-2945.53 [†]	-2948.18 [†]	-2782.37 [†]	-2930.49 [†]	-3076.08 [†]	-3053.41 [†]	-3074.04 [†]
Floyd (pygmtools) [33, 56]	-2926.98 [†]	-3283.94 [†]	-2755.15 [†]	-2945.53 [†]	-2948.18 [†]	-2782.37 [†]	-2930.49 [†]	-3076.08 [†]	-3053.41 [†]	-3074.04 [†]
MST [10]	-3265.30	-3004.91	-2764.88	-2939.39	-2661.98	-3052.46	-3311.39	-3405.55	-3143.44	-3180.28
MST-Star [22]	-3365.75	-3192.73	-2955.23	-3061.61	-2966.90	-3242.39	-3342.76	-3446.40	-3221.82	-3286.74
PPI ($\alpha = -1$) [36]	-3233.86	-3169.99	-2523.90	-2794.69	-3170.51	-2667.36	-3179.77	-3285.59	-2731.85	-3222.89
PPI ($\alpha = -10$) [36]	-3233.86	-3169.99	-2523.90	-2794.69	-3170.51	-2667.36	-3179.77	-3285.59	-2731.85	-3222.89
PPI ($\alpha = -50$) [36]	-3233.86	-3169.99	-2523.90	-2794.69	-3170.51	-2667.36	-3179.77	-3285.59	-2731.85	-3222.89
PPI ($\alpha = 0$) [36]	-3233.86	-3169.99	-2523.90	-2794.69	-3170.51	-2667.36	-3179.77	-3285.59	-2731.85	-3222.89
Stiefel ($\alpha = -1$) [25]	-3245.99	-3260.37	-2636.23	-3105.51	-3355.47	-2695.95	-3348.00	-3453.57	-3028.20	-3236.25
Stiefel ($\alpha = -10$) [25]	-3245.99	-3260.37	-2636.23	-3105.51	-3355.47	-2695.95	-3348.00	-3453.57	-3028.20	-3236.25
Stiefel ($\alpha = -50$) [25]	-3245.99	-3260.37	-2636.23	-3105.51	-3355.47	-2695.95	-3348.00	-3453.57	-3028.20	-3236.25
Stiefel ($\alpha = 0$) [25]	-3245.99	-3260.37	-2636.23	-3105.51	-3355.47	-2695.95	-3348.00	-3453.57	-3028.20	-3236.25
Spectral ($\alpha = -1$) [27, 53]	-3117.42	-3270.11	-2593.10	-3030.34	-3039.64	-2562.85	-3322.22	-3070.68	-2929.86	-3247.91
Spectral ($\alpha = -10$) [27, 53]	-3117.42	-3270.11	-2593.10	-3030.34	-3039.64	-2562.85	-3322.22	-3070.68	-2929.86	-3247.91
Spectral ($\alpha = -50$) [27, 53]	-3117.42	-3270.11	-2593.10	-3030.34	-3039.64	-2562.85	-3322.22	-3070.68	-2929.86	-3247.91
Spectral ($\alpha = 0$) [27, 53]	-3117.42	-3270.11	-2593.10	-3030.34	-3039.64	-2562.85	-3322.22	-3070.68	-2929.86	-3247.91

TABLE H.8
SYNTHETIC DEFORM (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-6507.40	-6569.81	-6164.47	-6278.85	-6272.24	-6349.59	-6231.85	-6615.35	-6411.24	-6561.48
GREEDA (worst)	-6491.59	-6569.81	-5651.26	-5850.96	-6272.24	-5989.36	-5831.64	-6615.35	-5979.06	-6557.37
GREEDA (Sync)(best)	-5625.95	-6476.52	-4905.23	-5571.68	-6021.39	-5111.74	-5788.84	-6186.30	-5819.38	-6436.94
GREEDA (Sync)(worst)	-5407.12	-6548.72	-4644.90	-5078.29	-6021.39	-4826.56	-5430.70	-6341.15	-5576.68	-6330.97
GREEDA-C (best)	-6507.40	-6569.81	-6150.18	-6208.06	-6205.33	-6137.90	-6178.78	-6615.35	-6400.05	-6449.65
GREEDA-C (worst)	-5826.41	-5720.11	-5572.70	-5653.17	-5437.93	-5961.16	-5707.70	-5905.28	-5862.12	-5929.07
MP-T [40]	-4596.00	-4586.45	-4462.58	-4206.68	-4587.89	-4515.56	-4594.57	-4897.13	-4556.19	-4596.19
MP-T (Sync)(best)	-5295.20	-6476.52	-5152.01	-5171.10	-6099.21	-4956.19	-5583.72	-6159.58	-5837.58	-6404.32
DS* [19]	-6102.13	-6569.81	-6150.22	-6267.58	-6272.24	-6042.47	-5603.83	-6462.57	-6410.54	-6552.52
Cao (pygmtools) [34, 56]	-5260.78 [†]	-6601.87 [†]	-5111.68 [†]	-5360.90 [†]	-5175.76 [†]	-5125.31 [†]	-5254.41 [†]	-5580.08 [†]	-5576.47 [†]	-6137.39 [†]
Floyd (pygmtools) [33, 56]	-5260.78 [†]	-6601.87 [†]	-5111.68 [†]	-5360.90 [†]	-5175.76 [†]	-5125.31 [†]	-5254.41 [†]	-5580.08 [†]	-5576.47 [†]	-6137.39 [†]
MST [10]	-5846.32	-6437.54	-5565.08	-5187.35	-4855.57	-5415.33	-5822.75	-6157.15	-5563.87	-5746.87
MST-Star [22]	-5891.28	-6234.64	-5611.47	-5631.62	-5407.07	-5699.01	-5955.77	-5955.59	-5803.71	-5795.19
PPI ($\alpha = -1$) [36]	-5351.94	-5719.55	-4652.30	-5195.43	-6105.31	-4806.49	-5244.93	-6184.40	-5625.49	-6279.53
PPI ($\alpha = -10$) [36]	-5351.94	-5719.55	-4652.30	-5195.43	-6105.31	-4806.49	-5244.93	-6184.40	-5625.49	-6279.53
PPI ($\alpha = -50$) [36]	-5351.94	-5719.55	-4652.30	-5195.43	-6105.31	-4806.49	-5244.93	-6184.40	-5625.49	-6279.53
PPI ($\alpha = 0$) [36]	-5351.94	-5719.55	-4652.30	-5195.43	-6105.31	-4806.49	-5244.93	-6184.40	-5625.49	-6279.53
Stiefel ($\alpha = -1$) [25]	-5401.39	-6429.82	-4856.57	-5220.50	-6083.29	-4804.07	-5854.60	-6280.60	-5424.93	-6177.22
Stiefel ($\alpha = -10$) [25]	-5401.39	-6429.82	-4856.57	-5220.50	-6083.29	-4804.07	-5854.60	-6280.60	-5424.93	-6177.22
Stiefel ($\alpha = -50$) [25]	-5401.39	-6429.82	-4856.57	-5220.50	-6083.29	-4804.07	-5854.60	-6280.60	-5424.93	-6177.22
Stiefel ($\alpha = 0$) [25]	-5401.39	-6429.82	-4856.57	-5220.50	-6083.29	-4804.07	-5854.60	-6280.60	-5424.93	-6177.22
Spectral ($\alpha = -1$) [27, 53]	-5632.37	-6418.48	-4734.20	-5176.75	-5583.30	-4755.40	-5956.49	-5566.09	-5390.31	-6153.14
Spectral ($\alpha = -10$) [27, 53]	-5632.37	-6418.48	-4734.20	-5176.75	-5583.30	-4755.40	-5956.49	-5566.09	-5390.31	-6153.14
Spectral ($\alpha = -50$) [27, 53]	-5632.37	-6418.48	-4734.20	-5176.75	-5583.30	-4755.40	-5956.49	-5566.09	-5390.31	-6153.14
Spectral ($\alpha = 0$) [27, 53]	-5632.37	-6418.48	-4734.20	-5176.75	-5583.30	-4755.40	-5956.49	-5566.09	-5390.31	-6153.14

TABLE H.9
SYNTHETIC DENSITY (4 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-180.66	-222.06	-202.26	-232.06	-176.74	-213.24	-208.91	-228.06	-210.06	-232.46
GREEDA (worst)	-176.58	-211.97	-196.23	-215.53	-171.30	-213.24	-198.67	-218.64	-203.48	-232.46
GREEDA (Sync)(best)	-144.55	-222.06	-182.26	-206.73	-131.47	-199.49	-186.17	-221.14	-172.43	-232.46
GREEDA (Sync)(worst)	-141.05	-222.06	-176.11	-206.73	-126.68	-199.49	-186.17	-221.14	-154.56	-215.52
GREEDA-C (best)	-180.66	-222.06	-202.26	-231.14	-172.88	-213.24	-208.91	-228.06	-204.68	-232.46
GREEDA-C (worst)	-171.51	-208.77	-194.25	-212.81	-171.30	-203.43	-193.44	-214.07	-202.48	-213.86
MP-T [40]	-126.18	-168.00	-162.30	-175.28	-131.48	-161.30	-149.75	-165.38	-159.68	-193.65
MP-T (Sync)(best)	-126.66	-222.06	-166.40	-219.06	-123.49	-200.77	-184.80	-216.94	-152.35	-232.46
DS* [19]	-127.32	-202.04	-195.01	-225.28	-128.95	-152.09	-142.68	-179.58	-204.61	-196.60
Cao (pygmtools) [34, 56]	-110.51 [†]	-211.35 [†]	-158.21 [†]	-214.77 [†]	-162.30 [†]	-143.51 [†]	-164.34 [†]	-177.42 [†]	-194.00 [†]	-179.48 [†]
Floyd (pygmtools) [33, 56]	-110.51 [†]	-211.35 [†]	-158.21 [†]	-214.77 [†]	-162.30 [†]	-143.51 [†]	-164.34 [†]	-177.42 [†]	-194.00 [†]	-179.48 [†]
MST [10]	-165.53	-208.46	-190.68	-228.04	-159.15	-208.36	-182.65	-214.93	-195.66	-228.02
MST-Star [22]	-174.25	-207.94	-186.90	-219.11	-167.78	-211.72	-193.38	-214.93	-195.66	-228.02
PPI ($\alpha = -1$) [36]	-121.80	-197.04	-177.46	-219.11	-123.09	-190.07	-159.78	-199.60	-170.95	-232.46
PPI ($\alpha = -10$) [36]	-121.80	-197.04	-177.46	-219.11	-123.09	-190.07	-159.78	-199.60	-170.95	-232.46
PPI ($\alpha = -50$) [36]	-121.80	-197.04	-177.46	-219.11	-123.09	-190.07	-159.78	-199.60	-170.95	-232.46
PPI ($\alpha = 0$) [36]	-121.80	-197.04	-177.46	-219.11	-123.09	-190.07	-159.78	-199.60	-170.95	-232.46
Stiefel ($\alpha = -1$) [25]	-144.55	-190.48	-169.00	-200.78	-145.28	-190.07	-197.18	-216.94	-163.16	-232.46
Stiefel ($\alpha = -10$) [25]	-144.55	-190.48	-169.00	-200.78	-145.28	-190.07	-197.18	-216.94	-163.16	-232.46
Stiefel ($\alpha = -50$) [25]	-144.55	-190.48	-169.00	-200.78	-145.28	-190.07	-197.18	-216.94	-163.16	-232.46
Stiefel ($\alpha = 0$) [25]	-144.55	-190.48	-169.00	-200.78	-145.28	-190.07	-197.18	-216.94	-163.16	-232.46
Spectral ($\alpha = -1$) [27, 53]	-167.12	-222.06	-185.66	-202.48	-131.91	-191.40	-193.77	-212.34	-184.66	-232.46
Spectral ($\alpha = -10$) [27, 53]	-167.12	-222.06	-185.66	-202.48	-131.91	-191.40	-193.77	-212.34	-184.66	-232.46
Spectral ($\alpha = -50$) [27, 53]	-167.12	-222.06	-185.66	-202.48	-131.91	-191.40	-193.77	-212.34	-184.66	-232.46
Spectral ($\alpha = 0$) [27, 53]	-167.12	-222.06	-185.66	-202.48	-131.91	-191.40	-193.77	-212.34	-184.66	-232.46

TABLE H.10
SYNTHETIC DENSITY (8 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-793.43	-970.22	-933.74	-920.62	-816.93	-865.71	-956.28	-1062.28	-938.28	-990.28
GREEDA (worst)	-764.35	-902.21	-892.44	-874.62	-766.18	-838.54	-904.83	-1062.28	-938.28	-894.50
GREEDA (Sync)(best)	-610.99	-930.85	-819.80	-902.99	-643.42	-774.07	-852.69	-1062.28	-785.49	-941.73
GREEDA (Sync)(worst)	-608.03	-930.85	-790.04	-877.26	-634.10	-691.23	-790.41	-1062.28	-816.06	-885.14
GREEDA-C (best)	-784.74	-969.81	-912.03	-919.80	-800.45	-855.02	-917.57	-1062.28	-915.77	-987.18
GREEDA-C (worst)	-746.59	-879.15	-846.54	-832.28	-761.45	-826.03	-870.25	-987.49	-850.59	-890.23
MP-T [40]	-549.26	-750.97	-685.10	-689.82	-596.83	-616.10	-646.45	-750.82	-649.43	-737.14
MP-T (Sync)(best)	-557.27	-916.15	-744.84	-861.38	-597.69	-524.84	-868.82	-1062.28	-752.18	-915.20
DS* [19]	-759.03	-969.81	-933.74	-790.95	-773.20	-716.29	-799.89	-1062.28	-771.63	-772.49
Cao (pygmtools) [34, 56]	-586.64 [†]	-876.16 [†]	-818.34 [†]	-854.17 [†]	-660.84 [†]	-676.36 [†]	-855.87 [†]	-1019.40	-822.56 [†]	-906.87 [†]
Floyd (pygmtools) [33, 56]	-586.64 [†]	-876.16 [†]	-818.34 [†]	-854.17 [†]	-660.84 [†]	-676.36 [†]	-855.87 [†]	-1019.40	-822.56 [†]	-906.87 [†]
MST [10]	-731.81	-945.83	-887.15	-851.44	-766.22	-821.40	-884.47	-1062.28	-790.34	-959.03
MST-Star [22]	-743.65	-933.68	-872.49	-842.11	-770.37	-820.22	-889.44	-1006.82	-847.90	-946.41
PPI ($\alpha = -1$) [36]	-546.31	-904.22	-769.58	-834.28	-555.39	-682.50	-717.21	-1062.28	-698.93	-869.19
PPI ($\alpha = -10$) [36]	-546.31	-904.22	-769.58	-834.28	-555.39	-682.50	-717.21	-1062.28	-698.93	-869.19
PPI ($\alpha = -50$) [36]	-546.31	-904.22	-769.58	-834.28	-555.39	-682.50	-717.21	-1062.28	-698.93	-869.19
PPI ($\alpha = 0$) [36]	-546.31	-904.22	-769.58	-834.28	-555.39	-682.50	-717.21	-1062.28	-698.93	-869.19
Stiefel ($\alpha = -1$) [25]	-562.69	-930.85	-815.66	-847.61	-641.71	-638.44	-800.21	-1062.28	-838.12	-946.36
Stiefel ($\alpha = -10$) [25]	-562.69	-930.85	-815.66	-847.61	-641.71	-638.44	-800.21	-1062.28	-838.12	-946.36
Stiefel ($\alpha = -50$) [25]	-562.69	-930.85	-815.66	-847.61	-641.71	-638.44	-800.21	-1062.28	-838.12	-946.36
Stiefel ($\alpha = 0$) [25]	-562.69	-930.85	-815.66	-847.61	-641.71	-638.44	-800.21	-1062.28	-838.12	-946.36
Spectral ($\alpha = -1$) [27, 53]	-595.65	-930.85	-808.79	-853.17	-606.34	-669.89	-853.03	-1062.28	-812.07	-916.63
Spectral ($\alpha = -10$) [27, 53]	-595.65	-930.85	-808.79	-853.17	-606.34	-669.89	-853.03	-1062.28	-812.07	-916.63
Spectral ($\alpha = -50$) [27, 53]	-595.65	-930.85	-808.79	-853.17	-606.34	-669.89	-853.03	-1062.28	-812.07	-916.63
Spectral ($\alpha = 0$) [27, 53]	-595.65	-930.85	-808.79	-853.17	-606.34	-669.89	-853.03	-1062.28	-812.07	-916.63

TABLE H.11
SYNTHETIC DENSITY (12 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-1941.66	-2326.68	-2075.42	-2174.66	-2006.66	-2023.55	-2162.66	-2374.66	-2298.66	-2350.66
GREEDA (worst)	-1825.29	-2326.68	-1939.62	-2071.96	-1855.79	-1905.93	-1994.70	-2374.66	-2029.63	-2350.66
GREEDA (Sync)(best)	-1757.44	-2326.68	-1761.23	-2146.04	-1742.02	-1788.76	-2142.07	-2374.66	-2214.01	-2308.66
GREEDA (Sync)(worst)	-1613.47	-2326.68	-1633.19	-2103.41	-1692.18	-1504.31	-2042.50	-2374.66	-2214.01	-2250.67
GREEDA-C (best)	-1891.98	-2326.68	-2011.05	-2088.83	-1954.22	-1981.40	-2151.95	-2374.66	-2229.22	-2350.66
GREEDA-C (worst)	-1795.56	-2095.35	-1925.60	-1947.70	-1841.22	-1865.75	-1941.44	-2046.83	-1968.98	-1817.98
MP-T [40]	-1312.45	-1685.81	-1554.37	-1487.44	-1442.00	-1462.78	-1522.89	-1529.82	-1785.80	-1509.32
MP-T (Sync)(best)	-1860.88	-2326.68	-1452.47	-2146.04	-1691.70	-1579.23	-2142.07	-2374.66	-2105.60	-2213.86
DS* [19]	-1930.30	-2326.68	-2044.00	-2174.62	-1828.16	-1938.15	-2162.66	-2374.66	-2298.66	-2350.66
Cao (pygmtools) [34, 56]	-1588.91 [†]	-2200.33 [†]	-1782.05 [†]	-2076.31 [†]	-1645.53 [†]	-1555.76 [†]	-1913.59 [†]	-2326.53 [†]	-2053.59 [†]	-2197.59 [†]
Floyd (pygmtools) [33, 56]	-1588.91 [†]	-2200.33 [†]	-1782.05 [†]	-2076.31 [†]	-1645.53 [†]	-1555.76 [†]	-1913.59 [†]	-2326.53 [†]	-2053.59 [†]	-2197.59 [†]
MST [10]	-1716.74	-2122.97	-1925.06	-1985.43	-1875.20	-1904.95	-2283.46	-1933.33	-2291.76	-2291.76
MST-Star [22]	-1759.23	-2253.90	-1890.06	-2004.62	-1847.32	-1949.72	-2009.43	-2263.79	-2025.74	-2223.52
PPI ($\alpha = -1$) [36]	-1418.20	-2326.68	-1696.74	-2062.63	-1706.97	-1457.52	-1672.87	-2353.32	-2231.85	-2277.12
PPI ($\alpha = -10$) [36]	-1418.20	-2326.68	-1696.74	-2062.63	-1706.97	-1457.52	-1672.87	-2353.32	-2231.85	-2277.12
PPI ($\alpha = -50$) [36]	-1418.20	-2326.68	-1696.74	-2062.63	-1706.97	-1457.52	-1672.87	-2353.32	-2231.85	-2277.12
PPI ($\alpha = 0$) [36]	-1418.20	-2326.68	-1696.74	-2062.63	-1706.97	-1457.52	-1672.87	-2353.32	-2231.85	-2277.12
Stiefel ($\alpha = -1$) [25]	-1583.65	-2259.81	-1659.84	-2048.72	-1731.20	-1585.42	-2055.75	-2374.66	-2233.78	-2286.28
Stiefel ($\alpha = -10$) [25]	-1583.65	-2259.81	-1659.84	-2048.72	-1731.20	-1585.42	-2055.75	-2374.66	-2233.78	-2286.28
Stiefel ($\alpha = -50$) [25]	-1583.65	-2259.81	-1659.84	-2048.72	-1731.20	-1585.42	-2055.75	-2374.66	-2233.78	-2286.28
Stiefel ($\alpha = 0$) [25]	-1583.65	-2259.81	-1659.84	-2048.72	-1731.20	-1585.42	-2055.75	-2374.66	-2233.78	-2286.28
Spectral ($\alpha = -1$) [27, 53]	-1563.41	-2274.60	-1678.91	-2037.94	-1490.46	-1565.99	-2056.52	-2374.66	-2236.16	-2322.34
Spectral ($\alpha = -10$) [27, 53]	-1563.41	-2274.60	-1678.91	-2037.94	-1490.46	-1565.99	-2056.52	-2374.66	-2236.16	-2322.34
Spectral ($\alpha = -50$) [27, 53]	-1563.41	-2274.60	-1678.91	-2037.94	-1490.46	-1565.99	-2056.52	-2374.66	-2236.16	-2322.34
Spectral ($\alpha = 0$) [27, 53]	-1563.41	-2274.60	-1678.91	-2037.94	-1490.46	-1565.99	-2056.52	-2374.66	-2236.16	-2322.34

TABLE H.12
SYNTHETIC DENSITY (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-3615.19	-4271.20	-3855.68	-3934.51	-3587.26	-3655.63	-3878.61	-4235.20	-4319.20	-4203.20
GREEDA (worst)	-3264.99	-4271.20	-3604.64	-3706.78	-3275.98	-3408.78	-3743.23	-4235.20	-4319.20	-4203.20
GREEDA (Sync)(best)	-3421.91	-4271.20	-3476.26	-3888.44	-3465.11	-3249.39	-3858.79	-4235.20	-4308.86	-4203.20
GREEDA (Sync)(worst)	-3402.87	-4271.20	-3321.71	-3846.02	-3152.67	-2711.32	-3858.79	-4235.20	-4308.86	-4203.20
GREEDA-C (best)	-3566.60	-4249.50	-3689.37	-3873.16	-3482.57	-3572.54	-3862.13	-4235.20	-4318.04	-4182.95
GREEDA-C (worst)	-3225.58	-3727.70	-3528.64	-3450.36	-3243.37	-3371.54	-3452.96	-3627.49	-3820.53	-3681.47
MP-T [40]	-2415.26	-3129.29	-2794.45	-2769.06	-2514.60	-2489.36	-2568.80	-2698.16	-2872.80	-2949.04
MP-T (Sync)(best)	-3418.92	-4271.20	-3406.42	-3797.15	-3087.34	-3008.98	-3858.79	-4220.86	-4308.86	-4203.20
DS* [19]	-3611.85	-4271.20	-3855.68	-3672.47	-3599.20	-3501.50	-3878.61	-4235.20	-4319.20	-4203.20
Cao (pygmtools) [34, 56]	-3021.38 [†]	-4239.30 [†]	-3405.61 [†]	-3761.75 [†]	-3084.22 [†]	-2892.22 [†]	-3504.96 [†]	-4171.92 [†]	-4329.05 [†]	-3991.81 [†]
Floyd (pygmtools) [33, 56]	-3021.38 [†]	-4239.30 [†]	-3405.61 [†]	-3761.75 [†]	-3084.22 [†]	-2892.22 [†]	-3504.96 [†]	-4171.92 [†]	-4329.05 [†]	-3991.81 [†]
MST [10]	-3214.17	-3601.05	-3560.12	-3573.63	-3387.10	-3080.99	-3424.63	-4050.58	-4218.91	-4071.88
MST-Star [22]	-3251.09	-3929.68	-3549.39	-3594.58	-3227.59	-3458.98	-3610.43	-3993.48	-4129.00	-3977.67
PPI ($\alpha = -1$) [36]	-3234.90	-4271.20	-3338.63	-3876.01	-3234.00	-2706.41	-3846.49	-4235.20	-4308.86	-4203.20
PPI ($\alpha = -10$) [36]	-3234.90	-4271.20	-3338.63	-3876.01	-3234.00	-2706.41	-3846.49	-4235.20	-4308.86	-4203.20
PPI ($\alpha = -50$) [36]	-3234.90	-4271.20	-3338.63	-3876.01	-3234.00	-2706.41	-3846.49	-4235.20	-4308.86	-4203.20
PPI ($\alpha = 0$) [36]	-3234.90	-4271.20	-3338.63	-3876.01	-3234.00	-2706.41	-3846.49	-4235.20	-4308.86	-4203.20
Stiefel ($\alpha = -1$) [25]	-3281.03	-4231.16	-3525.96	-3805.11	-3085.16	-2748.16	-3824.76	-4235.20	-4308.86	-4203.20
Stiefel ($\alpha = -10$) [25]	-3281.03	-4231.16	-3525.96	-3805.11	-3085.16	-2748.16	-3824.76	-4235.20	-4308.86	-4203.20
Stiefel ($\alpha = -50$) [25]	-3281.03	-4231.16	-3525.96	-3805.11	-3085.16	-2748.16	-3824.76	-4235.20	-4308.86	-4203.20
Stiefel ($\alpha = 0$) [25]	-3281.03	-4231.16	-3525.96	-3805.11	-3085.16	-2748.16	-3824.76	-4235.20	-4308.86	-4203.20
Spectral ($\alpha = -1$) [27, 53]	-3151.61	-4271.20	-3221.67	-3706.36	-2914.10	-2981.37	-3790.95	-4235.20	-4308.86	-4160.08
Spectral ($\alpha = -10$) [27, 53]	-3151.61	-4271.20	-3221.67	-3706.36	-2914.10	-2981.37	-3790.95	-4235.20	-4308.86	-4160.08
Spectral ($\alpha = -50$) [27, 53]	-3151.61	-4271.20	-3221.67	-3706.36	-2914.10	-2981.37	-3790.95	-4235.20	-4308.86	-4160.08
Spectral ($\alpha = 0$) [27, 53]	-3151.61	-4271.20	-3221.67	-3706.36	-2914.10	-2981.37	-3790.95	-4235.20	-4308.86	-4160.08

TABLE H.13
SYNTHETIC OUTLIER (4 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
GREEDA (worst)	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
GREEDA (Sync)(best)	114.27	110.54	103.01	101.21	126.88	84.20	123.17	110.09	116.72	100.86
GREEDA (Sync)(worst)	114.27	110.54	103.01	101.21	126.88	84.20	123.17	110.09	116.72	100.86
GREEDA-C (best)	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
GREEDA-C (worst)	-43.22	-43.22	-43.22	-43.22	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
MP-T [40]	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
MP-T (Sync)(best)	68.32	114.12	101.65	101.21	70.70	93.83	127.61	109.34	111.88	106.46
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
MST-Star [22]	-43.28	-43.22	-43.22	-43.28	-43.80	-43.93	-43.22	-43.22	-43.22	-43.22
PPI ($\alpha = -1$) [36]	151.31	123.79	126.37	114.25	91.40	138.68	107.06	115.97	109.48	95.50
PPI ($\alpha = -10$) [36]	151.31	123.79	126.37	114.25	91.40	138.68	107.06	115.97	109.48	95.50
PPI ($\alpha = -50$) [36]	151.31	123.79	126.37	114.25	91.40	138.68	107.06	115.97	109.48	95.50
PPI ($\alpha = 0$) [36]	151.31	123.79	126.37	114.25	91.40	138.68	107.06	115.97	109.48	95.50
Stiefel ($\alpha = -1$) [25]	107.49	116.87	123.31	120.32	110.56	119.03	113.74	117.90	111.75	110.41
Stiefel ($\alpha = -10$) [25]	107.49	116.87	123.31	120.32	110.56	119.03	113.74	117.90	111.75	110.41
Stiefel ($\alpha = -50$) [25]	107.49	116.87	123.31	120.32	110.56	119.03	113.74	117.90	111.75	110.41
Stiefel ($\alpha = 0$) [25]	107.49	116.87	123.31	120.32	110.56	119.03	113.74	117.90	111.75	110.41
Spectral ($\alpha = -1$) [27, 53]	59.17	72.86	59.69	31.01	37.28	67.33	76.65	62.67	64.79	70.23
Spectral ($\alpha = -10$) [27, 53]	59.17	72.86	59.69	31.01	37.28	67.33	76.65	62.67	64.79	70.23
Spectral ($\alpha = -50$) [27, 53]	59.17	72.86	59.69	31.01	37.28	67.33	76.65	62.67	64.79	70.23
Spectral ($\alpha = 0$) [27, 53]	59.17	72.86	59.69	31.01	37.28	67.33	76.65	62.67	64.79	70.23

TABLE H.14
SYNTHETIC OUTLIER (8 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-202.26	-201.70	-202.40	-202.31	-202.67	-204.36	-203.10	-202.33	-201.70	-201.75
GREEDA (worst)	-202.26	-201.70	-202.40	-202.31	-202.67	-204.36	-203.10	-202.33	-201.70	-201.75
GREEDA (Sync)(best)	548.77	513.28	522.24	534.65	552.76	450.82	534.26	524.21	582.42	539.49
GREEDA (Sync)(worst)	548.77	513.28	522.24	534.65	552.76	450.82	534.26	524.21	582.42	539.49
GREEDA-C (best)	-202.26	-201.70	-202.40	-202.13	-202.67	-204.21	-203.10	-202.33	-201.70	-201.70
GREEDA-C (worst)	-202.21	-201.70	-202.40	-202.13	-202.28	-203.75	-203.10	-202.33	-201.70	-201.70
MP-T [40]	-95.04	-36.38	-81.00	-60.70	-115.95	-36.83	-55.71	-115.21	-65.35	-160.65
MP-T (Sync)(best)	542.36	538.54	540.90	512.37	434.00	378.01	548.63	345.53	549.31	523.99
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-202.26	-201.70	-202.40	-202.19	-202.67	-204.36	-203.10	-202.33	-201.70	-201.70
MST-Star [22]	-202.21	-201.70	-202.40	-202.13	-202.28	-203.49	-203.10	-202.33	-201.70	-201.70
PPI ($\alpha = -1$) [36]	553.66	546.93	634.10	572.27	541.50	477.10	549.16	531.47	559.39	568.80
PPI ($\alpha = -10$) [36]	553.66	546.93	634.10	572.27	541.50	477.10	549.16	531.47	559.39	568.80
PPI ($\alpha = -50$) [36]	553.66	546.93	634.10	572.27	541.50	477.10	549.16	531.47	559.39	568.80
PPI ($\alpha = 0$) [36]	553.66	546.93	634.10	572.27	541.50	477.10	549.16	531.47	559.39	568.80
Stiefel ($\alpha = -1$) [25]	513.21	538.86	532.00	523.60	493.64	406.92	574.01	530.65	530.32	518.64
Stiefel ($\alpha = -10$) [25]	513.21	538.86	532.00	523.60	493.64	406.92	574.01	530.65	530.32	518.64
Stiefel ($\alpha = -50$) [25]	513.21	538.86	532.00	523.60	493.64	406.92	574.01	530.65	530.32	518.64
Stiefel ($\alpha = 0$) [25]	513.21	538.86	532.00	523.60	493.64	406.92	574.01	530.65	530.32	518.64
Spectral ($\alpha = -1$) [27, 53]	323.28	301.84	348.01	243.40	221.18	176.10	222.89	300.65	301.20	303.25
Spectral ($\alpha = -10$) [27, 53]	323.28	301.84	348.01	243.40	221.18	176.10	222.89	300.65	301.20	303.25
Spectral ($\alpha = -50$) [27, 53]	323.28	301.84	348.01	243.40	221.18	176.10	222.89	300.65	301.20	303.25
Spectral ($\alpha = 0$) [27, 53]	323.28	301.84	348.01	243.40	221.18	176.10	222.89	300.65	301.20	303.25

TABLE H.15
SYNTHETIC OUTLIER (12 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-476.28	-475.47	-476.74	-476.47	-477.54	-478.24	-477.09	-476.56	-475.89	-475.49
GREEDA (worst)	-476.28	-475.47	-476.74	-476.47	-477.54	-478.24	-477.09	-476.56	-475.89	-475.49
GREEDA (Sync)(best)	1288.41	1264.39	1243.46	1254.83	1262.77	1171.50	1240.49	1277.77	1254.45	1179.73
GREEDA (Sync)(worst)	1288.41	1264.39	1243.46	1254.83	1262.77	1171.50	1240.49	1277.77	1254.45	1179.73
GREEDA-C (best)	-476.28	-475.44	-476.50	-476.33	-477.43	-477.93	-477.09	-476.56	-475.89	-475.44
GREEDA-C (worst)	-475.95	-475.44	-476.14	-475.84	-477.04	-477.49	-476.84	-476.56	-475.89	-475.44
MP-T [40]	-190.32	-171.18	-332.10	-336.87	-315.53	-221.10	-220.39	-285.19	-200.63	-261.18
MP-T (Sync)(best)	863.37	853.79	788.24	726.51	981.94	1030.27	1286.33	1223.19	1178.51	1271.39
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-476.28	-475.47	-476.71	-474.23	-477.43	-478.24	-477.09	-476.56	-475.89	-475.44
MST-Star [22]	-475.95	-475.47	-476.14	-475.42	-476.13	-477.23	-477.09	-476.07	-475.89	-475.44
PPI ($\alpha = -1$) [36]	1320.46	1465.42	1232.40	1264.68	1282.66	1247.82	1253.24	1240.70	1257.74	1359.75
PPI ($\alpha = -10$) [36]	1320.46	1465.42	1232.40	1264.68	1282.66	1247.82	1253.24	1240.70	1257.74	1359.75
PPI ($\alpha = -50$) [36]	1320.46	1465.42	1232.40	1264.68	1282.66	1247.82	1253.24	1240.70	1257.74	1359.75
PPI ($\alpha = 0$) [36]	1320.46	1465.42	1232.40	1264.68	1282.66	1247.82	1253.24	1240.70	1257.74	1359.75
Stiefel ($\alpha = -1$) [25]	1280.12	1486.11	1207.86	1191.32	1242.04	1184.11	1226.88	1235.99	1276.20	1279.79
Stiefel ($\alpha = -10$) [25]	1280.12	1486.11	1207.86	1191.32	1242.04	1184.11	1226.88	1235.99	1276.20	1279.79
Stiefel ($\alpha = -50$) [25]	1280.12	1486.11	1207.86	1191.32	1242.04	1184.11	1226.88	1235.99	1276.20	1279.79
Stiefel ($\alpha = 0$) [25]	1280.12	1486.11	1207.86	1191.32	1242.04	1184.11	1226.88	1235.99	1276.20	1279.79
Spectral ($\alpha = -1$) [27, 53]	423.55	928.59	647.19	710.45	765.73	426.29	880.14	561.27	1022.84	861.40
Spectral ($\alpha = -10$) [27, 53]	423.55	928.59	647.19	710.45	765.73	426.29	880.14	561.27	1022.84	861.40
Spectral ($\alpha = -50$) [27, 53]	423.55	928.59	647.19	710.45	765.73	426.29	880.14	561.27	1022.84	861.40
Spectral ($\alpha = 0$) [27, 53]	423.55	928.59	647.19	710.45	765.73	426.29	880.14	561.27	1022.84	861.40

TABLE H.16
SYNTHETIC OUTLIER (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-866.71	-866.75	-866.49	-867.03	-867.90	-867.24	-866.42	-867.69	-864.88	-867.05
GREEDA (worst)	-866.71	-866.75	-866.49	-867.03	-867.90	-867.24	-866.42	-867.69	-864.88	-867.05
GREEDA (Sync)(best)	2331.70	2242.22	2226.52	2250.54	2299.76	2262.98	2345.80	2306.28	2332.84	2187.59
GREEDA (Sync)(worst)	2331.70	2242.22	2226.52	2250.54	2299.76	2262.98	2345.80	2306.28	2332.84	2187.59
GREEDA-C (best)	-866.60	-866.70	-866.27	-866.93	-867.79	-866.78	-866.42	-867.69	-864.88	-867.00
GREEDA-C (worst)	-866.32	-865.76	-865.89	-866.16	-866.92	-866.48	-865.84	-866.93	-864.88	-867.00
MP-T [40]	-305.00	-340.35	-364.98	-434.86	-491.93	-327.80	-320.58	-549.17	-407.05	-462.59
MP-T (Sync)(best)	1954.85	1430.73	1765.66	1925.45	1846.01	1859.53	1909.65	1536.94	2123.26	1817.73
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-866.71	-865.81	-866.46	-865.24	-867.79	-867.24	-866.42	-866.27	-864.88	-867.00
MST-Star [22]	-865.26	-866.16	-865.18	-865.95	-865.71	-866.22	-866.09	-866.30	-864.88	-865.99
PPI ($\alpha = -1$) [36]	2338.64	2292.25	2216.84	2312.93	2216.21	2127.46	2331.69	2300.79	2298.69	2256.67
PPI ($\alpha = -10$) [36]	2338.64	2292.25	2216.84	2312.93	2216.21	2127.46	2331.69	2300.79	2298.69	2256.67
PPI ($\alpha = -50$) [36]	2338.64	2292.25	2216.84	2312.93	2216.21	2127.46	2331.69	2300.79	2298.69	2256.67
PPI ($\alpha = 0$) [36]	2338.64	2292.25	2216.84	2312.93	2216.21	2127.46	2331.69	2300.79	2298.69	2256.67
Stiefel ($\alpha = -1$) [25]	2404.39	2343.06	2177.44	2276.56	2258.28	2434.65	2310.61	2294.85	2326.22	2278.10
Stiefel ($\alpha = -10$) [25]	2404.39	2343.06	2177.44	2276.56	2258.28	2434.65	2310.61	2294.85	2326.22	2278.10
Stiefel ($\alpha = -50$) [25]	2404.39	2343.06	2177.44	2276.56	2258.28	2434.65	2310.61	2294.85	2326.22	2278.10
Stiefel ($\alpha = 0$) [25]	2404.39	2343.06	2177.44	2276.56	2258.28	2434.65	2310.61	2294.85	2326.22	2278.10
Spectral ($\alpha = -1$) [27, 53]	1247.11	1344.17	709.74	1063.78	916.11	1258.55	1328.48	1389.94	1586.99	1270.14
Spectral ($\alpha = -10$) [27, 53]	1247.11	1344.17	709.74	1063.78	916.11	1258.55	1328.48	1389.94	1586.99	1270.14
Spectral ($\alpha = -50$) [27, 53]	1247.11	1344.17	709.74	1063.78	916.11	1258.55	1328.48	1389.94	1586.99	1270.14
Spectral ($\alpha = 0$) [27, 53]	1247.11	1344.17	709.74	1063.78	916.11	1258.55	1328.48	1389.94	1586.99	1270.14

TABLE H.17
HOTEL (4 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-13.87	-25.75	-25.06	-25.78	-12.36	-7.89	-20.20	-26.12	-16.67	-26.96
GREEDA (worst)	-13.71	-25.75	-24.99	-25.78	-10.20	-6.92	-20.20	-26.12	-16.38	-26.96
GREEDA (Sync)(best)	247.99	162.06	169.95	145.88	256.93	263.76	139.34	129.74	175.42	122.99
GREEDA (Sync)(worst)	266.43	162.06	166.81	145.88	229.98	249.72	139.34	129.74	175.42	122.99
GREEDA-C (best)	-13.87	-25.75	-25.06	-25.78	-12.36	-7.75	-20.20	-26.12	-16.38	-26.96
GREEDA-C (worst)	-8.47	-25.75	-24.99	-22.36	-9.29	-6.49	-15.31	-20.17	-12.38	-26.96
MP-T [40]	-9.54	-25.75	-25.06	-25.78	-10.94	-6.67	-20.20	-25.98	-14.94	-26.96
MP-T (Sync)(best)	203.23	166.88	118.55	46.48	136.59	129.98	132.08	71.46	115.36	145.93
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-13.80	-25.75	-23.17	-23.92	-11.12	-6.54	-20.20	-26.12	-15.27	-24.60
MST-Star [22]	-13.80	-25.75	-22.94	-23.92	-10.36	-5.93	-20.20	-26.12	-11.94	-22.70
PPI ($\alpha = -1$) [36]	266.72	201.60	177.05	204.26	304.33	322.57	146.02	146.19	247.50	183.10
PPI ($\alpha = -10$) [36]	266.72	201.60	177.05	204.26	304.33	322.57	146.02	146.19	247.50	183.10
PPI ($\alpha = -50$) [36]	266.72	201.60	177.05	204.26	304.33	322.57	146.02	146.19	247.50	183.10
PPI ($\alpha = 0$) [36]	266.72	201.60	177.05	204.26	304.33	322.57	146.02	146.19	247.50	183.10
Stiefel ($\alpha = -1$) [25]	253.98	168.45	165.85	147.66	283.59	274.00	235.13	152.52	224.35	147.68
Stiefel ($\alpha = -10$) [25]	253.98	168.45	165.85	147.66	283.59	274.00	235.13	152.52	224.35	147.68
Stiefel ($\alpha = -50$) [25]	253.98	168.45	165.85	147.66	283.59	274.00	235.13	152.52	224.35	147.68
Stiefel ($\alpha = 0$) [25]	253.98	168.45	165.85	147.66	283.59	274.00	235.13	152.52	224.35	147.68
Spectral ($\alpha = -1$) [27, 53]	265.72	138.86	124.54	86.58	155.98	205.09	227.20	83.68	102.20	57.94
Spectral ($\alpha = -10$) [27, 53]	265.72	138.86	124.54	86.58	155.98	205.09	227.20	83.68	102.20	57.94
Spectral ($\alpha = -50$) [27, 53]	265.72	138.86	124.54	86.58	155.98	205.09	227.20	83.68	102.20	57.94
Spectral ($\alpha = 0$) [27, 53]	265.72	138.86	124.54	86.58	155.98	205.09	227.20	83.68	102.20	57.94

TABLE H.18
HOTEL (8 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-97.33	-147.92	-81.44	-96.33	-61.84	-69.24	-101.97	-81.95	-69.69	-145.83
GREEDA (worst)	-97.25	-147.92	-81.44	-96.33	-59.91	-66.84	-101.97	-65.59	-69.69	-145.83
GREEDA (Sync)(best)	707.49	636.43	893.44	711.06	856.34	787.94	699.61	679.40	717.77	601.60
GREEDA (Sync)(worst)	770.43	636.43	793.40	744.61	856.34	887.12	699.61	679.40	834.52	628.61
GREEDA-C (best)	-97.25	-147.92	-81.37	-96.33	-60.40	-68.98	-101.97	-75.12	-68.97	-145.83
GREEDA-C (worst)	-62.78	-147.50	-68.87	-80.19	-38.61	-49.06	-65.60	-48.99	-43.03	-145.83
MP-T [40]	-70.24	-147.92	-72.00	-85.46	-43.72	-58.13	-100.78	-55.59	-44.61	-145.83
MP-T (Sync)(best)	240.77	497.75	472.98	293.62	458.72	400.17	358.32	389.02	453.47	432.01
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-87.15	-144.94	-72.17	-71.43	-34.63	-43.26	-97.64	-58.80	-48.29	-131.77
MST-Star [22]	-83.58	-137.08	-63.03	-74.19	-46.86	-57.42	-92.04	-50.30	-54.71	-120.12
PPI ($\alpha = -1$) [36]	970.66	871.59	861.77	986.47	840.45	937.18	728.11	966.45	998.48	705.68
PPI ($\alpha = -10$) [36]	970.66	871.59	861.77	986.47	840.45	937.18	728.11	966.45	998.48	705.68
PPI ($\alpha = -50$) [36]	970.66	871.59	861.77	986.47	840.45	937.18	728.11	966.45	998.48	705.68
PPI ($\alpha = 0$) [36]	970.66	871.59	861.77	986.47	840.45	937.18	728.11	966.45	998.48	705.68
Stiefel ($\alpha = -1$) [25]	931.64	727.24	994.98	869.29	930.37	1044.25	876.04	898.17	925.62	624.15
Stiefel ($\alpha = -10$) [25]	931.64	727.24	994.98	869.29	930.37	1044.25	876.04	898.17	925.62	624.15
Stiefel ($\alpha = -50$) [25]	931.64	727.24	994.98	869.29	930.37	1044.25	876.04	898.17	925.62	624.15
Stiefel ($\alpha = 0$) [25]	931.64	727.24	994.98	869.29	930.37	1044.25	876.04	898.17	925.62	624.15
Spectral ($\alpha = -1$) [27, 53]	673.72	589.10	538.22	628.80	412.79	783.59	230.58	391.64	891.15	202.24
Spectral ($\alpha = -10$) [27, 53]	673.72	589.10	538.22	628.80	412.79	783.59	230.58	391.64	891.15	202.24
Spectral ($\alpha = -50$) [27, 53]	673.72	589.10	538.22	628.80	412.79	783.59	230.58	391.64	891.15	202.24
Spectral ($\alpha = 0$) [27, 53]	673.72	589.10	538.22	628.80	412.79	783.59	230.58	391.64	891.15	202.24

TABLE H.19
HOTEL (12 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-232.45	-341.76	-200.29	-211.12	-140.78	-141.46	-277.02	-241.25	-189.06	-327.81
GREEDA (worst)	-231.43	-341.76	-200.01	-206.93	-99.92	-109.76	-277.02	-192.29	-189.02	-327.81
GREEDA (Sync)(best)	1560.06	1622.71	1896.38	1817.00	1917.43	2085.99	1670.05	1490.56	1648.95	1471.27
GREEDA (Sync)(worst)	1577.96	1596.57	1897.42	1810.71	1922.82	2094.53	1660.91	1641.96	1617.46	1526.63
GREEDA-C (best)	-230.78	-341.76	-200.19	-210.18	-139.29	-138.72	-277.02	-235.01	-188.28	-327.81
GREEDA-C (worst)	-155.77	-336.56	-170.80	-132.75	-78.64	-102.06	-241.55	-152.12	-142.71	-325.64
MP-T [40]	-157.08	-341.76	-183.58	-188.09	-95.82	-101.53	-277.02	-239.21	-143.58	-327.81
MP-T (Sync)(best)	591.34	863.61	596.38	983.26	596.41	714.15	700.49	440.14	472.57	563.21
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-128.48	-313.77	-171.91	-163.12	-59.77	-26.70	-256.78	-188.91	-111.27	-292.38
MST-Star [22]	-167.31	-299.59	-161.34	-171.24	-60.56	-87.77	-235.31	-183.26	-135.21	-272.65
PPI ($\alpha = -1$) [36]	2256.54	1739.01	2046.98	1933.65	2142.33	2166.98	1933.82	1739.19	1871.90	1941.30
PPI ($\alpha = -10$) [36]	2256.54	1739.01	2046.98	1933.65	2142.33	2166.98	1933.82	1739.19	1871.90	1941.30
PPI ($\alpha = -50$) [36]	2256.54	1739.01	2046.98	1933.65	2142.33	2166.98	1933.82	1739.19	1871.90	1941.30
PPI ($\alpha = 0$) [36]	2256.54	1739.01	2046.98	1933.65	2142.33	2166.98	1933.82	1739.19	1871.90	1941.30
Stiefel ($\alpha = -1$) [25]	2062.91	1680.36	2232.05	2288.69	2497.72	2833.54	1916.57	1905.15	1838.90	1616.33
Stiefel ($\alpha = -10$) [25]	2062.91	1680.36	2232.05	2288.69	2497.72	2833.54	1916.57	1905.15	1838.90	1616.33
Stiefel ($\alpha = -50$) [25]	2062.91	1680.36	2232.05	2288.69	2497.72	2833.54	1916.57	1905.15	1838.90	1616.33
Stiefel ($\alpha = 0$) [25]	2062.91	1680.36	2232.05	2288.69	2497.72	2833.54	1916.57	1905.15	1838.90	1616.33
Spectral ($\alpha = -1$) [27, 53]	1541.72	1006.73	1178.45	1642.52	2076.73	1969.04	1142.10	913.34	1433.70	728.80
Spectral ($\alpha = -10$) [27, 53]	1541.72	1006.73	1178.45	1642.52	2076.73	1969.04	1142.10	913.34	1433.70	728.80
Spectral ($\alpha = -50$) [27, 53]	1541.72	1006.73	1178.45	1642.52	2076.73	1969.04	1142.10	913.34	1433.70	728.80
Spectral ($\alpha = 0$) [27, 53]	1541.72	1006.73	1178.45	1642.52	2076.73	1969.04	1142.10	913.34	1433.70	728.80

TABLE H.20
HOTEL (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-377.28	-608.31	-391.78	-396.30	-281.84	-293.78	-501.24	-495.07	-320.94	-603.05
GREEDA (worst)	-371.22	-608.31	-388.53	-383.54	-226.71	-240.21	-501.24	-495.07	-320.94	-603.05
GREEDA (Sync)(best)	2992.76	2824.90	3362.97	3249.48	3436.20	3664.47	2960.99	2743.87	2986.11	2934.82
GREEDA (Sync)(worst)	2992.76	3069.38	3410.39	3247.56	3571.95	3844.94	2923.45	3042.92	3188.31	2853.44
GREEDA-C (best)	-373.12	-608.31	-389.42	-395.70	-252.29	-281.14	-501.24	-494.56	-319.09	-603.05
GREEDA-C (worst)	-250.94	-600.71	-361.52	-308.42	-144.20	-196.41	-487.06	-457.67	-247.00	-600.53
MP-T [40]	-168.51	-608.31	-358.35	-376.98	-248.73	-198.39	-495.75	-490.30	-205.27	-603.05
MP-T (Sync)(best)	2155.56	999.03	654.11	994.31	1383.27	992.42	693.12	288.88	1641.60	573.36
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-146.31	-559.87	-339.27	-296.91	-99.88	-93.65	-443.38	-403.01	-100.37	-533.47
MST-Star [22]	-268.90	-527.84	-318.06	-294.79	-133.34	-162.09	-394.38	-395.85	-164.61	-495.69
PPI ($\alpha = -1$) [36]	3102.56	2938.13	3469.10	3565.96	3369.28	3574.85	3002.15	2851.17	3825.12	3023.43
PPI ($\alpha = -10$) [36]	3102.56	2938.13	3469.10	3565.96	3369.28	3574.85	3002.15	2851.17	3825.12	3023.43
PPI ($\alpha = -50$) [36]	3102.56	2938.13	3469.10	3565.96	3369.28	3574.85	3002.15	2851.17	3825.12	3023.43
PPI ($\alpha = 0$) [36]	3102.56	2938.13	3469.10	3565.96	3369.28	3574.85	3002.15	2851.17	3825.12	3023.43
Stiefel ($\alpha = -1$) [25]	4017.13	3129.24	3466.01	3794.04	4234.95	4671.77	3095.22	3004.76	3461.87	3014.59
Stiefel ($\alpha = -10$) [25]	4017.13	3129.24	3466.01	3794.04	4234.95	4671.77	3095.22	3004.76	3461.87	3014.59
Stiefel ($\alpha = -50$) [25]	4017.13	3129.24	3466.01	3794.04	4234.95	4671.77	3095.22	3004.76	3461.87	3014.59
Stiefel ($\alpha = 0$) [25]	4017.13	3129.24	3466.01	3794.04	4234.95	4671.77	3095.22	3004.76	3461.87	3014.59
Spectral ($\alpha = -1$) [27, 53]	2826.93	2333.57	1830.32	2011.08	3195.24	2997.73	1937.30	1254.97	1765.99	1992.59
Spectral ($\alpha = -10$) [27, 53]	2826.93	2333.57	1830.32	2011.08	3195.24	2997.73	1937.30	1254.97	1765.99	1992.59
Spectral ($\alpha = -50$) [27, 53]	2826.93	2333.57	1830.32	2011.08	3195.24	2997.73	1937.30	1254.97	1765.99	1992.59
Spectral ($\alpha = 0$) [27, 53]	2826.93	2333.57	1830.32	2011.08	3195.24	2997.73	1937.30	1254.97	1765.99	1992.59

TABLE H.21
HOUSE (4 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-31.18	-20.40	-33.91	-32.64	-13.36	-25.03	-24.46	-40.35	-23.09	-15.90
GREEDA (worst)	-31.18	-20.40	-33.91	-32.64	-12.56	-25.03	-24.46	-40.35	-23.09	-15.90
GREEDA (Sync)(best)	114.88	136.59	98.01	83.17	191.64	73.96	129.39	128.77	144.31	169.15
GREEDA (Sync)(worst)	114.88	136.59	98.01	112.98	212.53	146.00	129.39	112.26	120.06	191.41
GREEDA-C (best)	-31.18	-20.40	-33.91	-32.64	-13.36	-25.03	-24.46	-40.35	-23.09	-15.90
GREEDA-C (worst)	-31.07	-18.49	-31.32	-23.67	-8.79	-21.87	-24.46	-40.35	-16.24	-13.23
MP-T [40]	-31.07	-19.17	-33.91	-32.64	-11.09	-25.03	-24.46	-40.35	-19.34	-12.63
MP-T (Sync)(best)	57.43	125.16	131.14	102.22	152.14	58.07	101.24	57.74	105.42	92.02
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-31.07	-20.21	-33.17	-31.77	-11.16	-21.74	-24.46	-39.65	-15.86	-15.90
MST-Star [22]	-31.07	-20.21	-33.17	-28.93	-9.19	-21.13	-24.46	-39.65	-14.39	-15.90
PPI ($\alpha = -1$) [36]	136.23	125.16	180.12	92.32	282.87	177.24	164.95	207.87	197.94	246.69
PPI ($\alpha = -10$) [36]	136.23	125.16	180.12	92.32	282.87	177.24	164.95	207.87	197.94	246.69
PPI ($\alpha = -50$) [36]	136.23	125.16	180.12	92.32	282.87	177.24	164.95	207.87	197.94	246.69
PPI ($\alpha = 0$) [36]	136.23	125.16	180.12	92.32	282.87	177.24	164.95	207.87	197.94	246.69
Stiefel ($\alpha = -1$) [25]	137.69	166.61	119.83	124.77	198.25	73.96	148.23	127.88	211.53	192.28
Stiefel ($\alpha = -10$) [25]	137.69	166.61	119.83	124.77	198.25	73.96	148.23	127.88	211.53	192.28
Stiefel ($\alpha = -50$) [25]	137.69	166.61	119.83	124.77	198.25	73.96	148.23	127.88	211.53	192.28
Stiefel ($\alpha = 0$) [25]	137.69	166.61	119.83	124.77	198.25	73.96	148.23	127.88	211.53	192.28
Spectral ($\alpha = -1$) [27, 53]	86.19	70.10	29.19	83.02	229.73	96.58	73.79	71.50	151.85	126.87
Spectral ($\alpha = -10$) [27, 53]	86.19	70.10	29.19	83.02	229.73	96.58	73.79	71.50	151.85	126.87
Spectral ($\alpha = -50$) [27, 53]	86.19	70.10	29.19	83.02	229.73	96.58	73.79	71.50	151.85	126.87
Spectral ($\alpha = 0$) [27, 53]	86.19	70.10	29.19	83.02	229.73	96.58	73.79	71.50	151.85	126.87

TABLE H.22
HOUSE (8 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-147.52	-145.68	-113.22	-116.08	-102.53	-145.90	-93.64	-116.73	-134.16	-89.10
GREEDA (worst)	-146.75	-145.68	-113.22	-115.34	-102.53	-145.37	-93.64	-116.73	-134.16	-85.24
GREEDA (Sync)(best)	530.60	541.07	627.59	567.80	721.03	566.02	712.76	681.97	536.48	695.94
GREEDA (Sync)(worst)	575.84	541.07	627.59	613.39	774.06	430.82	736.46	708.22	529.14	671.64
GREEDA-C (best)	-147.52	-145.68	-113.22	-116.08	-102.53	-145.90	-93.64	-116.73	-134.16	-88.60
GREEDA-C (worst)	-118.27	-142.31	-92.98	-92.36	-83.09	-122.36	-71.89	-99.05	-102.85	-70.66
MP-T [40]	-146.75	-145.68	-101.38	-112.17	-99.85	-144.82	-89.98	-116.46	-132.30	-79.20
MP-T (Sync)(best)	157.73	169.53	322.91	200.01	457.37	215.69	417.01	333.47	206.87	261.92
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-114.60	-134.33	-96.38	-105.50	-97.23	-126.63	-83.42	-102.53	-105.37	-70.95
MST-Star [22]	-134.26	-129.32	-88.49	-104.37	-83.84	-128.46	-78.53	-100.60	-94.23	-75.18
PPI ($\alpha = -1$) [36]	584.87	607.67	799.10	682.35	763.97	645.70	812.72	806.16	618.87	704.57
PPI ($\alpha = -10$) [36]	584.87	607.67	799.10	682.35	763.97	645.70	812.72	806.16	618.87	704.57
PPI ($\alpha = -50$) [36]	584.87	607.67	799.10	682.35	763.97	645.70	812.72	806.16	618.87	704.57
PPI ($\alpha = 0$) [36]	584.87	607.67	799.10	682.35	763.97	645.70	812.72	806.16	618.87	704.57
Stiefel ($\alpha = -1$) [25]	573.68	601.70	767.40	622.72	745.64	645.58	877.95	722.57	650.21	978.02
Stiefel ($\alpha = -10$) [25]	573.68	601.70	767.40	622.72	745.64	645.58	877.95	722.57	650.21	978.02
Stiefel ($\alpha = -50$) [25]	573.68	601.70	767.40	622.72	745.64	645.58	877.95	722.57	650.21	978.02
Stiefel ($\alpha = 0$) [25]	573.68	601.70	767.40	622.72	745.64	645.58	877.95	722.57	650.21	978.02
Spectral ($\alpha = -1$) [27, 53]	263.27	234.54	218.85	383.70	190.60	279.34	652.03	419.61	421.87	378.20
Spectral ($\alpha = -10$) [27, 53]	263.27	234.54	218.85	383.70	190.60	279.34	652.03	419.61	421.87	378.20
Spectral ($\alpha = -50$) [27, 53]	263.27	234.54	218.85	383.70	190.60	279.34	652.03	419.61	421.87	378.20
Spectral ($\alpha = 0$) [27, 53]	263.27	234.54	218.85	383.70	190.60	279.34	652.03	419.61	421.87	378.20

TABLE H.23
HOUSE (12 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-363.43	-391.51	-260.92	-250.44	-236.53	-353.46	-257.68	-355.49	-283.75	-223.04
GREEDA (worst)	-363.43	-391.51	-260.92	-249.89	-236.50	-352.24	-257.68	-350.77	-283.75	-215.66
GREEDA (Sync)(best)	1240.93	1251.69	1474.73	1489.89	1814.45	1465.15	1697.28	1318.11	1378.84	1490.50
GREEDA (Sync)(worst)	1256.15	1234.83	1355.09	1537.52	1752.48	1393.58	1638.37	1440.03	1314.52	1506.79
GREEDA-C (best)	-363.43	-390.92	-260.92	-247.02	-236.53	-353.46	-257.59	-355.49	-283.75	-214.52
GREEDA-C (worst)	-351.00	-359.92	-194.92	-174.93	-198.69	-339.26	-169.38	-345.36	-242.23	-195.80
MP-T [40]	-363.43	-390.92	-260.92	-249.22	-231.75	-351.03	-228.90	-355.49	-277.57	-192.37
MP-T (Sync)(best)	426.65	261.60	341.70	500.71	907.48	165.56	594.58	741.60	88.08	551.45
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-289.71	-330.79	-173.15	-217.05	-204.84	-286.46	-193.13	-327.71	-250.49	-176.06
MST-Star [22]	-322.89	-324.23	-185.50	-215.73	-190.46	-313.64	-194.31	-302.50	-224.37	-176.63
PPI ($\alpha = -1$) [36]	1319.68	1342.21	1574.82	1588.12	1968.44	1460.50	1789.96	1470.12	1329.92	2042.21
PPI ($\alpha = -10$) [36]	1319.68	1342.21	1574.82	1588.12	1968.44	1460.50	1789.96	1470.12	1329.92	2042.21
PPI ($\alpha = -50$) [36]	1319.68	1342.21	1574.82	1588.12	1968.44	1460.50	1789.96	1470.12	1329.92	2042.21
PPI ($\alpha = 0$) [36]	1319.68	1342.21	1574.82	1588.12	1968.44	1460.50	1789.96	1470.12	1329.92	2042.21
Stiefel ($\alpha = -1$) [25]	1350.64	1353.99	1852.22	1904.55	2224.61	1377.13	1877.88	1429.20	1341.02	2178.41
Stiefel ($\alpha = -10$) [25]	1350.64	1353.99	1852.22	1904.55	2224.61	1377.13	1877.88	1429.20	1341.02	2178.41
Stiefel ($\alpha = -50$) [25]	1350.64	1353.99	1852.22	1904.55	2224.61	1377.13	1877.88	1429.20	1341.02	2178.41
Stiefel ($\alpha = 0$) [25]	1350.64	1353.99	1852.22	1904.55	2224.61	1377.13	1877.88	1429.20	1341.02	2178.41
Spectral ($\alpha = -1$) [27, 53]	496.81	753.09	557.44	1207.69	1066.84	466.91	1065.15	975.07	677.24	1173.69
Spectral ($\alpha = -10$) [27, 53]	496.81	753.09	557.44	1207.69	1066.84	466.91	1065.15	975.07	677.24	1173.69
Spectral ($\alpha = -50$) [27, 53]	496.81	753.09	557.44	1207.69	1066.84	466.91	1065.15	975.07	677.24	1173.69
Spectral ($\alpha = 0$) [27, 53]	496.81	753.09	557.44	1207.69	1066.84	466.91	1065.15	975.07	677.24	1173.69

TABLE H.24
HOUSE (16 OBJECTS). OBJECTIVE VALUES PER INSTANCE.

Method	Instance									
	1	2	3	4	5	6	7	8	9	10
GREEDA (best)	-667.56	-715.49	-502.76	-489.41	-479.29	-649.10	-480.40	-639.27	-450.88	-444.46
GREEDA (worst)	-664.91	-715.49	-496.10	-485.91	-479.29	-644.28	-480.40	-637.95	-450.88	-421.20
GREEDA (Sync)(best)	2346.69	2213.55	2624.82	2708.58	3151.27	2166.15	2816.14	2608.47	2752.78	2498.18
GREEDA (Sync)(worst)	2310.27	2406.13	2752.27	3262.84	2433.73	2433.73	2750.72	2738.45	2888.04	3010.58
GREEDA-C (best)	-667.36	-715.49	-502.46	-484.01	-479.16	-645.38	-475.24	-637.32	-450.14	-444.33
GREEDA-C (worst)	-589.33	-684.13	-433.77	-360.92	-389.41	-552.89	-388.60	-563.75	-312.46	-355.07
MP-T [40]	-667.38	-715.49	-486.26	-469.58	-451.45	-649.10	-428.67	-638.35	-392.39	-384.21
MP-T (Sync)(best)	729.77	705.93	448.54	397.13	853.03	235.58	851.25	1152.72	391.73	579.71
DS* [19]	—	—	—	—	—	—	—	—	—	—
Cao (pygmtools) [34, 56]	—	—	—	—	—	—	—	—	—	—
Floyd (pygmtools) [33, 56]	—	—	—	—	—	—	—	—	—	—
MST [10]	-536.05	-559.74	-415.77	-417.18	-393.44	-520.41	-351.99	-559.70	-309.78	-345.32
MST-Star [22]	-573.66	-567.98	-347.24	-403.03	-336.81	-572.01	-337.31	-532.08	-272.68	-347.34
PPI ($\alpha = -1$) [36]	2433.95	2263.70	2613.44	2676.20	3280.38	2491.27	3532.05	2891.44	2566.02	2963.33
PPI ($\alpha = -10$) [36]	2433.95	2263.70	2613.44	2676.20	3280.38	2491.27	3532.05	2891.44	2566.02	2963.33
PPI ($\alpha = -50$) [36]	2433.95	2263.70	2613.44	2676.20	3280.38	2491.27	3532.05	2891.44	2566.02	2963.33
PPI ($\alpha = 0$) [36]	2433.95	2263.70	2613.44	2676.20	3280.38	2491.27	3532.05	2891.44	2566.02	2963.33
Stiefel ($\alpha = -1$) [25]	2371.13	2269.42	3205.86	2994.48	3700.07	2539.10	2681.12	2758.95	3063.26	3376.15
Stiefel ($\alpha = -10$) [25]	2371.13	2269.42	3205.86	2994.48	3700.07	2539.10	2681.12	2758.95	3063.26	3376.15
Stiefel ($\alpha = -50$) [25]	2371.13	2269.42	3205.86	2994.48	3700.07	2539.10	2681.12	2758.95	3063.26	3376.15
Stiefel ($\alpha = 0$) [25]	2371.13	2269.42	3205.86	2994.48	3700.07	2539.10	2681.12	2758.95	3063.26	3376.15
Spectral ($\alpha = -1$) [27, 53]	1159.75	1807.76	1909.72	2071.93	2113.20	1096.22	1372.29	1770.04	1737.46	1780.06
Spectral ($\alpha = -10$) [27, 53]	1159.75	1807.76	1909.72	2071.93	2113.20	1096.22	1372.29	1770.04	1737.46	1780.06
Spectral ($\alpha = -50$) [27, 53]	1159.75	1807.76	1909.72	2071.93	2113.20	1096.22	1372.29	1770.04	1737.46	1780.06
Spectral ($\alpha = 0$) [27, 53]	1159.75	1807.76	1909.72	2071.93	2113.20	1096.22	1372.29	1770.04	1737.46	1780.06

APPENDIX I
LIST OF MATHEMATICAL SYMBOLS

Symbol	Definition	Page
d	Number of objects to be matched	1, 4
$C_{is,jt}^{p,q}$	Single cost factor between objects V^p and V^q	1, 4
V^1, V^2, V^p	Vertex sets/objects that need to be matched	1, 4
V	Union over all vertex sets of the matching problem	3, 4
$[d]$	Set of object indices. Abbreviation for the interval $[1, d] \cap \mathbb{N}$	4
$\bar{E}$	Edge set, specifying all possible matchings	4
$\bar{G}$	d -partite Graph	4
$C(i, s, j, t)$	Cost function	4
$C_{is,jt}$	Matrix representation of cost factor $C(i, s, j, t)$	4
E	Subset of edges constituting a matching	4
Q	Clique	4
$\mathcal{Q}$	Vertex/clique partition; A feasible solution	4
$\mathbb{Q}$	Set of all feasible solutions/vertex partitions	4
$\mathbb{Q}^D$	Set of all feasible solutions/vertex partitions, restricted to set of objects D	4
$D(Q)$	set of object indices covered by clique Q	4
Q^p	single vertex, belonging to object p , contained in clique Q	4
$C(\mathcal{Q})$	Total cost of clique partition/solution $\mathcal{Q}$	4